

CHAPTER 12



PROBLEM 12.1

The value of g at any latitude ϕ may be obtained from the formula

$$g = 9.78(1 + 0.0053 \sin^2 \phi) \text{ m/s}^2$$

which takes into account the effect of the rotation of the earth, as well as the fact that the earth is not truly spherical. Determine to four significant figures (a) the weight in N, (b) the mass in kg, at the latitudes of 0° , 45° , and 60° , of a silver bar, the mass of which has been officially designated as 5 kg.

SOLUTION

$$g = 9.78(1 + 0.0053 \sin^2 \phi) \text{ m/s}^2$$

$$\phi = 0^\circ: \quad g = 9.78 \text{ m/s}^2$$

$$\phi = 45^\circ: \quad g = 9.8059 \text{ m/s}^2$$

$$\phi = 60^\circ: \quad g = 9.8189 \text{ m/s}^2$$

(a) Weight: $W = mg$

$$\phi = 0^\circ: \quad W = (5 \text{ kg})(9.78 \text{ m/s}^2) = 48.9 \text{ N} \quad \blacktriangleleft$$

$$\phi = 45^\circ: \quad W = (5 \text{ kg})(9.8059 \text{ m/s}^2) = 49.0 \text{ N} \quad \blacktriangleleft$$

$$\phi = 60^\circ: \quad W = (5 \text{ kg})(9.8189 \text{ m/s}^2) = 49.1 \text{ N} \quad \blacktriangleleft$$

(b) Mass: At all latitudes:

$$m = 5.000 \text{ kg} \quad \blacktriangleleft$$

PROBLEM 12.2

The acceleration due to gravity on the moon is 1.62 m/s^2 . Determine (a) the weight in newtons, (b) the mass in kilograms, on the moon, of a gold bar, the mass of which has been officially designated as 2 kg.

SOLUTION

(a) Weight: $W = mg = (2 \text{ kg})(1.62 \text{ m/s}^2)$ $W = 3.24 \text{ N} \blacktriangleleft$

(b) Mass: Same as on earth: $m = 2.00 \text{ kg} \blacktriangleleft$

PROBLEM 12.3

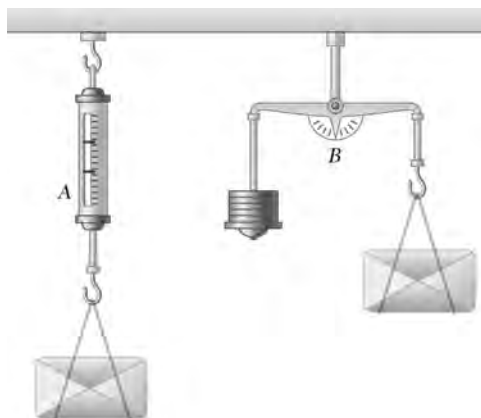
A 200-kg satellite is in a circular orbit 1500 km above the surface of Venus. The acceleration due to the gravitational attraction of Venus at this altitude is 5.52 m/s^2 . Determine the magnitude of the linear momentum of the satellite knowing that its orbital speed is $23.4 \times 10^3 \text{ km/h}$.

SOLUTION

First note $v = 23.4 \times 10^3 \text{ km/h} = 6500 \text{ m/s}$

Now $L = mv = 200 \text{ kg} \times 6500 \text{ m/s}$

or $L = 1.300 \times 10^6 \text{ kg} \cdot \text{m/s} \quad \blacktriangleleft$



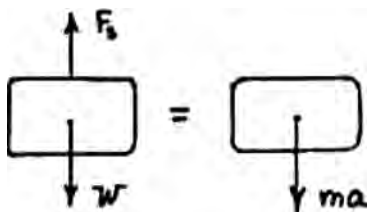
PROBLEM 12.4

A spring scale A and a lever scale B having equal lever arms are fastened to the roof of an elevator, and identical packages are attached to the scales as shown. Knowing that when the elevator moves downward with an acceleration of 1.2 m/s^2 the spring scale indicates a load of 70 N , determine (a) the weight of the packages, (b) the load indicated by the spring scale and the mass needed to balance the lever scale when the elevator moves upward with an acceleration of 1.2 m/s^2 .

SOLUTION

Assume $g = 9.81 \text{ m/s}^2$

(a)



or

$$m = \frac{W}{g}$$

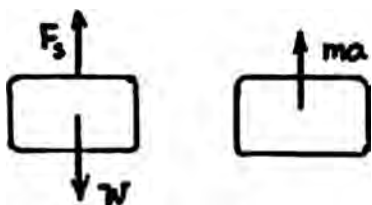
$$+\downarrow \Sigma F = ma: W - F_s = \frac{W}{g} a$$

$$W \left(1 - \frac{a}{g} \right) = F_s$$

$$W = \frac{F_s}{1 - \frac{a}{g}} = \frac{70}{1 - \frac{1.2}{9.81}}$$

$$W = 79.8 \text{ N} \blacktriangleleft$$

(b)



$$\Sigma F = ma: F_s - W = \frac{W}{g} a$$

$$F_s = W \left(1 + \frac{a}{g} \right)$$

$$= 79.756 \left(1 + \frac{1.2}{9.81} \right) \quad F_s = 89.5 \text{ N} \blacktriangleleft$$

PROBLEM 12.4 (Continued)

For the balance system B ,

$$\Sigma M_0 = 0: \quad bF_w - bF_p = 0$$

$$F_w = F_p$$

But

$$F_w = W_w \left(1 + \frac{a}{g} \right)$$

and

$$F_p = W_p \left(1 + \frac{a}{g} \right)$$

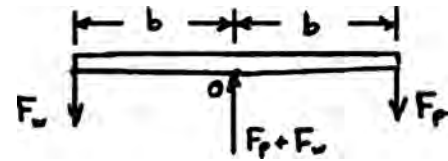
so that

$$W_w = W_p$$

and

$$m_w = \frac{W_p}{g} = \frac{79.756}{9.81}$$

$$m_w = 8.13 \text{ kg} \quad \blacktriangleleft$$



PROBLEM 12.5

A hockey player hits a puck so that it comes to rest in 9 s after sliding 30 m on the ice. Determine (a) the initial velocity of the puck, (b) the coefficient of friction between the puck and the ice.

SOLUTION

(a) Assume uniformly decelerated motion.

Then $v = v_0 + at$

At $t = 9$ s: $0 = v_0 + a(9)$

or $a = -\frac{v_0}{9}$

Also $v^2 = v_0^2 + 2a(x - 0)$

At $t = 9$ s: $0 = v_0^2 + 2a(30)$

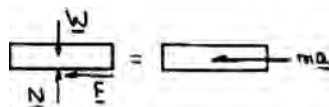
Substituting for a $0 = v_0^2 + 2\left(-\frac{v_0}{9}\right)(30) = 0$

or $v_0 = 6.6667$ m/s

or $v_0 = 6.67$ m/s ◀

and $a = -\frac{6.6667}{9} = -0.74074$ m/s²

(b) We have



$+\uparrow \Sigma F_y = 0: N - W = 0$

or $N = W = mg$

Sliding: $F = \mu_k N = \mu_k mg$

$+\rightarrow \Sigma F_x = ma: -F = ma$

or $-\mu_k mg = ma$

or $\mu_k = -\frac{a}{g} = -\frac{-0.74074 \text{ m/s}^2}{9.81 \text{ m/s}^2}$

or $\mu_k = 0.0755$ ◀

PROBLEM 12.6

Determine the maximum theoretical speed that an automobile starting from rest can reach after traveling 400 m. Assume that the coefficient of static friction is 0.80 between the tires and the pavement and that (a) the automobile has front-wheel drive and the front wheels support 62 percent of the automobile's weight, (b) the automobile has rear-wheel drive and the rear wheels support 43 percent of the automobile's weight.

SOLUTION

(a) For maximum acceleration

$$\begin{aligned} F_F &= F_{\max} = \mu_s N_F = 0.8(0.62 W) \\ &= 0.496 W = 0.496 mg \end{aligned}$$

$$\text{Now } \quad \overset{+}{\rightarrow} \Sigma F_x = ma: \quad F_F = ma$$

$$\text{or} \quad 0.496 mg = ma$$

$$\text{Then} \quad a = 0.496(9.81 \text{ m/s}^2) = 4.86576 \text{ m/s}^2$$

Since a is constant, we have

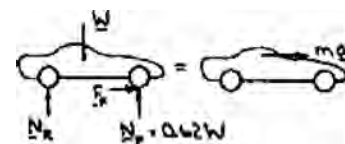
$$v^2 = 0 + 2a(x - 0)$$

$$\text{When } \quad x = 400 \text{ m: } v_{\max}^2 = 2(4.86576 \text{ m/s}^2)(400 \text{ m})$$

$$\text{or} \quad v_{\max} = 62.391 \text{ m/s}$$

or

$$v_{\max} = 225 \text{ km/h} \quad \blacktriangleleft$$



(b) For maximum acceleration

$$\begin{aligned} F_R &= F_{\max} = \mu_s N_R = 0.8(0.43 W) \\ &= 0.344 W = 0.344 mg \end{aligned}$$

$$\text{Now } \quad \overset{+}{\rightarrow} \Sigma F_x = ma: \quad F_R = ma$$

$$\text{or} \quad 0.344 mg = ma$$

$$\text{Then} \quad a = 0.344(9.81 \text{ m/s}^2) = 3.37464 \text{ m/s}^2$$

Since a is constant, we have

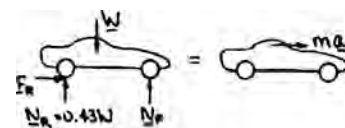
$$v^2 = 0 + 2a(x - 0)$$

$$\text{When } \quad x = 400 \text{ m: } v_{\max}^2 = 2(3.37464 \text{ m/s}^2)(400 \text{ m})$$

$$\text{or} \quad v_{\max} = 51.959 \text{ m/s}$$

or

$$v_{\max} = 187.1 \text{ km/h} \quad \blacktriangleleft$$



PROBLEM 12.7

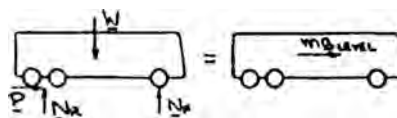
In anticipation of a long 7° upgrade, a bus driver accelerates at a constant rate of 0.9 m/s^2 while still on a level section of the highway. Knowing that the speed of the bus is 100 km/h as it begins to climb the grade and that the driver does not change the setting of his throttle or shift gears, determine the distance traveled by the bus up the grade when its speed has decreased to 80 km/h .

SOLUTION

First consider when the bus is on the level section of the highway.

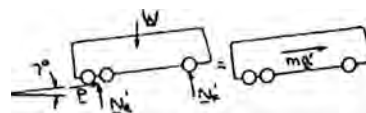
$$a_{\text{level}} = 0.9 \text{ m/s}^2$$

We have
$$+\rightarrow \Sigma F_x = ma: \quad P = \frac{W}{g} a_{\text{level}}$$



Now consider when the bus is on the upgrade.

We have
$$+\nearrow \Sigma F_x = ma: \quad P - W \sin 7^\circ = \frac{W}{g} a'$$



Substituting for P
$$\frac{W}{g} a_{\text{level}} - W \sin 7^\circ = \frac{W}{g} a'$$

or
$$\begin{aligned} a' &= a_{\text{level}} - g \sin 7^\circ \\ &= (0.9 - 9.81 \sin 7^\circ) \text{ m/s}^2 \\ &= -0.29554 \text{ m/s}^2 \end{aligned}$$

For the uniformly decelerated motion

$$v^2 = (v_0)_{\text{upgrade}}^2 + 2a'(x_{\text{upgrade}} - 0)$$

Noting that $100 \text{ km/h} = \frac{250}{9} \text{ m/s}$, then when $v = 80 \text{ km/h} \left(= \frac{5}{0.8} v_0 \right)$, we have

$$\left(0.8 \times \frac{250}{9} \text{ m/s} \right)^2 = \left(\frac{250}{9} \text{ m/s} \right)^2 + 2(-0.29554 \text{ m/s}^2) x_{\text{upgrade}}$$

or
$$x_{\text{upgrade}} = 469.949 \text{ m}$$

or
$$x_{\text{upgrade}} = 0.470 \text{ km} \quad \blacktriangleleft$$

PROBLEM 12.8

If an automobile's braking distance from 100 km/h is 50 m on level pavement, determine the automobile's braking distance from 100 km/h when it is (a) going up a 5° incline, (b) going down a 3-percent incline. Assume the braking force is independent of grade.

SOLUTION

Assume uniformly decelerated motion in all cases.

For braking on the level surface,

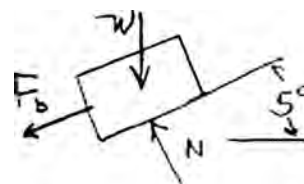
$$\begin{aligned} v_0 &= 100 \text{ km/h} = \frac{250}{9} \text{ m/s}, \quad v_f = 0 \\ x_f - x_0 &= 50 \text{ m} \\ v_f^2 &= v_0^2 + 2a(x_f - x_0) \\ a &= \frac{v_f^2 - v_0^2}{2(x_f - x_0)} \\ &= \frac{0 - \left(\frac{250}{9}\right)^2}{2(50)} \\ &= -7.7160 \text{ m/s}^2 \end{aligned}$$

Braking force.

$$\begin{aligned} F_b &= ma \\ &= \frac{W}{g} a \\ &= -\frac{7.7160}{9.81} W \\ &= -0.78654 W \end{aligned}$$

(a) Going up a 5° incline.

$$\begin{aligned} \sum F &= ma \\ -F_b - W \sin 5^\circ &= \frac{W}{g} a \\ a &= -\frac{F_b + W \sin 5^\circ}{W} g \\ &= -(0.78654 + \sin 5^\circ)(9.81) \\ &= -8.5621 \text{ m/s}^2 \end{aligned}$$

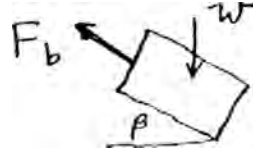


PROBLEM 12.8 (Continued)

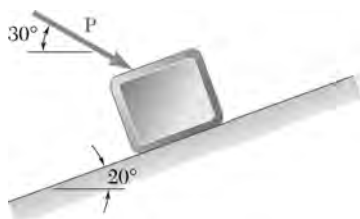
$$\begin{aligned}
 x_f - x_0 &= \frac{v_f^2 - v_0^2}{2a} \\
 &= \frac{0 - \left(\frac{250}{9}\right)^2}{(2)(-8.5621)}
 \end{aligned}
 \qquad x_f - x_0 = 45.1 \text{ m} \blacktriangleleft$$

(b) Going down a 3 percent incline.

$$\begin{aligned}
 \tan \beta &= \frac{3}{100} \quad \beta = 1.71835^\circ \\
 -F_b + W \sin \beta &= \frac{W}{g} a \\
 a &= -(0.78654 - \sin \beta)(9.81) \\
 &= -7.4130 \text{ m/s}^2
 \end{aligned}$$



$$\begin{aligned}
 x_f - x_0 &= \frac{0 - \left(\frac{250}{9}\right)^2}{2(-7.4130)}
 \end{aligned}
 \qquad x_f - x_0 = 52.0 \text{ m} \blacktriangleleft$$



PROBLEM 12.9

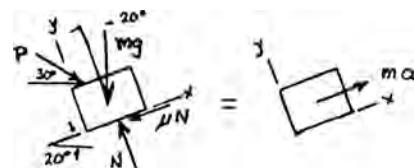
A 20-kg package is at rest on an incline when a force **P** is applied to it. Determine the magnitude of **P** if 10 s is required for the package to travel 5 m up the incline. The static and kinetic coefficients of friction between the package and the incline are both equal to 0.3.

SOLUTION

Kinematics: Uniformly accelerated motion. ($x_0 = 0$, $v_0 = 0$)

$$x = x_0 + v_0 t + \frac{1}{2} a t^2,$$

or
$$a = \frac{2x}{t^2} = \frac{(2)(5)}{(10)^2} = 0.100 \text{ m/s}^2$$



$$+\nearrow \Sigma F_y = 0: N - P \sin 50^\circ - mg \cos 20^\circ = 0$$

$$N = P \sin 50^\circ + mg \cos 20^\circ$$

$$+\nearrow \Sigma F_x = ma: P \cos 50^\circ - mg \sin 20^\circ - \mu N = ma$$

or
$$P \cos 50^\circ - mg \sin 20^\circ - \mu(P \sin 50^\circ + mg \cos 20^\circ) = ma$$

$$P = \frac{ma + mg(\sin 20^\circ + \mu \cos 20^\circ)}{\cos 50^\circ - \mu \sin 50^\circ}$$

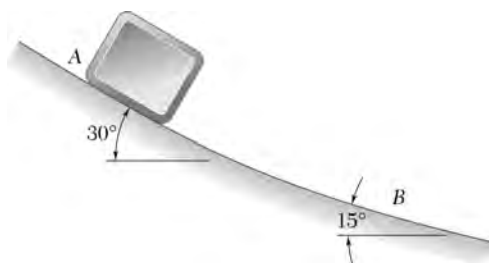
For motion impending, set $a = 0$ and $\mu = \mu_s = 0.4$

$$P = \frac{(20)(0) + (20)(9.81)(\sin 20^\circ + 0.4 \cos 20^\circ)}{\cos 50^\circ - 0.4 \sin 50^\circ}$$

$$= 419 \text{ N} \quad \blacktriangleleft$$

For motion with $a = 0.100 \text{ m/s}^2$, use $\mu = \mu_k = 0.3$.

$$P = \frac{(20)(0.100) + (20)(9.81)(\sin 20^\circ + 0.3 \cos 20^\circ)}{\cos 50^\circ - 0.3 \sin 50^\circ} \quad P = 301 \text{ N} \quad \blacktriangleleft$$



PROBLEM 12.10

The acceleration of a package sliding at Point A is 3 m/s^2 . Assuming that the coefficient of kinetic friction is the same for each section, determine the acceleration of the package at Point B.

SOLUTION

For any angle θ .

Use x and y coordinates as shown.

$$a_y = 0$$

$$+\nearrow \Sigma F_y = ma_y: N - mg \cos \theta = 0$$

$$N = mg \cos \theta$$

$$+\searrow \Sigma F_x = ma_x: mg \sin \theta - \mu_k N = ma_x$$

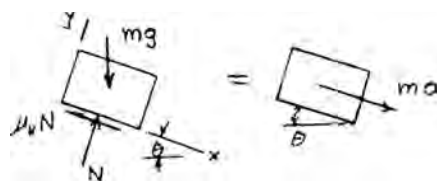
$$a_x = g(\sin \theta - \mu_k \cos \theta)$$

At Point A. $\theta = 30^\circ, a_x = 3 \text{ m/s}^2$

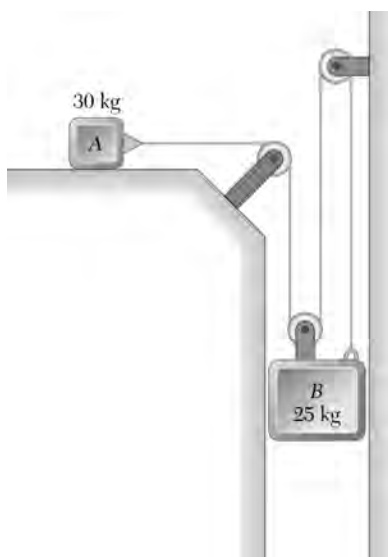
$$\begin{aligned} \mu_k &= \frac{g \sin 30^\circ - a_x}{g \cos 30^\circ} \\ &= \frac{9.81 \sin 30^\circ - 3}{9.81 \cos 30^\circ} \\ &= 0.22423 \end{aligned}$$

At Point B. $\theta = 15^\circ, \mu_k = 0.22423$

$$\begin{aligned} a_x &= 9.81(\sin 15^\circ - 0.22423 \cos 15^\circ) \\ &= 0.414 \text{ m/s}^2 \end{aligned}$$



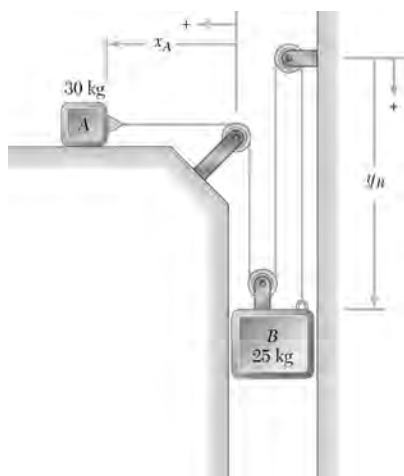
$$\mathbf{a} = 0.414 \text{ m/s}^2 \searrow 15^\circ \blacktriangleleft$$



PROBLEM 12.11

The two blocks shown are originally at rest. Neglecting the masses of the pulleys and the effect of friction in the pulleys and between block A and the horizontal surface, determine (a) the acceleration of each block, (b) the tension in the cable.

SOLUTION



From the diagram

$$x_A + 3y_B = \text{constant}$$

Then $v_A + 3v_B = 0$

and $a_A + 3a_B = 0$

or $a_A = -3a_B$ (1)

(a) A: $\leftarrow^+ \Sigma F_x = m_A a_A: -T = m_A a_B$

Using Eq. (1) $T = 3m_A a_B$

B: $+\downarrow \Sigma F_y = m_B a_B: W_B - 3T = m_B a_B$

Substituting for T

$$m_B g - 3(3m_A a_B) = m_B a_B$$

or
$$a_B = \frac{g}{1 + 9 \frac{m_A}{m_B}} = \frac{9.81 \text{ m/s}^2}{1 + 9 \frac{30 \text{ kg}}{25 \text{ kg}}} = 0.83136 \text{ m/s}^2$$

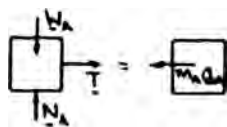
Then $a_A = 2.49 \text{ m/s}^2 \rightarrow \blacktriangleleft$

and $a_B = 0.831 \text{ m/s}^2 \downarrow \blacktriangleleft$

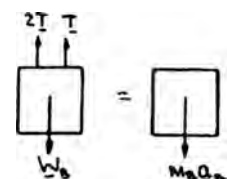
(b) We have $T = 3 \times 30 \text{ kg} \times 0.83136 \text{ m/s}^2$

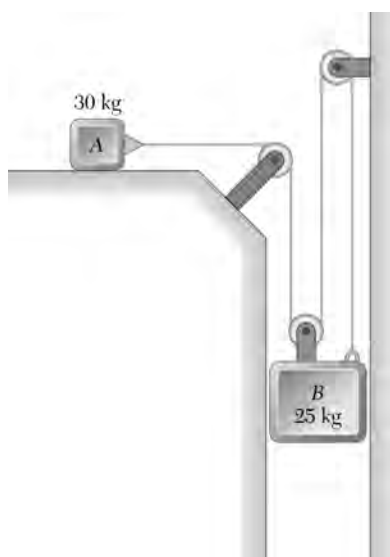
or $T = 74.8 \text{ N} \blacktriangleleft$

A:



B:

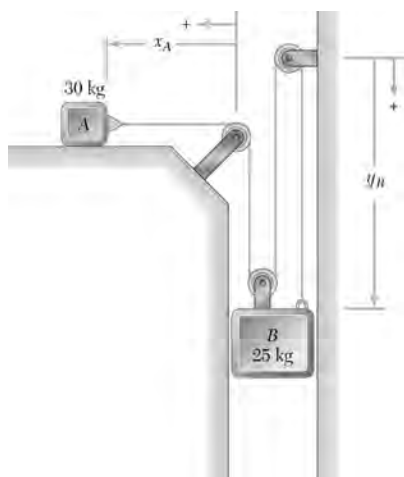




PROBLEM 12.12

The two blocks shown are originally at rest. Neglecting the masses of the pulleys and the effect of friction in the pulleys and assuming that the coefficients of friction between block A and the horizontal surface are $\mu_s = 0.25$ and $\mu_k = 0.20$, determine (a) the acceleration of each block, (b) the tension in the cable.

SOLUTION



From the diagram

$$x_A + 3y_B = \text{constant}$$

Then

$$v_A + 3v_B = 0$$

and

$$a_A + 3a_B = 0$$

or

$$a_A = -3a_B \quad (1)$$

First determine if the blocks will move with $a_A = a_B = 0$. We have

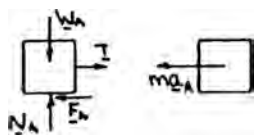
$$A: \quad +\downarrow \Sigma F_y = 0: \quad W_B - 3T = 0 \quad \text{or} \quad T = \frac{1}{3}m_B g$$

$$B: \quad \leftarrow \Sigma F_x = 0: \quad F_A - T = 0$$

A:

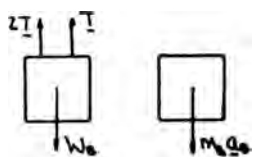
$$\text{Then} \quad F_A = \frac{1}{3} \times 25 \text{ kg} \times 9.81 \text{ m/s}^2 = 81.75 \text{ N}$$

$$+\downarrow \Sigma F_y = 0: \quad W_A - N_A = 0 \quad \text{or} \quad N_A = m_A g$$



B:

$$\begin{aligned} \text{Also,} \quad (F_A)_{\max} &= (\mu_s)_A N_A = (\mu_s)_A m_A g \\ &= 0.25 \times 30 \text{ kg} \times 9.81 \text{ m/s}^2 \\ &= 73.575 \text{ N} \end{aligned}$$



$F_A > (F_A)_{\max}$, which implies that the blocks will move.

PROBLEM 12.12 (Continued)

(a) A: $+\downarrow \Sigma F_y = 0: W_A - N_A = 0 \quad \text{or} \quad N_A = m_A g$

Sliding: $F_A = (\mu_k)_A N_A = 0.20 m_A g$

$\leftarrow +\Sigma F_x = m_A a_A: F_A - T = m_A a_A$

Using Eq. (1) $T = 0.20 m_A g + 3m_A a_B$

B: $+\downarrow \Sigma F_y = m_B a_B: W_B - 3T = m_B a_B$

or $m_B g - 3(0.20 m_A g + 3m_A a_B) = m_B a_B$

or
$$a_B = \frac{g \left(1 - 0.6 \frac{m_A}{m_B} \right)}{1 + 9 \frac{m_A}{m_B}}$$

$$= \frac{(9.81 \text{ m/s}^2) \left(1 - 0.6 \frac{30 \text{ kg}}{25 \text{ kg}} \right)}{1 + 9 \frac{30 \text{ kg}}{25 \text{ kg}}}$$

$$= 0.23278 \text{ m/s}^2$$

Then

$\mathbf{a}_A = 0.698 \text{ m/s}^2 \rightarrow \blacktriangleleft$

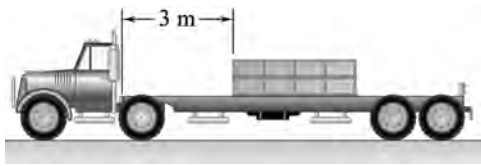
and

$\mathbf{a}_B = 0.233 \text{ m/s}^2 \downarrow \blacktriangleleft$

(b) We have $T = (30 \text{ kg})(0.20 \times 9.81 + 3 \times 0.23278) \text{ m/s}^2$

or

$T = 79.8 \text{ N} \blacktriangleleft$



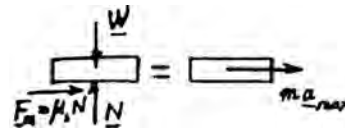
PROBLEM 12.13

The coefficients of friction between the load and the flat-bed trailer shown are $\mu_s = 0.40$ and $\mu_k = 0.30$. Knowing that the speed of the rig is 70 km/h, determine the shortest distance in which the rig can be brought to a stop if the load is not to shift.

SOLUTION

Load: We assume that sliding of load relative to trailer is impending:

$$F = F_m \\ = \mu_s N$$



Deceleration of load is same as deceleration of trailer, which is the maximum allowable deceleration $a_{\max}$.

$$+\uparrow \Sigma F_y = 0: \quad N - W = 0 \quad N = W$$

$$F_m = \mu_s N = 0.40 W$$

$$+\rightarrow \Sigma F_x = ma: \quad F_m = ma_{\max}$$

$$0.40 W = \frac{W}{g} a_{\max}$$

$$a_{\max} = 3.92 \text{ m/s}^2$$

$$a_{\max} = 0.4 \times 9.81 \text{ m/s}^2 = 3.924 \text{ m/s}^2$$

$$a_{\max} = 3.92 \text{ m/s}^2 \rightarrow$$

Uniformly accelerated motion.

$$v^2 = v_0^2 + 2ax \quad \text{with} \quad v = 0$$

$$v_0 = 70 \text{ km/h} = \frac{175}{9} \text{ m/s}$$

$$a = -a_{\max} = -3.924 \text{ m/s}^2$$

$$0 = \left(\frac{175}{9} \right)^2 + 2(-3.924)x$$

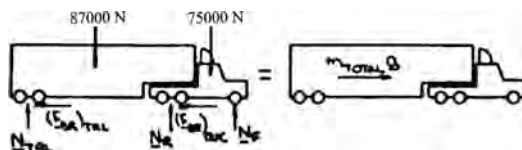
$$x = 48.2 \text{ m} \quad \blacktriangleleft$$



PROBLEM 12.14

A tractor-trailer is traveling at 90 km/h when the driver applies his brakes. Knowing that the braking forces of the tractor and the trailer are 18 kN and 70 kN, respectively, determine (a) the distance traveled by the tractor-trailer before it comes to a stop, (b) the horizontal component of the force in the hitch between the tractor and the trailer while they are slowing down.

SOLUTION



$$(a) \quad \rightarrow \Sigma F_x = ma: \quad -(F_{br})_{trac} - (F_{br})_{trl} = \frac{W_{total}}{g} a$$

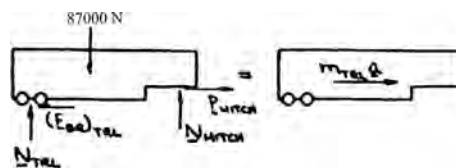
$$\text{or} \quad a = -\frac{9.81 \text{ m/s}^2}{(87,000 + 75,000) \text{ N}} (18,000 + 70,000) \text{ N} \\ = -5.3289 \text{ m/s}^2$$

For uniformly decelerated motion

$$v^2 = v_0^2 + 2a(x - x_0) \quad v_0 = 90 \text{ km/h} = 25 \text{ m/s}$$

$$\text{When } v = 0: \quad 0 = (25 \text{ m/s})^2 + 2(-5.3289 \text{ m/s}^2)(\Delta x)$$

$$\text{or} \quad \Delta x = 58.6 \text{ m} \quad \blacktriangleleft$$

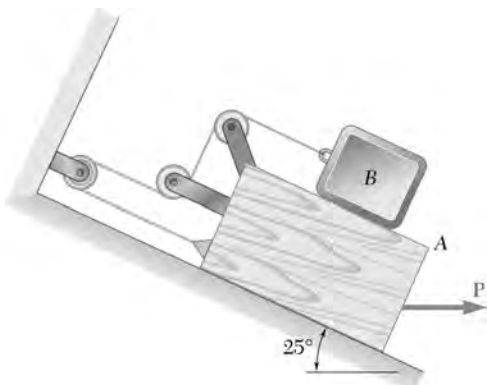


$$(b) \quad \rightarrow \Sigma F_x = m_{trl} a: \quad -(F_{br})_{trl} + P_{hitch} = \frac{W_{trl}}{g} a$$

$$\text{Then} \quad P_{hitch} = 70,000 \text{ N} + \frac{87,000 \text{ N}}{9.81 \text{ m/s}^2} \times (-5.3289 \text{ m/s}^2) \\ = 22,740 \text{ N}$$

$$\text{or} \quad P_{hitch} = 22,700 \text{ N (tension)} \quad \blacktriangleleft$$

PROBLEM 12.15



Block A has a mass of 40 kg, and block B has a mass of 8 kg. The coefficients of friction between all surfaces of contact are $\mu_s = 0.20$ and $\mu_k = 0.15$. If $P = 0$, determine (a) the acceleration of block B, (b) the tension in the cord.

SOLUTION

From the diagram

$$2x_A + x_{B/A} = \text{constant}$$

Then

$$2v_A + v_{B/A} = 0$$

and

$$2a_A + a_{B/A} = 0$$

Now

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$

Then

$$a_B = a_A + (-2a_A)$$

or

$$a_B = -a_A \quad (1)$$

First we determine if the blocks will move for the given value of θ . Thus, we seek the value of θ for which the blocks are in impending motion, with the impending motion of A down the incline.

$$B: \quad +\uparrow \Sigma F_y = 0: \quad N_{AB} - W_B \cos \theta = 0$$

B:

or

$$N_{AB} = m_B g \cos \theta$$

Now

$$F_{AB} = \mu_s N_{AB} \\ = 0.2 m_B g \cos \theta$$

$$+\searrow \Sigma F_x = 0: \quad -T + F_{AB} + W_B \sin \theta = 0$$

or

$$T = m_B g (0.2 \cos \theta + \sin \theta)$$

$$A: \quad +\uparrow \Sigma F_y = 0: \quad N_A - N_{AB} - W_A \cos \theta = 0$$

A:

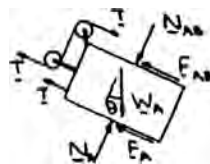
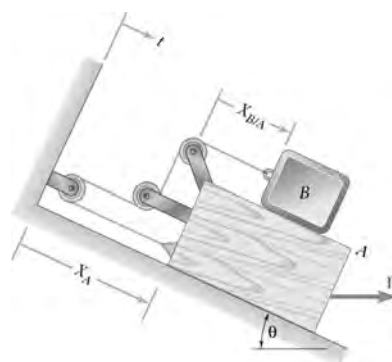
or

$$N_A = (m_A + m_B) g \cos \theta$$

Now

$$F_A = \mu_s N_A \\ = 0.2 (m_A + m_B) g \cos \theta$$

$$+\searrow \Sigma F_x = 0: \quad -T - F_A - F_{AB} + W_A \sin \theta = 0$$



PROBLEM 12.15 (Continued)

or
$$T = m_A g \sin \theta - 0.2(m_A + m_B)g \cos \theta - 0.2m_B g \cos \theta$$
$$= g[m_A \sin \theta - 0.2(m_A + 2m_B) \cos \theta]$$

Equating the two expressions for T

$$m_B g(0.2 \cos \theta + \sin \theta) = g[m_A \sin \theta - 0.2(m_A + 2m_B) \cos \theta]$$

or
$$8(0.2 + \tan \theta) = [40 \tan \theta - 0.2(40 + 2 \times 8)]$$

or
$$\tan \theta = 0.4$$

or $\theta = 21.8^\circ$. For impending motion. Since $\theta < 25^\circ$, the blocks will move. Now consider the motion of the blocks.

(a) $+\nearrow \Sigma F_y = 0: N_{AB} - W_B \cos 25^\circ = 0$

or
$$N_{AB} = m_B g \cos 25^\circ$$

Sliding:
$$F_{AB} = \mu_k N_{AB} = 0.15 m_B g \cos 25^\circ$$

$$+\searrow \Sigma F_x = m_B a_B: -T + F_{AB} + W_B \sin 25^\circ = m_B a_B$$

or
$$T = m_B [g(0.15 \cos 25^\circ + \sin 25^\circ) - a_B]$$
$$= 8[9.81(0.15 \cos 25^\circ + \sin 25^\circ) - a_B]$$
$$= 8(5.47952 - a_B) \quad (\text{N})$$

$$+\nearrow \Sigma F_y = 0: N_A - N_{AB} - W_A \cos 25^\circ = 0$$

or
$$N_A = (m_A + m_B)g \cos 25^\circ$$

Sliding:
$$F_A = \mu_k N_A = 0.15(m_A + m_B)g \cos 25^\circ$$

$$+\searrow \Sigma F_x = m_A a_A: -T - F_A - F_{AB} + W_A \sin 25^\circ = m_A a_A$$

Substituting and using Eq. (1)

$$T = m_A g \sin 25^\circ - 0.15(m_A + m_B)g \cos 25^\circ$$
$$- 0.15m_B g \cos 25^\circ - m_A(-a_B)$$
$$= g[m_A \sin 25^\circ - 0.15(m_A + 2m_B) \cos 25^\circ] + m_A a_B$$
$$= 9.81[40 \sin 25^\circ - 0.15(40 + 2 \times 8) \cos 25^\circ] + 40a_B$$
$$= 91.15202 + 40a_B \quad (\text{N})$$

Equating the two expressions for T

$$8(5.47952 - a_B) = 91.15202 + 40a_B$$

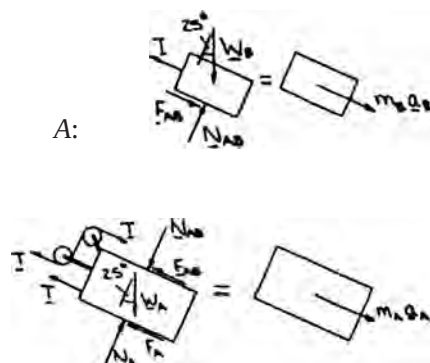
or
$$a_B = -0.98575 \text{ m/s}^2$$

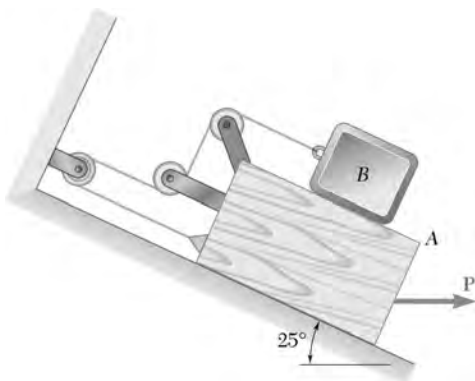
$$a_B = 0.986 \text{ m/s}^2 \nearrow 25^\circ \blacktriangleleft$$

(b) We have
$$T = 8[5.47952 - (-0.98575)]$$

or

$$T = 51.7 \text{ N} \blacktriangleleft$$





PROBLEM 12.16

Block A has a mass of 40 kg, and block B has a mass of 8 kg. The coefficients of friction between all surfaces of contact are $\mu_s = 0.20$ and $\mu_k = 0.15$. If $\mathbf{P} = 40 \text{ N} \rightarrow$, determine (a) the acceleration of block B, (b) the tension in the cord.

SOLUTION

From the diagram

$$2x_A + x_{B/A} = \text{constant}$$

Then

$$2v_A + v_{B/A} = 0$$

and

$$2a_A + a_{B/A} = 0$$

Now

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$

Then

$$a_B = a_A + (-2a_A)$$

or

$$a_B = -a_A \quad (1)$$

First we determine if the blocks will move for the given value of $\mathbf{P}$. Thus, we seek the value of $\mathbf{P}$ for which the blocks are in impending motion, with the impending motion of a down the incline.

$$B: \quad +\nearrow \Sigma F_y = 0: \quad N_{AB} - W_B \cos 25^\circ = 0$$

or

$$N_{AB} = m_B g \cos 25^\circ$$

Now

$$\begin{aligned} F_{AB} &= \mu_s N_{AB} \\ &= 0.2 m_B g \cos 25^\circ \end{aligned}$$

$$+\searrow \Sigma F_x = 0: \quad -T + F_{AB} + W_B \sin 25^\circ = 0$$

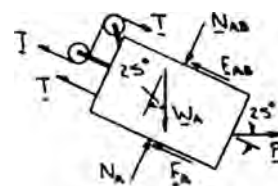
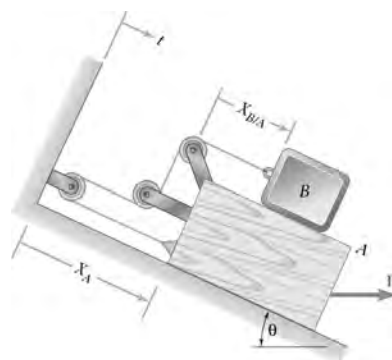
or

$$\begin{aligned} T &= 0.2 m_B g \cos 25^\circ + m_B g \sin 25^\circ \\ &= (8 \text{ kg})(9.81 \text{ m/s}^2)(0.2 \cos 25^\circ + \sin 25^\circ) \\ &= 47.39249 \text{ N} \end{aligned}$$

$$A: \quad +\nearrow \Sigma F_y = 0: \quad N_A - N_{AB} - W_A \cos 25^\circ + P \sin 25^\circ = 0$$

or

$$N_A = (m_A + m_B)g \cos 25^\circ - P \sin 25^\circ$$



PROBLEM 12.16 (Continued)

Now

$$F_A = \mu_s N_A$$

or

$$F_A = 0.2[(m_A + m_B)g \cos 25^\circ - P \sin 25^\circ]$$

$$\sum F_x = 0: -T - F_A - F_{AB} + W_A \sin 25^\circ + P \cos 25^\circ = 0$$

or

$$-T - 0.2[(m_A + m_B)g \cos 25^\circ - P \sin 25^\circ] - 0.2m_B g \cos 25^\circ + m_A g \sin 25^\circ + P \cos 25^\circ = 0$$

or

$$P(0.2 \sin 25^\circ + \cos 25^\circ) = T + 0.2[(m_A + 2m_B)g \cos 25^\circ] - m_A g \sin 25^\circ$$

Then

$$P(0.2 \sin 25^\circ + \cos 25^\circ) = 47.39249 \text{ N} + 9.81 \text{ m/s}^2 \{0.2[(40 + 2 \times 8) \cos 25^\circ - 40 \sin 25^\circ] \text{ kg}\}$$

or

$$P = -19.04 \text{ N for impending motion.}$$

Since $P < 40 \text{ N}$, the blocks will move. Now consider the motion of the blocks.

$$(a) \quad \sum F_y = 0: N_{AB} - W_B \cos 25^\circ = 0$$

or

$$N_{AB} = m_B g \cos 25^\circ$$

Sliding:

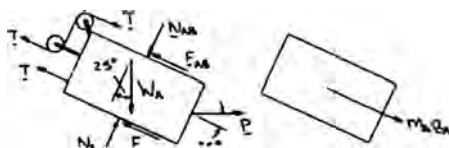
$$\begin{aligned} F_{AB} &= \mu_k N_{AB} \\ &= 0.15 m_B g \cos 25^\circ \end{aligned}$$

$$\sum F_x = m_B a_B: -T + F_{AB} + W_B \sin 25^\circ = m_B a_B$$

or

$$\begin{aligned} T &= m_B [g(0.15 \cos 25^\circ + \sin 25^\circ) - a_B] \\ &= 8[9.81(0.15 \cos 25^\circ + \sin 25^\circ) - a_B] \\ &= 8(5.47952 - a_B) \quad (\text{N}) \end{aligned}$$

A:



$$\sum F_y = 0: N_A - N_{AB} - W_A \cos 25^\circ + P \sin 25^\circ = 0$$

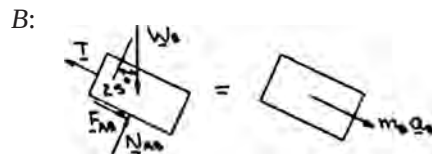
or

$$N_A = (m_A + m_B)g \cos 25^\circ - P \sin 25^\circ$$

Sliding:

$$\begin{aligned} F_A &= \mu_k N_A \\ &= 0.15[(m_A + m_B)g \cos 25^\circ - P \sin 25^\circ] \end{aligned}$$

$$\sum F_x = m_A a_A: -T - F_A - F_{AB} + W_A \sin 25^\circ + P \cos 25^\circ = m_A a_A$$



PROBLEM 12.16 (Continued)

Substituting and using Eq. (1)

$$\begin{aligned} T &= m_A g \sin 25^\circ - 0.15[(m_A + m_B)g \cos 25^\circ - P \sin 25^\circ] \\ &\quad - 0.15 m_B g \cos 25^\circ + P \cos 25^\circ - m_A(-a_B) \\ &= g[m_A \sin 25^\circ - 0.15(m_A + 2m_B) \cos 25^\circ] \\ &\quad + P(0.15 \sin 25^\circ + \cos 25^\circ) + m_A a_B \\ &= 9.81[40 \sin 25^\circ - 0.15(40 + 2 \times 8) \cos 25^\circ] \\ &\quad + 40(0.15 \sin 25^\circ + \cos 25^\circ) + 40a_B \\ &= 129.94004 a_B \quad (\text{N}) \end{aligned}$$

Equating the two expressions for T

$$8(5.47952 - a_B) = 129.94004 + 40a_B$$

or

$$a_B = -1.79383 \text{ m/s}^2$$

$$a_B = 1.794 \text{ m/s}^2 \nearrow 25^\circ \blacktriangleleft$$

(b) We have

$$T = 8[5.47952 - (-1.79383)]$$

or

$$T = 58.2 \text{ N} \blacktriangleleft$$



PROBLEM 12.17

Boxes *A* and *B* are at rest on a conveyor belt that is initially at rest. The belt is suddenly started in an upward direction so that slipping occurs between the belt and the boxes. Knowing that the coefficients of kinetic friction between the belt and the boxes are $(\mu_k)_A = 0.30$ and $(\mu_k)_B = 0.32$, determine the initial acceleration of each box.

SOLUTION

Assume that $a_B > a_A$ so that the normal force N_{AB} between the boxes is zero.

$$A: \quad \uparrow \Sigma F_y = 0: \quad N_A - W_A \cos 15^\circ = 0$$

or

$$N_A = W_A \cos 15^\circ$$

Slipping:

$$F_A = (\mu_k)_A N_A \\ = 0.3 W_A \cos 15^\circ$$

$$+\nearrow \Sigma F_x = m_A a_A: \quad F_A - W_A \sin 15^\circ = m_A a_A$$

$$\text{or} \quad 0.3 W_A \cos 15^\circ - W_A \sin 15^\circ = \frac{W_A}{g} a_A$$

or

$$a_A = (9.81 \text{ m/s}^2)(0.3 \cos 15^\circ - \sin 15^\circ) \\ = 0.3037 \text{ m/s}^2$$

$$B: \quad +\uparrow \Sigma F_y = 0: \quad N_B - W_B \cos 15^\circ = 0$$

or

$$N_B = W_B \cos 15^\circ$$

Slipping:

$$F_B = (\mu_k)_B N_B \\ = 0.32 W_B \cos 15^\circ$$

$$+\nearrow \Sigma F_x = m_B a_B: \quad F_B - W_B \sin 15^\circ = m_B a_B$$

$$\text{or} \quad 0.32 W_B \cos 15^\circ - W_B \sin 15^\circ = \frac{W_B}{g} a_B$$

or

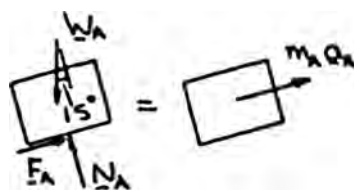
$$a_B = (9.81 \text{ m/s}^2)(0.32 \cos 15^\circ - \sin 15^\circ) = 0.49322 \text{ m/s}^2$$

$$a_B > a_A \Rightarrow \text{assumption is correct}$$

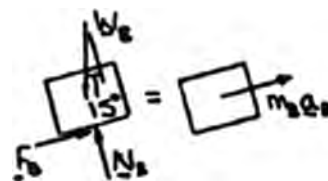
$$a_A = 0.304 \text{ m/s}^2 \nearrow 15^\circ \blacktriangleleft$$

$$a_B = 0.493 \text{ m/s}^2 \nearrow 15^\circ \blacktriangleleft$$

A:



B:



PROBLEM 12.17 (Continued)

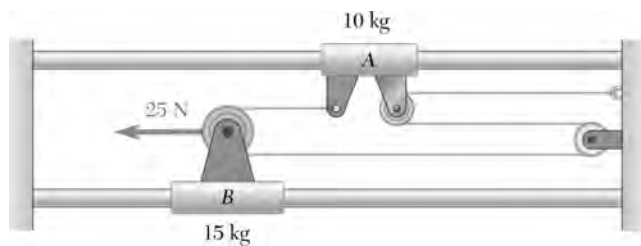
Note: If it is assumed that the boxes remain in contact ($N_{AB} \neq 0$), then

$$a_A = a_B \quad \text{and find} \quad (\Sigma F_x = ma)$$

$$A: \quad 0.3W_A \cos 15^\circ - W_A \sin 15^\circ - N_{AB} = \frac{W_A}{g} a$$

$$B: \quad 0.32W_B \cos 15^\circ - W_B \sin 15^\circ + N_{AB} = \frac{W_B}{g} a$$

Solving yields $a = 0.3878 \text{ m/s}^2$ and $N_{AB} = -4.2168 \text{ N}$, which contradicts the assumption.



PROBLEM 12.18

Knowing that the system shown starts from rest, find the velocity at $t = 1.2$ s of (a) collar A, (b) collar B. Neglect the masses of the pulleys and the effect of friction.

SOLUTION

Referring to solution of Problem 11.51 we note that

$$a_B = -\frac{1}{2}a_A \quad (1)$$

where minus sign indicates that $\mathbf{a}_A$ and $\mathbf{a}_B$ have opposite sense.

Block B.

$$\leftarrow \Sigma F = ma: 25 - 2T = 7.5a_A \quad (2)$$

Block A.

$$\begin{aligned} \rightarrow \Sigma F = ma: 2T - T &= 10a_A \\ T &= 10a_A \end{aligned} \quad (3)$$

(a) Substituting for T from (3) into (2):

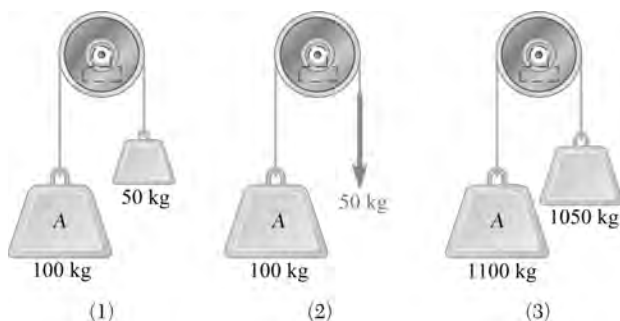
$$\begin{aligned} 25 - 2(10 a_A) &= 7.5a_A \\ 25 &= 27.5a_A \end{aligned} \quad \mathbf{a}_A = 0.909 \text{ m/s}^2 \rightarrow$$

$$v_A = (v_A)_0 + a_A t = 0 + 0.909(1.2), \quad \mathbf{v}_A = 1.091 \text{ m/s} \rightarrow \blacktriangleleft$$

(b) Substituting $a = 0.909$ into Eq. (1):

$$\mathbf{a}_B = 0.455 \text{ m/s}^2 \leftarrow$$

$$v_B = (v_B)_0 + a_B t = 0 + 0.455(1.2) \quad \mathbf{v}_B = 0.545 \text{ m/s} \leftarrow \blacktriangleleft$$



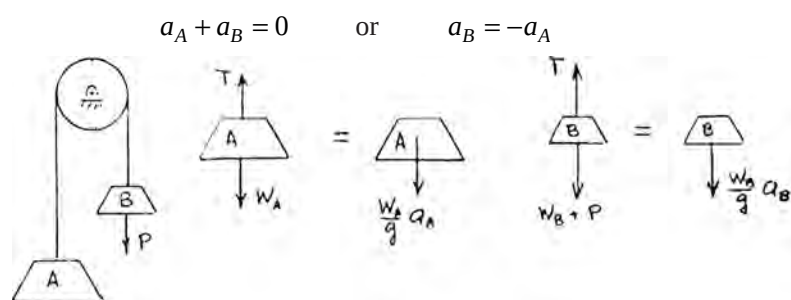
PROBLEM 12.19

Each of the systems shown is initially at rest. Neglecting axle friction and the masses of the pulleys, determine for each system (a) the acceleration of block A, (b) the velocity of block A after it has moved through 3 m, (c) the time required for block A to reach a velocity of 6 m/s.

SOLUTION

Let y be positive downward for both blocks.

Constraint of cable: $y_A + y_B = \text{constant}$



For blocks A and B, $\downarrow \Sigma F = ma$:

Block A:
$$W_A - T = \frac{W_A}{g} a_A \quad \text{or} \quad T = W_A - \frac{W_A}{g} a_A$$

Block B:
$$P + W_B - T = \frac{W_B}{g} a_B = -\frac{W_B}{g} a_A$$

$$P + W_B - W_A + \frac{W_A}{g} a_A = -\frac{W_B}{g} a_A$$

Solving for a_A ,
$$a_A = \frac{W_A - W_B - P}{W_A + W_B} g \quad (1)$$

$$v_A^2 - (v_A)_0^2 = 2a_A[y_A - (y_A)_0] \quad \text{with} \quad (v_A)_0 = 0$$

$$v_A = \sqrt{2a_A[y_A - (y_A)_0]} \quad (2)$$

$$v_A - (v_A)_0 = a_A t \quad \text{with} \quad (v_A)_0 = 0$$

$$t = \frac{v_A}{a_A} \quad (3)$$

PROBLEM 12.19 (Continued)

(a) Acceleration of block A.

System (1): $W_A = 100 \text{ g}, \quad W_B = 50 \text{ g}, \quad P = 0$

By formula (1), $(a_A)_1 = \frac{100 - 50}{100 + 50}(9.81) \qquad (a_A)_1 = 3.27 \text{ m/s}^2 \downarrow \blacktriangleleft$

System (2): $W_A = 100 \text{ g}, \quad W_B = 0, \quad P = 50 \text{ g N}$

By formula (1), $(a_A)_2 = \frac{100 - 50}{100}(9.81) = 4.905 \text{ m/s}^2 \qquad (a_A)_2 = 4.91 \text{ m/s}^2 \downarrow \blacktriangleleft$

System (3): $W_A = 1100 \text{ g}, \quad W_B = 1050 \text{ g}, \quad P = 0$

By formula (1), $(a_A)_3 = \frac{1100 - 1050}{1100 + 1050}(9.81) \qquad (a_A)_3 = 0.228 \text{ m/s}^2 \downarrow \blacktriangleleft$

(b) v_A at $y_A - (y_A)_0 = 3 \text{ m}$. Use formula (2).

System (1): $(v_A)_1 = \sqrt{(2)(3.27)(3)} \qquad (v_A)_1 = 4.43 \text{ m/s} \downarrow \blacktriangleleft$

System (2): $(v_A)_2 = \sqrt{(2)(4.905)(3)} \qquad (v_A)_2 = 5.42 \text{ m/s} \downarrow \blacktriangleleft$

System (3): $(v_A)_3 = \sqrt{(2)(0.228)(3)} \qquad (v_A)_3 = 1.170 \text{ m/s} \downarrow \blacktriangleleft$

(c) Time at $v_A = 6 \text{ m/s}$. Use formula (3).

System (1): $t_1 = \frac{6}{3.27} \qquad t_1 = 1.835 \text{ s} \blacktriangleleft$

System (2): $t_2 = \frac{6}{4.905} \qquad t_2 = 1.223 \text{ s} \blacktriangleleft$

System (3): $t_3 = \frac{6}{0.228} \qquad t_3 = 26.3 \text{ s} \blacktriangleleft$



PROBLEM 12.20

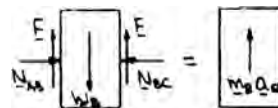
A man standing in an elevator that is moving with a constant acceleration holds a 3-kg block B between two other blocks in such a way that the motion of B relative to A and C is impending. Knowing that the coefficients of friction between all surfaces are $\mu_s = 0.30$ and $\mu_k = 0.25$, determine (a) the acceleration of the elevator if it is moving upward and each of the forces exerted by the man on blocks A and C has a horizontal component equal to twice the weight of B , (b) the horizontal components of the forces exerted by the man on blocks A and C if the acceleration of the elevator is 2.0 m/s^2 downward.

SOLUTION

First we observe that because B is not moving relative to A and to C that $\mathbf{a}_B = \mathbf{a}_{EL}$.

(a) We have

$$\begin{aligned} F &= \mu_s N \\ &= 0.30(2W_B) \\ &= 0.6W_B = 0.6m_B g \end{aligned}$$



For $\mathbf{a}_{EL}$ to be $\uparrow$, the net vertical force must be $\uparrow$, which requires that the frictional forces be acting as shown. It then follows that the impending motion of B relative to A and C is downward. Then

$$+\uparrow \Sigma F_y = m_B a_{EL}: 2F - W_B = m_B a_{EL}$$

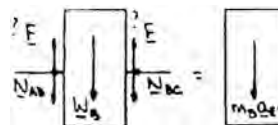
$$\text{or} \quad 2(0.6m_B g) - m_B g = m_B a_{EL}$$

$$\text{or} \quad a_{EL} = 0.2 \times 9.81 \text{ m/s}^2$$

$$\text{or} \quad a_{EL} = 1.962 \text{ m/s}^2 \uparrow \blacktriangleleft$$

(b) We have

$$\begin{aligned} F &= \mu_s N \\ &= 0.30N \end{aligned}$$



Now we observe that because the direction of the impending motion is unknown, the directions of the frictional forces is also unknown (although $\mathbf{F}_{\text{net}}$ must be downward).

$$+\uparrow \Sigma F_y = m_B a_{EL}: \pm 2F - W_B = -m_B |a_{EL}|$$

PROBLEM 12.20 (Continued)

or

$$\begin{aligned}\pm 2F &= m_B(g - |a_{EL}|) \\ &= 3 \text{ kg} \times (9.81 - 2) \text{ m/s}^2\end{aligned}$$

Since the magnitude of $\mathbf{F}$ must be positive, it then follows that $\mathbf{F} \uparrow$, and that the impending motion of B relative to A and C is downward. Finally

$$2(0.30 \text{ N}) = 3 \text{ kg} \times (9.81 - 2) \text{ m/s}^2$$

or

$$N_{AB} = N_{BC} = 39.1 \text{ N} \quad \blacktriangleleft$$

PROBLEM 12.21



A package is at rest on a conveyor belt, which is initially at rest. The belt is started and moves to the right for 1.3 s with a constant acceleration of 2 m/s^2 . The belt then moves with a constant deceleration $\mathbf{a}_2$ and comes to a stop after a total displacement of 2.2 m. Knowing that the coefficients of friction between the package and the belt are $\mu_s = 0.35$ and $\mu_k = 0.25$, determine (a) the deceleration $\mathbf{a}_2$ of the belt, (b) the displacement of the package relative to the belt as the belt comes to a stop.

SOLUTION

(a) Kinematics of the belt. $v_o = 0$

1. Acceleration phase: with $a_1 = 2 \text{ m/s}^2 \rightarrow$

$$v_1 = v_o + a_1 t_1 = 0 + (2)(1.3) = 2.6 \text{ m/s}$$

$$x_1 = x_o + v_o t_1 + \frac{1}{2} a_1 t_1^2 = 0 + 0 + \frac{1}{2} (2)(1.3)^2 = 1.69 \text{ m}$$

2. Deceleration phase: $v_2 = 0$ since the belt stops.

$$v_2^2 - v_1^2 = 2a_2(x_2 - x_1)$$

$$a_2 = \frac{v_2^2 - v_1^2}{2(x_2 - x_1)} = \frac{0 - (2.6)^2}{2(2.2 - 1.69)} = -6.63$$

$$\mathbf{a}_2 = 6.63 \text{ m/s}^2 \leftarrow \blacktriangleleft$$

$$t_2 - t_1 = \frac{v_2 - v_1}{a_2} = \frac{0 - 2.6}{-6.63} = 0.3923 \text{ s}$$

(b) Motion of the package

1. Acceleration phase. Assume no slip. $(a_p)_1 = 2 \text{ m/s}^2 \rightarrow$

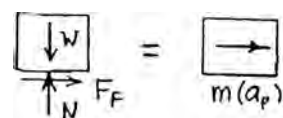
$$\Sigma F_y = 0: N - W = 0 \text{ or } N = W = mg$$

$$\rightarrow \Sigma F_x = ma: F_f = m(a_p)_1$$

The required friction force is F_f .

The available friction force is $\mu_s N = 0.35W = 0.35mg$

$$\frac{F_f}{m} = (a_p)_1 < \frac{\mu_s N}{m} = \mu_s g = (0.35)(9.81) = 3.43 \text{ m/s}^2$$



PROBLEM 12.21 (Continued)

Since $2.0 \text{ m/s}^2 < 3.43 \text{ m/s}^2$, the package does not slip.

$$(v_p)_1 = v_1 = 2.6 \text{ m/s} \quad \text{and} \quad (x_p)_1 = 1.69 \text{ m.}$$

2. Deceleration phase. Assume no slip. $(a_p)_2 = -11.52 \text{ m/s}^2$

$$\rightarrow \Sigma F_x = ma: \quad -F_f = m(a_p)_2$$

$$\frac{F_f}{m} = (a_p)_2 = -6.63 \text{ m/s}^2$$

$$\frac{\mu_s N}{m} = \frac{\mu_s mg}{m} = \mu_s g = 3.43 \text{ m/s}^2 < 6.63 \text{ m/s}^2$$

Since the available friction force $\mu_s N$ is less than the required friction force F_f for no slip, the package does slip.

$$(a_p)_2 < 6.63 \text{ m/s}^2, \quad F_f = \mu_k N$$

$$\rightarrow \Sigma F_x = m(a_p)_2: \quad -\mu_k N = m(a_p)_2$$

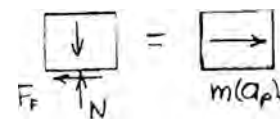
$$\begin{aligned} (a_p)_2 &= -\frac{\mu_k N}{m} = -\mu_k g \\ &= -(0.25)(9.81) \\ &= -2.4525 \text{ m/s}^2 \end{aligned}$$

$$\begin{aligned} (v_p)_2 &= (v_p)_1 + (a_p)_2(t_2 - t_1) \\ &= 2.6 + (-2.4525)(0.3923) \\ &= 1.638 \text{ m/s}^2 \end{aligned}$$

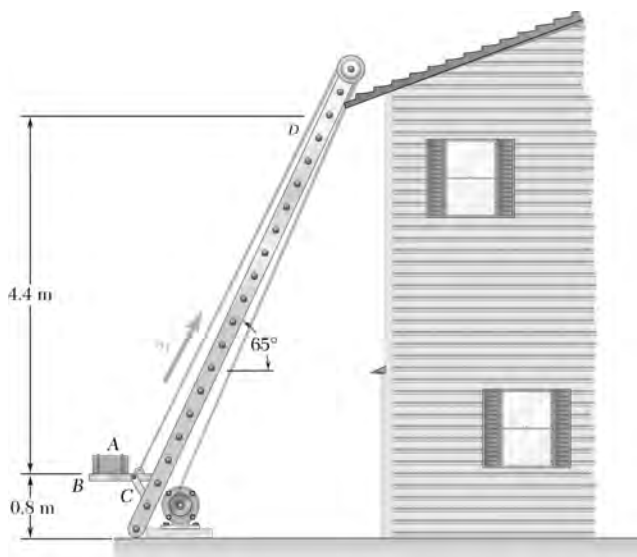
$$\begin{aligned} (x_p)_2 &= (x_p)_1 + (v_p)_1(t_2 - t_1) + \frac{1}{2}(a_p)_2(t_2 - t_1)^2 \\ &= 1.69 + (2.6)(0.3923) + \frac{1}{2}(-2.4525)(0.3923)^2 \\ &= 2.521 \text{ m} \end{aligned}$$

Position of package relative to the belt

$$(x_p)_2 - x_2 = 2.521 - 2.2 = 0.321$$



$$x_{p/\text{belt}} = 0.321 \text{ m} \rightarrow \blacktriangleleft$$



PROBLEM 12.22

To transport a series of bundles of shingles A to a roof, a contractor uses a motor-driven lift consisting of a horizontal platform BC which rides on rails attached to the sides of a ladder. The lift starts from rest and initially moves with a constant acceleration $\mathbf{a}_1$ as shown. The lift then decelerates at a constant rate $\mathbf{a}_2$ and comes to rest at D , near the top of the ladder. Knowing that the coefficient of static friction between a bundle of shingles and the horizontal platform is 0.30, determine the largest allowable acceleration $\mathbf{a}_1$ and the largest allowable deceleration $\mathbf{a}_2$ if the bundle is not to slide on the platform.

SOLUTION

Acceleration $\mathbf{a}_1$: Impending slip. $F_1 = \mu_s N_1 = 0.30 N_1$

$$\begin{aligned}\Sigma F_y = m_A a_y: \quad N_1 - W_A &= m_A a_1 \sin 65^\circ \\ N_1 &= W_A + m_A a_1 \sin 65^\circ \\ &= m_A (g + a_1 \sin 65^\circ)\end{aligned}$$

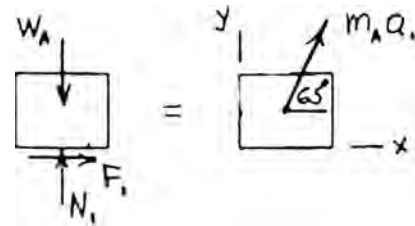
$$\begin{aligned}\rightarrow \Sigma F_x = m_A a_x: \quad F_1 &= m_A a_1 \cos 65^\circ \\ F_1 &= \mu_s N\end{aligned}$$

or $m_A a_1 \cos 65^\circ = 0.30 m_A (g + a_1 \sin 65^\circ)$

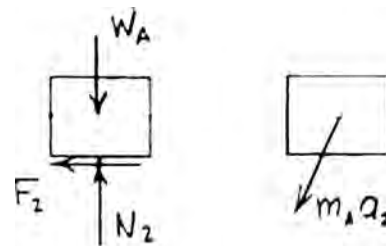
$$\begin{aligned}a_1 &= \frac{0.30g}{\cos 65^\circ - 0.30 \sin 65^\circ} \\ &= (1.990)(9.81) \\ &= 19.53 \text{ m/s}^2\end{aligned}$$

Deceleration $\mathbf{a}_2$: Impending slip. $F_2 = \mu_s N_2 = 0.30 N_2$

$$\begin{aligned}\Sigma F_y = m a_y: \quad N_1 - W_A &= -m_A a_2 \sin 65^\circ \\ N_1 &= W_A - m_A a_2 \sin 65^\circ \\ \leftarrow \Sigma F_x = m a_x: \quad F_2 &= m_A a_2 \cos 65^\circ \\ F_2 &= \mu_s N_2\end{aligned}$$



$$\mathbf{a}_1 = 19.53 \text{ m/s}^2 \nearrow 65^\circ \blacktriangleleft$$



PROBLEM 12.22 (Continued)

or

$$m_A a_2 \cos 65^\circ = 0.30 m_A (g - a_2 \cos 65^\circ)$$

$$a_2 = \frac{0.30g}{\cos 65^\circ + 0.30 \sin 65^\circ}$$

$$= (0.432)(9.81)$$

$$= 4.24 \text{ m/s}^2$$

$$\mathbf{a}_2 = 4.24 \text{ m/s}^2 \nearrow 65^\circ \blacktriangleleft$$

PROBLEM 12.23



To unload a bound stack of plywood from a truck, the driver first tilts the bed of the truck and then accelerates from rest. Knowing that the coefficients of friction between the bottom sheet of plywood and the bed are $\mu_s = 0.40$ and $\mu_k = 0.30$, determine (a) the smallest acceleration of the truck which will cause the stack of plywood to slide, (b) the acceleration of the truck which causes corner A of the stack to reach the end of the bed in 0.9 s.

SOLUTION

Let $\mathbf{a}_P$ be the acceleration of the plywood, $\mathbf{a}_T$ be the acceleration of the truck, and $\mathbf{a}_{P/T}$ be the acceleration of the plywood relative to the truck.

(a) Find the value of $\mathbf{a}_T$ so that the relative motion of the plywood with respect to the truck is impending.

$$a_P = a_T \text{ and } F_1 = \mu_s N_1 = 0.40 N_1$$

$$+\nearrow \Sigma F_y = m_P a_y: N_1 - W_P \cos 20^\circ = -m_P a_T \sin 20^\circ$$

$$N_1 = m_P (g \cos 20^\circ - a_T \sin 20^\circ)$$

$$+\nearrow \Sigma F_x = m a_x: F_1 - W_P \sin 20^\circ = m_P a_T \cos 20^\circ$$

$$F_1 = m_P (g \sin 20^\circ + a_T \cos 20^\circ)$$

$$m_P (g \sin 20^\circ + a_T \cos 20^\circ) = 0.40 m_P (g \cos 20^\circ - a_T \sin 20^\circ)$$

$$\begin{aligned} a_T &= \frac{(0.40 \cos 20^\circ - \sin 20^\circ)}{\cos 20^\circ + 0.40 \sin 20^\circ} g \\ &= (0.03145)(9.81) \\ &= 0.309 \end{aligned}$$

$$\mathbf{a}_T = 0.309 \text{ m/s}^2 \rightarrow \blacktriangleleft$$

$$(b) \quad x_{P/T} = (x_{P/T})_0 + (v_{P/T})t + \frac{1}{2} a_{P/T} t^2 = 0 + 0 + \frac{1}{2} a_{P/T} t^2$$

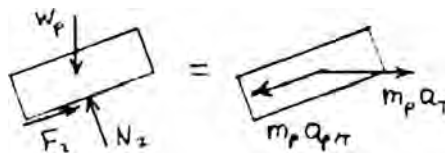
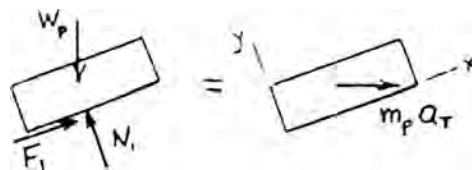
$$a_{P/T} = \frac{2x_{P/T}}{t^2} = \frac{(2)(2)}{(0.9)^2} = 4.94 \text{ m/s}^2$$

$$\mathbf{a}_{P/T} = 4.94 \text{ m/s}^2 \nearrow 20^\circ$$

$$\mathbf{a}_P = \mathbf{a}_T + \mathbf{a}_{P/T} = (a_T \rightarrow) + (4.94 \text{ m/s}^2 \nearrow 20^\circ)$$

$$+\nearrow F_y = m_P a_y: N_2 - W_P \cos 20^\circ = -m_P a_T \sin 20^\circ$$

$$N_2 = m_P (g \cos 20^\circ - a_T \sin 20^\circ)$$



PROBLEM 12.23 (Continued)

$$+\nearrow \Sigma F_x = \Sigma ma_x: F_2 - W_P \sin 20^\circ = m_P a_T \cos 20^\circ - m_P a_{P/T}$$

$$F_2 = m_P (g \sin 20^\circ + a_T \cos 20^\circ - a_{P/T})$$

For sliding with friction $F_2 = \mu_k N_2 = 0.30 N_2$

$$m_P (g \sin 20^\circ + a_T \cos 20^\circ - a_{P/T}) = 0.30 m_P (g \cos 20^\circ + a_T \sin 20^\circ)$$

$$a_T = \frac{(0.30 \cos 20^\circ - \sin 20^\circ)g + a_{P/T}}{\cos 20^\circ + 0.30 \sin 20^\circ}$$

$$= (-0.05767)(9.81) + (0.9594)(4.94)$$

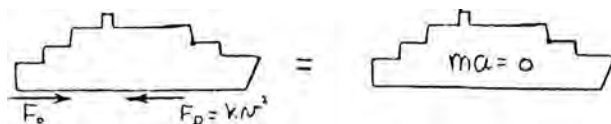
$$= 4.17$$

$$\mathbf{a_T = 4.17 \text{ m/s}^2 \rightarrow \blacktriangleleft}$$

PROBLEM 12.24

The propellers of a ship of weight W can produce a propulsive force F_0 ; they produce a force of the same magnitude but of opposite direction when the engines are reversed. Knowing that the ship was proceeding forward at its maximum speed v_0 when the engines were put into reverse, determine the distance the ship travels before coming to a stop. Assume that the frictional resistance of the water varies directly with the square of the velocity.

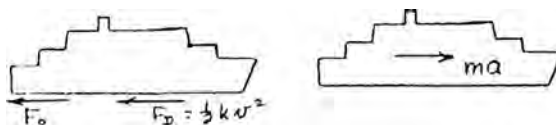
SOLUTION



At maximum speed $a = 0$.

$$F_0 = kv_0^2 \quad k = \frac{F_0}{v_0^2}$$

When the propellers are reversed, F_0 is reversed.



$$\sum F_x = ma: -F_0 - kv^2 = ma$$

$$-F_0 - F_0 \frac{v^2}{v_0^2} = ma \quad a = -\frac{F_0}{mv_0^2} (v_0^2 + v^2)$$

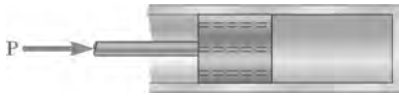
$$dx = \frac{v dv}{a} = -\frac{mv_0^2 v dv}{F_0 (v_0^2 + v^2)}$$

$$\int_0^x dx = -\frac{mv_0^2}{F_0} \int_{v_0}^0 \frac{v dv}{v_0^2 + v^2}$$

$$x = -\frac{mv_0^2}{F_0} \left[\frac{1}{2} \ln(v_0^2 + v^2) \right]_{v_0}^0$$

$$= -\frac{mv_0^2}{2F_0} [\ln v_0^2 - \ln(2v_0^2)] = \frac{mv_0^2}{2F_0} \ln 2 \quad x = 0.347 \frac{m_0 v_0^2}{F_0} \blacktriangleleft$$

PROBLEM 12.25



A constant force $\mathbf{P}$ is applied to a piston and rod of total mass m to make them move in a cylinder filled with oil. As the piston moves, the oil is forced through orifices in the piston and exerts on the piston a force of magnitude kv in a direction opposite to the motion of the piston. Knowing that the piston starts from rest at $t = 0$ and $x = 0$, show that the equation relating x , v , and t , where x is the distance traveled by the piston and v is the speed of the piston, is linear in each of the variables.

SOLUTION

$$\rightarrow \Sigma F = ma: \quad P - kv = ma$$

$$\frac{dv}{dt} = a = \frac{P - kv}{m}$$

$$\int_0^t dt = \int_0^v \frac{m dv}{P - kv}$$

$$= -\frac{m}{k} \ln(P - kv) \Big|_0^v$$

$$= -\frac{m}{k} [\ln(P - kv) - \ln P]$$

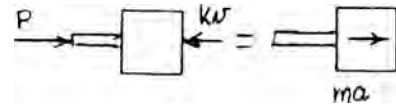
$$t = -\frac{m}{k} \ln \frac{P - kv}{P} \quad \text{or} \quad \ln \frac{P - kv}{P} = -\frac{kt}{m}$$

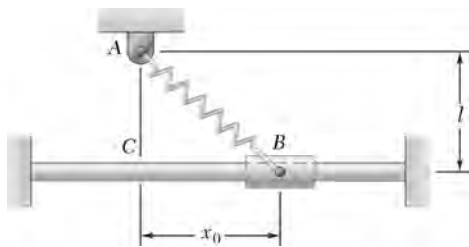
$$\frac{P - kv}{P} = e^{-kt/m} \quad \text{or} \quad v = \frac{P}{k} (1 - e^{-kt/m})$$

$$x = \int_0^t v dt = \frac{Pt}{k} \Big|_0^t - \frac{P}{k} \left(-\frac{k}{m} e^{-kt/m} \right) \Big|_0^t$$

$$= \frac{Pt}{k} + \frac{P}{m} (e^{-kt/m} - 1) = \frac{Pt}{k} - \frac{P}{m} (1 - e^{-kt/m})$$

$$x = \frac{Pt}{k} - \frac{kv}{m}, \text{ which is linear.}$$





PROBLEM 12.26

A spring AB of constant k is attached to a support at A and to a collar of mass m . The unstretched length of the spring is ℓ . Knowing that the collar is released from rest at $x = x_0$ and neglecting friction between the collar and the horizontal rod, determine the magnitude of the velocity of the collar as it passes through Point C .

SOLUTION

Choose the origin at Point C and let x be positive to the right. Then x is a position coordinate of the slider B and x_0 is its initial value. Let L be the stretched length of the spring. Then, from the right triangle

$$L = \sqrt{\ell^2 + x^2}$$

The elongation of the spring is $e = L - \ell$, and the magnitude of the force exerted by the spring is

$$F_s = ke = k(\sqrt{\ell^2 + x^2} - \ell)$$

By geometry,

$$\cos \theta = \frac{x}{\sqrt{\ell^2 + x^2}}$$

$$+\rightarrow \Sigma F_x = ma_x: -F_s \cos \theta = ma$$

$$-k(\sqrt{\ell^2 + x^2} - \ell) \frac{x}{\sqrt{\ell^2 + x^2}} = ma$$

$$a = -\frac{k}{m} \left(x - \frac{\ell x}{\sqrt{\ell^2 + x^2}} \right)$$

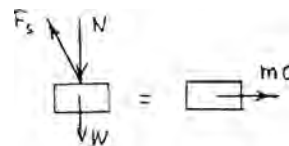
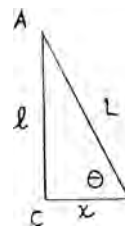
$$\int_0^v v \, dv = \int_{x_0}^0 a \, dx$$

$$\frac{1}{2} v^2 \Big|_0^v = -\frac{k}{m} \int_{x_0}^0 \left(x - \frac{\ell x}{\sqrt{\ell^2 + x^2}} \right) dx = -\frac{k}{m} \left(\frac{1}{2} x^2 - \ell \sqrt{\ell^2 + x^2} \right) \Big|_{x_0}^0$$

$$\frac{1}{2} v^2 = -\frac{k}{m} \left(0 - \ell^2 - \frac{1}{2} x_0^2 + \ell \sqrt{\ell^2 + x_0^2} \right)$$

$$\begin{aligned} v^2 &= \frac{k}{m} \left(2\ell^2 + x_0^2 - 2\ell \sqrt{\ell^2 + x_0^2} \right) \\ &= \frac{k}{m} \left[(\ell^2 + x_0^2) - 2\ell \sqrt{\ell^2 + x_0^2} + \ell^2 \right] \end{aligned}$$

$$\text{answer: } v = \sqrt{\frac{k}{m} (\sqrt{\ell^2 + x_0^2} - \ell)} \quad \blacktriangleleft$$



PROBLEM 12.27

Determine the maximum theoretical speed that a 1350 kg automobile starting from rest can reach after traveling a 400 m if air resistance is considered. Assume that the coefficient of static friction between the tires and the pavement is 0.70, that the automobile has front-wheel drive, that the front wheels support 62 percent of the automobile's weight, and that the aerodynamic drag **D** has a magnitude $D = 0.667 v^2$, where D and v are expressed in N and m/s, respectively.

SOLUTION

$$F = F_{\max} \text{ for } v = v_{\max}$$

$$F = \mu_s N_F = 0.70(0.62 mg) \\ = 0.434 mg$$

$$\rightarrow \Sigma F_x = ma: F - D = ma$$

$$\text{or} \quad a = \frac{(0.434 mg - 0.667 v^2)}{m}$$

$$\text{Now} \quad v \frac{dv}{dx} = a = \frac{(0.434 mg - 0.667 v^2)}{m}$$

$$\text{At } x = 0, v = 0: \quad \frac{1}{m} \int_0^x dx = \int_0^v \frac{v dv}{0.434 mg - 0.667 v^2}$$

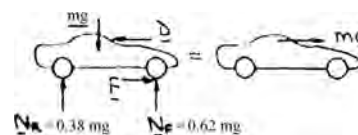
$$\text{or} \quad \frac{1}{m} x = \frac{-1}{2(0.667)} \ln [0.434 mg - 0.667 v^2]_0^v$$

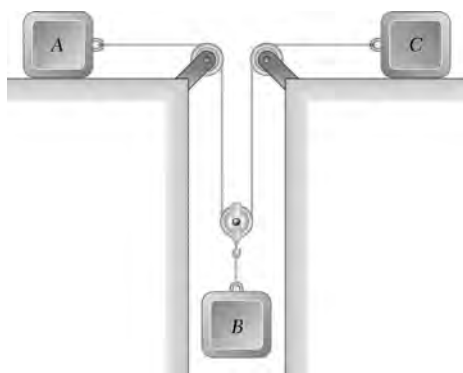
$$\frac{1}{m} x = \frac{-1}{1.334} \ln \left(\frac{0.434 mg - 0.667 v^2}{0.434 mg} \right)$$

$$\text{or} \quad \frac{0.434 mg - 0.667 v^2}{0.434 mg} = \exp \left(\frac{-1.334 x}{m} \right)$$

$$\text{Substituting } m = 1350 \text{ kg}, g = 9.81 \text{ m/s}^2, x = 400 \text{ m} \Rightarrow v_{\max} = 53.042 \text{ m/s} = 190.951 \text{ km/h}$$

$$\text{or} \quad v_{\max} = 191.0 \text{ km/h} \quad \blacktriangleleft$$





PROBLEM 12.28

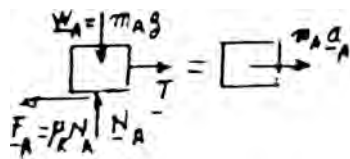
The coefficients of friction between blocks A and C and the horizontal surfaces are $\mu_s = 0.24$ and $\mu_k = 0.20$. Knowing that $m_A = 5$ kg, $m_B = 10$ kg, and $m_C = 10$ kg, determine (a) the tension in the cord, (b) the acceleration of each block.

SOLUTION

We first check that static equilibrium is not maintained:

$$\begin{aligned}(F_A)_m + (F_C)_m &= \mu_s(m_A + m_C)g \\ &= 0.24(5 + 10)g \\ &= 3.6g\end{aligned}$$

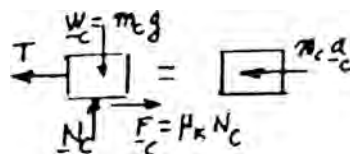
Since $W_B = m_B g = 10g > 3.6g$, equilibrium is *not* maintained.



Block A: $\Sigma F_y: N_A = m_A g$

$$F_A = \mu_k N_A = 0.2m_A g$$

$$\rightarrow \Sigma F_x = m_A a_A: T - 0.2m_A g = m_A a_A \quad (1)$$



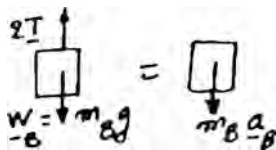
Block C: $\Sigma F_y: N_C = m_C g$

$$F_C = \mu_k N_C = 0.2m_C g$$

$$\leftarrow \Sigma F_x = m_C a_C: T - 0.2m_C g = m_C a_C \quad (2)$$

Block B: $\downarrow \Sigma F_y = m_B a_B$

$$m_B g - 2T = m_B a_B \quad (3)$$



From kinematics: $a_B = \frac{1}{2}(a_A + a_C) \quad (4)$

(a) Tension in cord. Given data: $m_A = 5$ kg

$$m_B = m_C = 10 \text{ kg}$$

$$\text{Eq. (1): } T - 0.2(5)g = 5a_A \quad a_A = 0.2T - 0.2g \quad (5)$$

$$\text{Eq. (2): } T - 0.2(10)g = 10a_C \quad a_C = 0.1T - 0.2g \quad (6)$$

$$\text{Eq. (3): } 10g - 2T = 10a_B \quad a_B = g - 0.2T \quad (7)$$

PROBLEM 12.28 (Continued)

Substitute into (4):

$$g - 0.2T = \frac{1}{2}(0.2T - 0.2g + 0.1T - 0.2g)$$

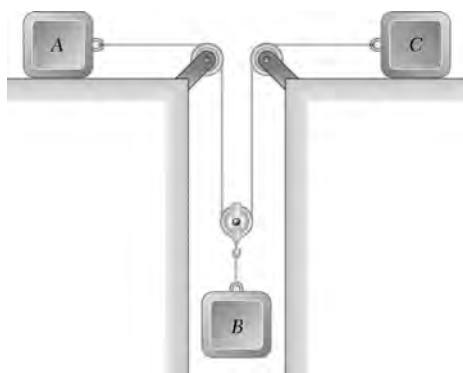
$$1.2g = 0.35T \quad T = \frac{24}{7}g = \frac{24}{7}(9.81 \text{ m/s}^2) \quad T = 33.6 \text{ N} \blacktriangleleft$$

(b) Substitute for T into (5), (7), and (6):

$$a_A = 0.2\left(\frac{24}{7}g\right) - 0.2g = 0.4857(9.81 \text{ m/s}^2) \quad \mathbf{a}_A = 4.76 \text{ m/s}^2 \rightarrow \blacktriangleleft$$

$$a_B = g - 0.2\left(\frac{24}{7}g\right) = 0.3143(9.81 \text{ m/s}^2) \quad \mathbf{a}_B = 3.08 \text{ m/s}^2 \downarrow \blacktriangleleft$$

$$a_C = 0.1\left(\frac{24}{7}g\right) - 0.2g = 0.14286(9.81 \text{ m/s}^2) \quad \mathbf{a}_C = 1.401 \text{ m/s}^2 \leftarrow \blacktriangleleft$$



PROBLEM 12.29

Solve Problem 12.28, assuming that $m_A = 5$ kg, $m_B = 10$ kg, and $m_C = 20$ kg.

PROBLEM 12.28 The coefficients of friction between blocks A and C and the horizontal surfaces are $\mu_s = 0.24$ and $\mu_k = 0.20$. Knowing that $m_A = 5$ kg, $m_B = 10$ kg, and $m_C = 10$ kg, determine (a) the tension in the cord, (b) the acceleration of each block.

SOLUTION

We first check that static equilibrium is not maintained:

$$(F_A)_m + (F_C)_m = \mu_s(m_A + m_C)g = 0.24(5 + 20)g = 6g$$

Since

$$W_B = m_B g = 10g > 6g, \text{ equilibrium is not maintained.}$$

We shall assume that all 3 blocks move and use Eqs. (1), (2), (3), (4) derived in solution of Problem 12.28.

(a) Tension in cord.

Given data: $m_A = 5$ kg, $m_B = 10$ kg, $m_C = 20$ kg

$$\text{Eq. (1): } T - 0.2(5)g = 5a_A \quad a_A = 0.2T - 0.2g \quad (5)$$

$$\text{Eq. (2): } T - 0.2(20)g = 20a_C \quad a_C = 0.05T - 0.2g \quad (6)$$

$$\text{Eq. (3): } 10g - 2T = 10a_B \quad a_B = g - 0.2T \quad (7)$$

Substituting into (4):

$$\begin{aligned} g - 0.2T &= \frac{1}{2}(0.2T - 0.2g + 0.05T - 0.2g) \\ 1.2g &= 0.325T \quad T = \frac{48}{13}g = \frac{48}{13}(9.81 \text{ m/s}^2) \quad T = 36.2 \text{ N} < \end{aligned}$$

$$\text{Substituting into (6): } a_C = 0.05\left(\frac{48}{13}g\right) - 0.2g = -0.0154g \text{ (impossible)}$$

This means that our assumption that block C moves was wrong.

PROBLEM 12.29 (Continued)

Assuming now that C does not move, we have

$$a_C = 0 \quad \blacktriangleleft$$

Substituting $a_C = 0$ and from (5) and (6) into Eq. (4):

$$g - 0.2T = \frac{1}{2}(0.2T - 0.2g)$$
$$1.1g = 0.3T \quad T = \frac{11}{3}g = \frac{11}{3}(9.81 \text{ m/s}^2) = 35.97 \text{ N}$$

$$T = 36.0 \text{ N} \quad \blacktriangleleft$$

Substituting for T into Eqs. (5) and (7):

$$a_A = 0.2\left(\frac{11}{3}g\right) - 0.2g = 0.5333(9.81 \text{ m/s}^2) \quad \mathbf{a}_A = 5.23 \text{ m/s}^2 \rightarrow \quad \blacktriangleleft$$

$$a_B = g - 0.2\left(\frac{11}{3}g\right) = 0.2667(9.81 \text{ m/s}^2) \quad \mathbf{a}_B = 2.62 \text{ m/s}^2 \downarrow \quad \blacktriangleleft$$

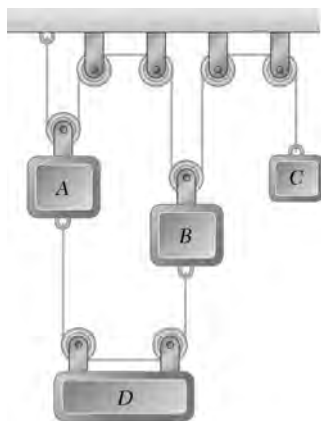
We also check that $T = \frac{11}{3}g$ is less than $(F_C)_m = 0.24(20g) = 4.8g$.

The value assumed for a_C is thus correct:

$$\mathbf{a}_C = 0 \quad \blacktriangleleft$$

as well as that obtained for T :

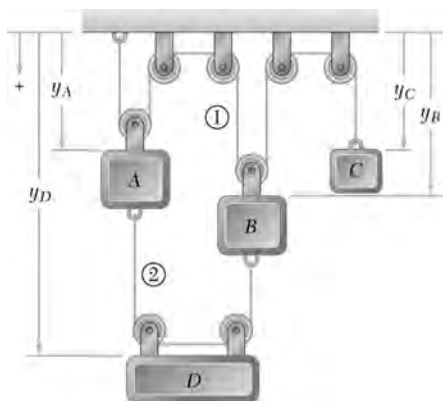
$$T = 36.0 \text{ N} \quad \blacktriangleleft$$



PROBLEM 12.30

Blocks A and B weigh 10 kg each, block C weighs 7 kg, and block D weighs 8 kg. Knowing that a downward force of magnitude 120 N is applied to block D, determine (a) the acceleration of each block, (b) the tension in cord ABC. Neglect the weights of the pulleys and the effect of friction.

SOLUTION



Note: As shown, the system is in equilibrium.

From the diagram:

$$\text{Cord 1:} \quad 2y_A + 2y_B + y_C = \text{constant}$$

$$\text{Then} \quad 2v_A + 2v_B + v_C = 0$$

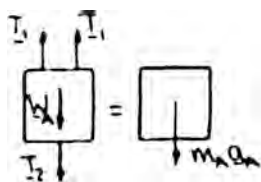
$$\text{and} \quad 2a_A + 2a_B + a_C = 0 \quad (1)$$

$$\text{Cord 2:} \quad (y_D - y_A) + (y_D - y_B) = \text{constant}$$

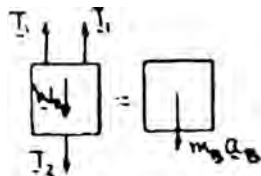
$$\text{Then} \quad 2v_D - v_A - v_B = 0$$

$$\text{and} \quad 2a_D - a_A - a_B = 0 \quad (2)$$

A:



B:



$$(a) \quad +\downarrow \Sigma F_y = m_A a_A: \quad m_A g - 2T_1 + T_2 = m_A a_A$$

$$\text{or} \quad 10g - 2T_1 + T_2 = 10a_A \quad (3)$$

$$+\downarrow \Sigma F_y = m_B a_B: \quad m_B g - 2T_1 + T_2 = m_B a_B$$

$$\text{or} \quad 10g - 2T_1 + T_2 = 10a_B \quad (4)$$

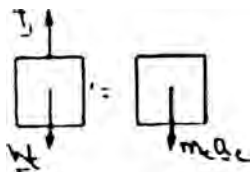
$$\text{Note: Eqs. (3) and (4)} \Rightarrow \mathbf{a_A = a_B}$$

$$\text{Then} \quad \text{Eq. (1)} \Rightarrow a_C = -4a_A$$

$$\text{Eq. (2)} \Rightarrow a_D = a_A$$

PROBLEM 12.30 (Continued)

C:



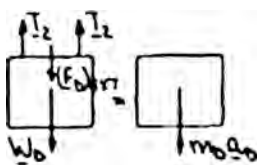
$$+\downarrow \Sigma F_y = m_C a_C: m_C g - T_1 = m_C a_C$$

$$\text{or} \quad T_1 = 7(g - a_C)$$

$$T_1 = 7(g + 4a_A) \quad (5)$$

$$+\downarrow \Sigma F_y = m_D a_D: m_D g - 2T_2 + (F_D)_{\text{ext}} = m_D a_D$$

D:



$$\text{or} \quad T_2 = \frac{1}{2}[8(g - a_D) + 120]$$

$$T_2 = 4g - 4a_A + 60$$

Substituting for T_1 [Eq. (5)] and T_2 [Eq. (6)] in Eq. (3)

$$10g - 14(g + 4a_A) + (4g - 4a_A + 60) = 10a_A$$

$$\Rightarrow -56a_A - 4a_A + 60 = 10a_A$$

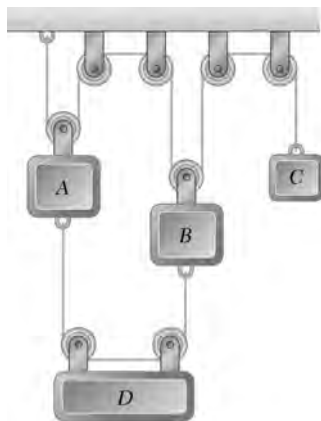
$$\text{or} \quad a_A = \frac{6}{7} \text{ m/s}^2$$

$$\mathbf{a_A = a_B = a_D = 0.857 \text{ m/s}^2 \downarrow \blacktriangleleft}$$

$$\text{and} \quad a_C = -\frac{24}{7} \text{ m/s}^2 \quad \text{or} \quad \mathbf{a_C = 3.43 \text{ m/s}^2 \uparrow \blacktriangleleft}$$

(b) Substituting into Eq. (5)

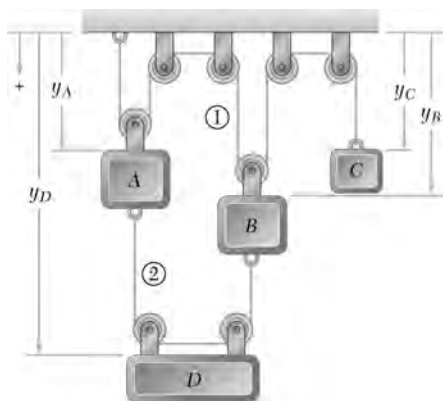
$$T_1 = 7\left(9.81 + \frac{24}{7}\right) \quad \text{or} \quad \mathbf{T_1 = 92.7 \text{ N} \blacktriangleleft}$$



PROBLEM 12.31

Blocks *A* and *B* weigh 10 kg each, block *C* weighs 7 kg, and block *D* weighs 8 kg. Knowing that a downward force of magnitude 120 N is applied to block *B* and that the system starts from rest, determine at $t = 3$ s the velocity (*a*) of *D* relative to *A*, (*b*) of *C* relative to *D*. Neglect the weights of the pulleys and the effect of friction.

SOLUTION



Note: As shown, the system is in equilibrium.

From the diagram:

Cord 1: $2y_A + 2y_B + y_C = \text{constant}$

Then $2v_A + 2v_B + v_C = 0$

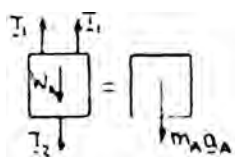
and $2a_A + 2a_B + a_C = 0$ (1)

Cord 2: $(y_D - y_A) + (y_D - y_B) = \text{constant}$

Then $2v_D - v_A - v_B = 0$

and $2a_D - a_A - a_B = 0$ (2)

A:



(a) $+\downarrow \Sigma F_y = m_A a_A: m_A g - 2T_1 + T_2 = m_A a_A$

or $10g - 2T_1 + T_2 = 10a_A$ (3)

$+\downarrow \Sigma F_y = m_B a_B: m_B g - 2T_1 + T_2 + (F_B)_{\text{ext}} = m_B a_B$

or $10g - 2T_1 + T_2 + 50 = 10a_B$ (4)

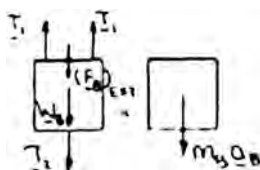
Forming (3) - (4) $\Rightarrow -50 = 10(a_A - a_B)$

or $a_B = a_A + 5$

$-5 = a_A - a_B$

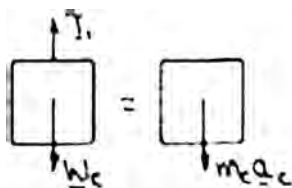
$a_B = a_A + 5$

B:



PROBLEM 12.31 (Continued)

C:



Then Eq. (1): $2a_A + 2(a_A + 5) + a_C = 0$

or $a_C = -4a_A - 10$

Eq. (2): $2a_D - a_A - (a_A + 5) = 0$

or $a_D = a_A + \frac{5}{2}$

$\downarrow \Sigma F_y = m_C a_C: m_C g - T_1 = m_C a_C$

or $T_1 = 7(g - a_C) = 7(g + 4a_A + 10)$ (5)

$\downarrow \Sigma F_y = m_D a_D: m_D g - 2T_2 = m_D a_D$

or $T_2 = \frac{1}{2} \times 8(g - a_D) = 4\left(g - a_A - \frac{5}{2}\right)$ (6)

Substituting for T_1 [Eq. (5)] and T_2 [Eq. (6)] in Eq. (3)

$$10g - 14(g + 4a_A + 10) + 4\left(g - a_A - \frac{5}{2}\right) = 10a_A$$

$$\Rightarrow 10g - 14g - 56a_A - 140 + 4g - 4a_A - 10 = 10a_A$$

or $a_A = -\frac{15}{7} \text{ m/s}^2$

Then $a_C = -4 \times \left(-\frac{15}{7}\right) - 10 = \frac{-10}{7} \text{ m/s}^2$

$$a_D = \frac{-15}{7} + \frac{5}{2} = \frac{5}{14} \text{ m/s}^2$$

Note: We have uniformly accelerated motion, so that

$$v = 0 + at$$

(a) We have $\mathbf{v}_{D/A} = \mathbf{v}_D - \mathbf{v}_A$

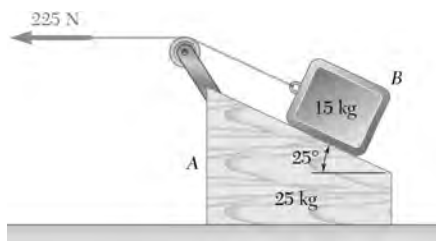
or $\mathbf{v}_{D/A} = a_D t - a_A t = \left[\frac{5}{14} - \left(-\frac{15}{7}\right)\right] \frac{\text{m}}{\text{s}^2} \times 3 \text{ s}$

or $\mathbf{v}_{D/A} = 7.50 \text{ m/s} \downarrow \blacktriangleleft$

(b) And $\mathbf{v}_{C/D} = \mathbf{v}_C - \mathbf{v}_D$

or $\mathbf{v}_{C/D} = a_C t - a_D t = \left(\frac{-10}{7} - \frac{5}{14}\right) \frac{\text{m}}{\text{s}^2} \times 3 \text{ s} = \frac{-75}{14} \text{ m/s}$

or $\mathbf{v}_{C/D} = 5.36 \text{ m/s} \uparrow \blacktriangleleft$



PROBLEM 12.32

The 15-kg block B is supported by the 25-kg block A and is attached to a cord to which a 225-N horizontal force is applied as shown. Neglecting friction, determine (a) the acceleration of block A, (b) the acceleration of block B relative to A.

SOLUTION

(a) First we note $\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$, where $\mathbf{a}_{B/A}$ is directed along the inclined surface of A.

B: $\leftarrow^+ \Sigma F_x = m_B a_x: P - W_B \sin 25^\circ = m_B a_A \cos 25^\circ + m_B a_{B/A}$

or $225 - 15g \sin 25^\circ = 15(a_A \cos 25^\circ + a_{B/A})$

or $15 - g \sin 25^\circ = a_A \cos 25^\circ + a_{B/A}$

$\nearrow^+ \Sigma F_y = m_B a_y: N_{AB} - W_B \cos 25^\circ = -m_B a_A \sin 25^\circ$

or $N_{AB} = 15(g \cos 25^\circ - a_A \sin 25^\circ)$

A: $\leftarrow^+ \Sigma F_x = m_A a_A: P - P \cos 25^\circ + N_{AB} \sin 25^\circ = m_A a_A$

or $N_{AB} = [25a_A - 225(1 - \cos 25^\circ)] / \sin 25^\circ$

Equating the two expressions for N_{AB}

$$15(g \cos 25^\circ - a_A \sin 25^\circ) = \frac{25a_A - 225(1 - \cos 25^\circ)}{\sin 25^\circ}$$

or
$$a_A = \frac{3(9.81) \cos 25^\circ \sin 25^\circ + 45(1 - \cos 25^\circ)}{5 + 3 \sin^2 25^\circ}$$

$$= 2.7979 \text{ m/s}^2$$

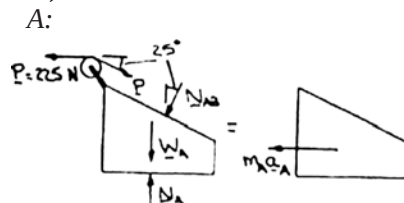
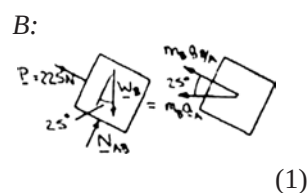
$\mathbf{a}_A = 2.80 \text{ m/s}^2 \leftarrow$

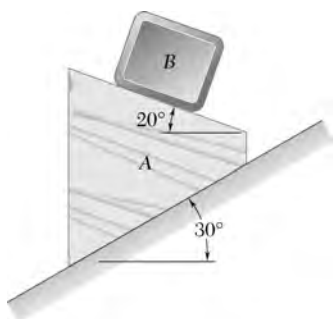
(b) From Eq. (1)

$$a_{B/A} = 15 - (9.81) \sin 25^\circ - 2.7979 \cos 25^\circ$$

or

$\mathbf{a}_{B/A} = 8.32 \text{ m/s}^2 \nearrow 25^\circ$



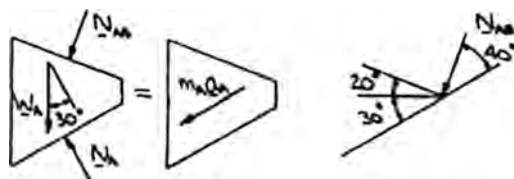


PROBLEM 12.33

Block B of mass 10 kg rests as shown on the upper surface of a 22-kg wedge A . Knowing that the system is released from rest and neglecting friction, determine (a) the acceleration of B , (b) the velocity of B relative to A at $t = 0.5$ s.

SOLUTION

A:

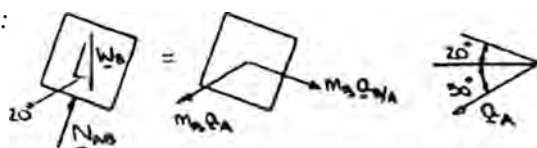


$$(a) \quad \sum F_x = m_A a_A: \quad W_A \sin 30^\circ + N_{AB} \cos 40^\circ = m_A a_A$$

$$\text{or} \quad N_{AB} = \frac{22 \left(a_A - \frac{1}{2} g \right)}{\cos 40^\circ}$$

Now we note: $\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$, where $\mathbf{a}_{B/A}$ is directed along the top surface of A .

B:



$$+\uparrow \sum F_y = m_B a_y: \quad N_{AB} - W_B \cos 20^\circ = -m_B a_A \sin 50^\circ$$

$$\text{or} \quad N_{AB} = 10 (g \cos 20^\circ - a_A \sin 50^\circ)$$

Equating the two expressions for N_{AB}

$$\frac{22 \left(a_A - \frac{1}{2} g \right)}{\cos 40^\circ} = 10 (g \cos 20^\circ - a_A \sin 50^\circ)$$

$$\text{or} \quad a_A = \frac{(9.81)(1.1 + \cos 20^\circ \cos 40^\circ)}{2.2 + \cos 40^\circ \sin 50^\circ} = 6.4061 \text{ m/s}^2$$

$$+\searrow \sum F_x = m_B a_x: \quad W_B \sin 20^\circ = m_B a_{B/A} - m_B a_A \cos 50^\circ$$

PROBLEM 12.33 (Continued)

$$\begin{aligned}\text{or} \quad a_{B/A} &= g \sin 20^\circ + a_A \cos 50^\circ \\ &= (9.81 \sin 20^\circ + 6.4061 \cos 50^\circ) \text{ m/s}^2 \\ &= 7.4730 \text{ m/s}^2\end{aligned}$$

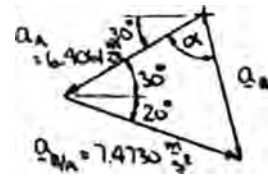
$$\text{Finally} \quad \mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$

$$\text{We have} \quad a_B^2 = 6.4061^2 + 7.4730^2 - 2(6.4061 \times 7.4730) \cos 50^\circ$$

$$\text{or} \quad a_B = 5.9447 \text{ m/s}^2$$

$$\text{and} \quad \frac{7.4730}{\sin \alpha} = \frac{5.9447}{\sin 50^\circ}$$

$$\text{or} \quad \alpha = 74.4^\circ$$



$$\mathbf{a}_B = 5.94 \text{ m/s}^2 \nearrow 75.6^\circ \blacktriangleleft$$

(b) *Note:* We have uniformly accelerated motion, so that

$$\mathbf{v} = \mathbf{0} + \mathbf{a}t$$

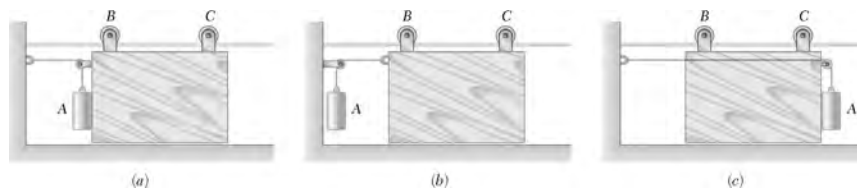
$$\text{Now} \quad \mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = \mathbf{a}_B t - \mathbf{a}_A t = \mathbf{a}_{B/A} t$$

$$\text{At } t = 0.5 \text{ s:} \quad v_{B/A} = 7.4730 \text{ m/s}^2 \times 0.5 \text{ s}$$

$$\text{or} \quad \mathbf{v}_{B/A} = 3.74 \text{ m/s} \nearrow 20^\circ \blacktriangleleft$$

PROBLEM 12.34

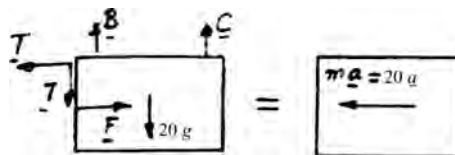
A 20-kg sliding panel is supported by rollers at B and C . A 12.5-kg counterweight A is attached to a cable as shown and, in cases a and c , is initially in contact with a vertical edge of the panel. Neglecting friction, determine in each case shown the acceleration of the panel and the tension in the cord immediately after the system is released from rest.



SOLUTION

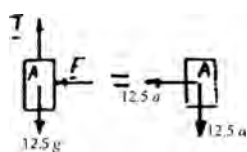
(a) Panel:

F = Force exerted by counterweight



$$\leftarrow^+ \Sigma F_x = ma: \quad T - F = 20a \quad (1)$$

Counterweight A: Its acceleration has two components



$$\mathbf{a}_A = \mathbf{a}_P + \mathbf{a}_{A/P} = a \rightarrow + a \downarrow$$

$$\leftarrow^+ \Sigma F_x = ma_x: \quad F = 12.5a \quad (2)$$

$$+\downarrow \Sigma F_y = ma_y: \quad 12.5g - T = 12.5a \quad (3)$$

Adding (1), (2), and (3):

$$T - F + F + 12.5g - T = (20 + 12.5 + 12.5)a$$

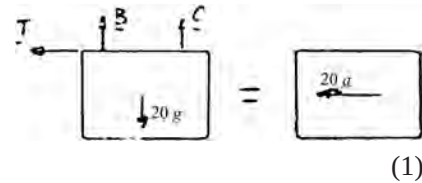
$$a = \frac{12.5g}{45} = \frac{12.5 \times 9.81}{45} = 2.725 \text{ m/s}^2 \quad \mathbf{a = 2.73 \text{ m/s}^2 \blacktriangleleft}$$

Substituting for a into (3):

$$12.5 \times 9.81 - T = 12.5 \times (2.725) \quad T = 12.5 \times (9.81 - 2.725) \quad \mathbf{T = 88.6 \text{ N} \blacktriangleleft}$$

PROBLEM 12.34 (Continued)

(b) Panel:



$$\leftarrow^+ \Sigma F_x = ma: \quad T = 20a \quad (1)$$

Counterweight A:

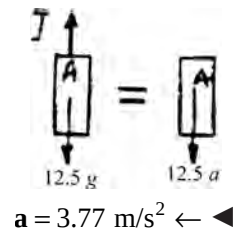
$$+\downarrow \Sigma F_y = ma: \quad 12.5g - T = 12.5a \quad (2)$$

Adding (1) and (2): $T + 12.5g - T = 32.5a$

$$a = \frac{5}{13}g$$

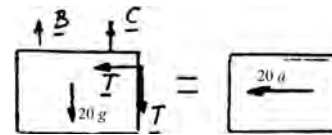
Substituting for a into (1):

$$T = 20 \left(\frac{5}{13}g \right) \quad T = 75.5 \text{ N} \blacktriangleleft$$



(c) Since panel is accelerated to the left, there is no force exerted by panel on counterweight and vice versa.

Panel:



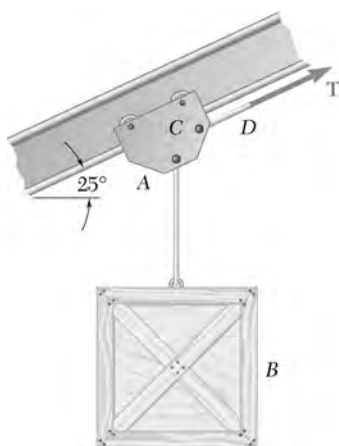
$$\leftarrow^+ \Sigma F_x = ma: \quad T = 20a \quad (1)$$

Counterweight A: Same free body as in Part (b):

$$+\downarrow \Sigma F_y = ma: \quad 12.5g - T = 12.5a \quad (2)$$

Since Eqs. (1) and (2) are the same as in (b), we get the same answers:

$$a = 3.77 \text{ m/s}^2 \leftarrow; T = 75.5 \text{ N} \blacktriangleleft$$



PROBLEM 12.35

A 250-kg crate B is suspended from a cable attached to a 20-kg trolley A which rides on an inclined I-beam as shown. Knowing that at the instant shown the trolley has an acceleration of 0.4 m/s^2 up and to the right, determine (a) the acceleration of B relative to A , (b) the tension in cable CD .

SOLUTION

- (a) First we note: $\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$, where $\mathbf{a}_{B/A}$ is directed perpendicular to cable AB

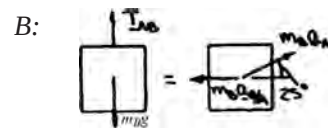
$$\rightarrow \Sigma F_x = m_B a_x: 0 = -m_B a_{B/A} + m_B a_A \cos 25^\circ$$

or

$$a_{B/A} = (0.4 \text{ m/s}^2) \cos 25^\circ$$

or

$$\mathbf{a}_{B/A} = 0.363 \text{ m/s}^2 \leftarrow \blacktriangleleft$$



- (b) For crate B

$$\uparrow \Sigma F_y = m_B a_y: T_{AB} - m_B g = m_B a_A \sin 25^\circ$$

$$(\text{Not that } (a_B)_v = a_A \sin 25^\circ)$$

or

$$\begin{aligned} T_{AB} &= 250(g + a_A \sin 25^\circ) \\ &= 2494.76 \text{ N} \end{aligned}$$

For trolley A

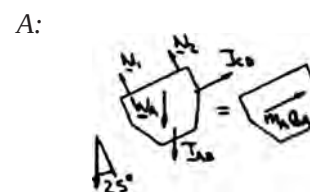
$$\nearrow \Sigma F_x = m_A a_A: T_{CD} - T_{AB} \sin 25^\circ - m_A g \sin 25^\circ = m_A a_A$$

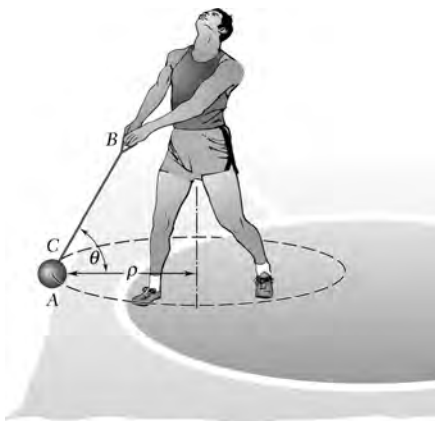
or

$$T_{CD} = (2494.76 \text{ N}) \sin 25^\circ + 20(9.81 \times \sin 25^\circ + 0.4)$$

or

$$T_{CD} = 1145 \text{ N} \blacktriangleleft$$





PROBLEM 12.36

During a hammer thrower's practice swings, the 7.1-kg head A of the hammer revolves at a constant speed v in a horizontal circle as shown. If $\rho = 0.93$ m and $\theta = 60^\circ$, determine (a) the tension in wire BC, (b) the speed of the hammer's head.

SOLUTION

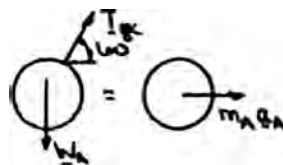
First we note

$$a_A = a_n = \frac{v_A^2}{\rho}$$

$$(a) \quad +\uparrow \Sigma F_y = 0: \quad T_{BC} \sin 60^\circ - W_A = 0$$

or

$$T_{BC} = \frac{7.1 \text{ kg} \times 9.81 \text{ m/s}^2}{\sin 60^\circ} \\ = 80.426 \text{ N}$$



$$T_{BC} = 80.4 \text{ N} \quad \blacktriangleleft$$

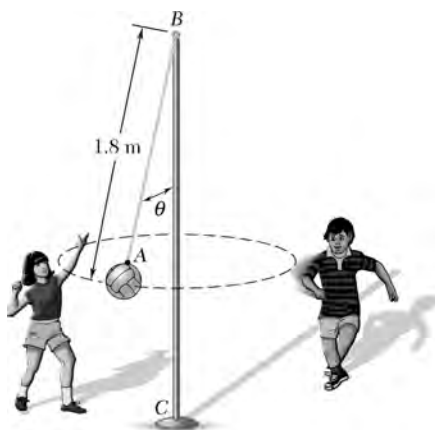
$$(b) \quad +\rightarrow \Sigma F_x = m_A a_A: \quad T_{BC} \cos 60^\circ = m_A \frac{v_A^2}{\rho}$$

or

$$v_A^2 = \frac{(80.426 \text{ N}) \cos 60^\circ \times 0.93 \text{ m}}{7.1 \text{ kg}}$$

or

$$v_A = 2.30 \text{ m/s} \quad \blacktriangleleft$$



PROBLEM 12.37

A 450-g tetherball A is moving along a horizontal circular path at a constant speed of 4 m/s. Determine (a) the angle θ that the cord forms with pole BC, (b) the tension in the cord.

SOLUTION

First we note

$$a_A = a_n = \frac{v_A^2}{\rho}$$

where

$$\rho = l_{AB} \sin \theta$$

$$(a) \quad +\uparrow \Sigma F_y = 0: \quad T_{AB} \cos \theta - W_A = 0$$

or

$$T_{AB} = \frac{m_A g}{\cos \theta}$$

$$+\rightarrow \Sigma F_x = m_A a_A: \quad T_{AB} \sin \theta = m_A \frac{v_A^2}{\rho}$$

Substituting for T_{AB} and ρ

$$\frac{m_A g}{\cos \theta} \sin \theta = m_A \frac{v_A^2}{l_{AB} \sin \theta} \quad \sin^2 \theta = 1 - \cos^2 \theta$$

$$1 - \cos^2 \theta = \frac{(4 \text{ m/s})^2}{1.8 \text{ m} \times 9.81 \text{ m/s}^2} \cos \theta$$

or

$$\cos^2 \theta + 0.906105 \cos \theta - 1 = 0$$

Solving

$$\cos \theta = 0.64479$$

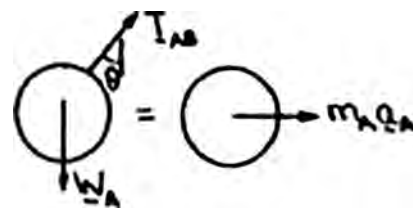
or

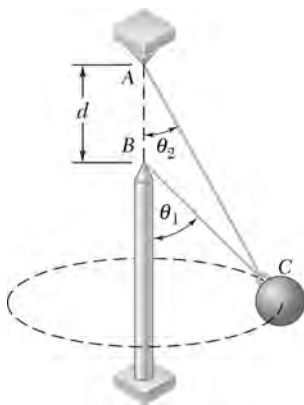
$$\theta = 49.9^\circ \quad \blacktriangleleft$$

$$(b) \quad \text{From above} \quad T_{AB} = \frac{m_A g}{\cos \theta} = \frac{0.450 \text{ kg} \times 9.81 \text{ m/s}^2}{0.64479}$$

or

$$T_{AB} = 6.85 \text{ N} \quad \blacktriangleleft$$





PROBLEM 12.38

A single wire ACB of length 2 m in. passes through a ring at C that is attached to a sphere which revolves at a constant speed v in the horizontal circle shown. Knowing that $\theta_1 = 60^\circ$ and $\theta_2 = 30^\circ$ and that the tension is the same in both portions of the wire, determine the speed v .

SOLUTION

First we note

$$a_C = a_N = \frac{v_C^2}{\rho}$$

where

$$\rho = L_{AC} \sin 30^\circ = L_{BC} \sin 60^\circ$$

Now

$$L_{AC} + L_{BC} = L_{ABC}$$

or

$$\rho \left(\frac{1}{\sin 30^\circ} + \frac{1}{\sin 60^\circ} \right) = 2 \text{ m}$$

or

$$\rho = 0.633975 \text{ m}$$

$$+\uparrow \Sigma F_y = 0: T_{CA} \cos 30^\circ + T_{CB} \cos 60^\circ - m_C g = 0$$

or

$$T = \frac{m_C g}{\cos 30^\circ + \cos 60^\circ} = 0.73205 m_C g$$

$$\leftarrow \Sigma F_x = m_C a_C: T_{CA} \sin 30^\circ + T_{CB} \sin 60^\circ = m_C \frac{v_C^2}{\rho}$$

or

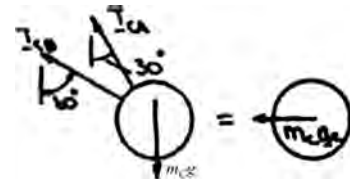
$$0.73205 m_C g (\sin 30^\circ + \sin 60^\circ) = m_C \frac{v_C^2}{\rho}$$

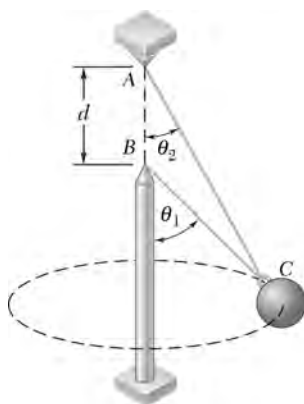
or

$$v_C^2 = 0.73205 (9.81 \text{ m/s}^2) (\sin 30^\circ + \sin 60^\circ) (0.633975 \text{ m})$$

or

$$v_C = 2.49 \text{ m/s} \quad \blacktriangleleft$$





PROBLEM 12.39

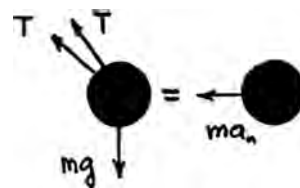
A single wire ACB passes through a ring at C that is attached to a 6 kg sphere which revolves at a constant speed v in the horizontal circle shown. Knowing that $\theta_1 = 50^\circ$ and $d = 750$ mm and that the tension in both portions of the wire is 40 N, determine (a) the angle θ_2 , (b) the speed v .

SOLUTION

$$(a) \quad +\uparrow \Sigma F_y = 0: T \cos \theta_1 + T \cos \theta_2 - mg = 0$$

$$\begin{aligned} \cos \theta_2 &= \frac{mg}{T} - \cos \theta_1 \\ &= \frac{6 \times 9.81}{40} - \cos 50^\circ = 0.82871 \end{aligned}$$

$$\theta_2 = 34.033^\circ$$



$$\theta_2 = 34.0^\circ \blacktriangleleft$$

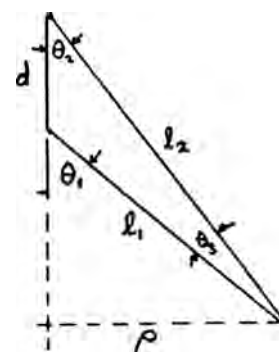
$$(b) \quad \theta_3 = \theta_1 - \theta_2 = 50^\circ - 34.033^\circ = 15.967^\circ$$

$$\frac{\ell_1}{\sin \theta_2} = \frac{d}{\sin \theta_3} \quad \text{or} \quad \ell_1 = \frac{d \sin \theta_2}{\sin \theta_3}$$

Radius of horizontal circle

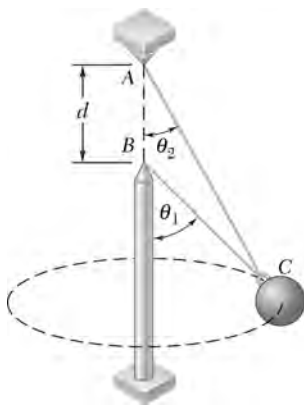
$$\begin{aligned} \rho &= \ell_1 \sin \theta_1 = \frac{d \sin \theta_2 \sin \theta_1}{\sin \theta_3} \\ &= \frac{(0.75)(\sin 34.033^\circ)(\sin 50^\circ)}{\sin(15.967^\circ)} \\ &= 1.16891 \text{ m} \end{aligned}$$

$$\begin{aligned} \leftarrow \Sigma F_x &= ma_n: T \sin \theta_1 + T \sin \theta_2 = \frac{mv^2}{\rho} \\ v^2 &= \frac{\rho T (\sin \theta_1 + \sin \theta_2)}{m} \\ &= \frac{(1.16891)(40)(\sin 50^\circ + \sin 34.033^\circ)}{6} \\ &= 10.3309 \text{ m}^2/\text{s}^2 \end{aligned}$$



$$v = 3.21 \text{ m/s} \blacktriangleleft$$

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PROBLEM 12.40

Two wires AC and BC are tied at C to a 7-kg sphere which revolves at a constant speed v in the horizontal circle shown. Knowing that $\theta_1 = 55^\circ$ and $\theta_2 = 30^\circ$ and that $d = 1.4$ m, determine the range of values of v for which both wires remain taut.

SOLUTION

First we note

$$a_C = a_n = \frac{v_C^2}{\rho}$$

where

$$\rho = l \tan 55^\circ \text{ and}$$

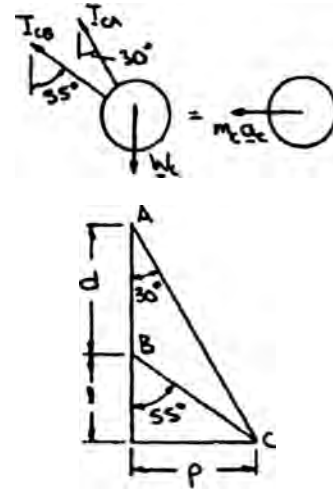
$$\rho = (d + l) \tan 30^\circ$$

Then

$$\rho = \left(d + \frac{\rho}{\tan 55^\circ} \right) \tan 30^\circ$$

or

$$\rho = \frac{1.4 \text{ m}}{\frac{1}{\tan 30^\circ} - \frac{1}{\tan 55^\circ}} = 1.35680 \text{ m}$$



$$\leftarrow \Sigma F_x = m_C a_C: T_{CA} \sin 30^\circ + T_{CB} \sin 55^\circ = m_C \frac{v_C^2}{\rho} \quad (1)$$

$$+\uparrow \Sigma F_y = 0: T_{CA} \cos 30^\circ + T_{CB} \cos 55^\circ - W_C = 0$$

or

$$T_{CA} \cos 30^\circ + T_{CB} \cos 55^\circ = m_C g \quad (2)$$

Case 1: $T_{CA} = 0$:

$$\text{Eq. (2)} \Rightarrow T_{CB} = \frac{m_C g}{\cos 55^\circ}$$

Substituting into Eq. (1)

$$\frac{m_C g}{\cos 55^\circ} \sin 55^\circ = m_C \frac{v_C^2}{\rho}$$

or

$$(v_C^2)_{T_{CA}=0} = (1.35680 \text{ m})(9.81 \text{ m}) \tan 55^\circ$$

or

$$(v_C)_{T_{CA}=0} = 4.36 \text{ m/s}$$

PROBLEM 12.40 (Continued)

Now we form

$$(\cos 30^\circ)(1) - (\sin 30^\circ)(2)$$

$$T_{CB} \sin 55^\circ \cos 30^\circ - T_{CB} \cos 55^\circ \sin 30^\circ = m_C \frac{v_C^2}{\rho} \cos 30^\circ - m_C g \sin 30^\circ$$

or

$$T_{CB} \sin 25^\circ = m_C \frac{v_C^2}{\rho} \cos 30^\circ - m_C g \sin 30^\circ$$

$(v_C)_{\max}$ occurs when $T_{CB} = (T_{CB})_{\max}$, which occurs when $T_{CA} = 0$.

$(v_C)_{\max} = 4.36$ m/s and wire AC will be taut if $v_C < 4.36$ m/s.

Case 2: $T_{CB} = 0$:

$$\text{Eq. (2)} \Rightarrow T_{CA} = \frac{m_C g}{\cos 30^\circ}$$

Substituting into Eq. (1)

$$\frac{m_C g}{\cos 30^\circ} \sin 30^\circ = m_C \frac{v_C^2}{\rho}$$

or

$$(v_C^2)_{T_{CB}=0} = (1.35680 \text{ m})(9.81 \text{ m/s}^2) \tan 30^\circ$$

or

$$(v_C)_{T_{CB}=0} = 2.77 \text{ m/s}$$

Now we form

$$(\cos 55^\circ)(1) - (\sin 55^\circ)(2)$$

$$T_{CA} \sin 30^\circ \cos 55^\circ - T_{CA} \cos 30^\circ \sin 55^\circ = m_C \frac{v_C^2}{\rho} \cos 55^\circ - m_C g \sin 55^\circ$$

or

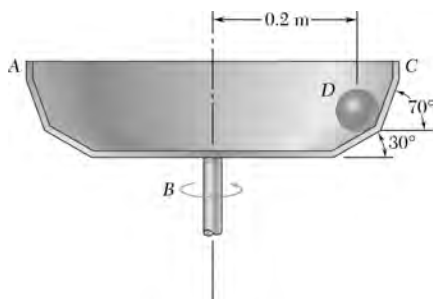
$$-T_{CA} \sin 25^\circ = m_C \frac{v_C^2}{\rho} \cos 55^\circ - m_C g \sin 55^\circ$$

$(v_C)_{\min}$ occurs when $T_{CA} = (T_{CA})_{\max}$, which occurs when $T_{CB} = 0$.

$(v_C)_{\min} = 2.77$ m/s and wire BC will be taut if $v_C > 2.77$ m/s.

Both wires are taut when

$$2.77 \text{ m/s} < v < 4.36 \text{ m/s} \quad \blacktriangleleft$$



PROBLEM 12.41

A 100-g sphere D is at rest relative to drum ABC , which rotates at a constant rate. Neglecting friction, determine the range of the allowable values of the velocity v of the sphere if neither of the normal forces exerted by the sphere on the inclined surfaces of the drum is to exceed 1.1 N.

SOLUTION

First we note

$$a_D = a_n = \frac{v_D^2}{\rho}$$

where

$$\rho = 0.2 \text{ m}$$

$$\leftarrow + \Sigma F_x = m_D a_D: N_1 \cos 60^\circ + N_2 \cos 20^\circ = m_D \frac{v_D^2}{\rho} \quad (1)$$

$$+\uparrow \Sigma F_y = 0: N_1 \sin 60^\circ + N_2 \sin 20^\circ - w_D = 0$$

or

$$N_1 \sin 60^\circ + N_2 \sin 20^\circ = m_D g \quad (2)$$

Case 1: N_1 is maximum.

Let

$$N_1 = 1.1 \text{ N}$$

Eq. (2)

$$(1.1 \text{ N}) \sin 60^\circ + N_2 \sin 20^\circ = (0.1 \text{ kg}) (9.81 \text{ m/s}^2)$$

or

$$N_2 = 0.082954 \text{ N}$$

$$(N_2)_{(N_1)_{\max}} < 1.1 \text{ N. O.K.}$$

Eq. (1)

$$(v_D^2)_{(N_1)_{\max}} = \frac{0.2 \text{ m}}{0.1 \text{ kg}} (1.1 \cos 60^\circ + 0.082954 \cos 20^\circ) \text{ N}$$

or

$$(v_D)_{(N_1)_{\max}} = 1.121 \text{ m/s}$$

Now we form

$$(\sin 20^\circ)(1) - (\cos 20^\circ)(2)$$

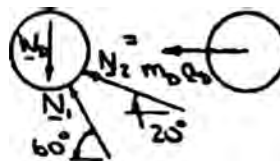
$$N_1 \cos 60^\circ \sin 20^\circ - N_1 \sin 60^\circ \cos 20^\circ = m_D \frac{v_D^2}{\rho} \sin 20^\circ - m_D g \cos 20^\circ$$

or

$$-N_1 \sin 40^\circ = m_D \frac{v_D^2}{\rho} \sin 20^\circ - m_D g \cos 20^\circ$$

$(v_D)_{\min}$ occurs when $N_1 = (N_1)_{\max}$

$$(v_D)_{\min} = 1.121 \text{ m/s}$$



PROBLEM 12.41 (Continued)

Case 2: N_2 is maximum.

Let

$$N_2 = 1.1 \text{ N}$$

Eq. (2)

$$N_1 \sin 60^\circ + (1.1 \text{ N}) \sin 20^\circ = (0.1 \text{ kg})(9.81 \text{ m/s}^2)$$

or

$$N_1 = 0.69834 \text{ N}$$

$$(N_1)_{(N_2)_{\max}} < 1.1 \text{ N.} \quad \text{O.K.}$$

Eq. (1)

$$(v_D^2)_{(N_2)_{\max}} = \frac{0.2 \text{ m}}{0.1 \text{ kg}} (0.69834 \cos 60^\circ + 1.1 \cos 20^\circ) \text{ N}$$

or

$$(v_D)_{(N_2)_{\max}} = 1.663 \text{ m/s}$$

Now we form

$$(\sin 60^\circ)(1) - (\cos 60^\circ)(2)$$

$$N_2 \cos 20^\circ \sin 60^\circ - N_2 \sin 20^\circ \cos 60^\circ = m_D \frac{v_D^2}{\rho} \sin 60^\circ - m_D g \cos 60^\circ$$

or

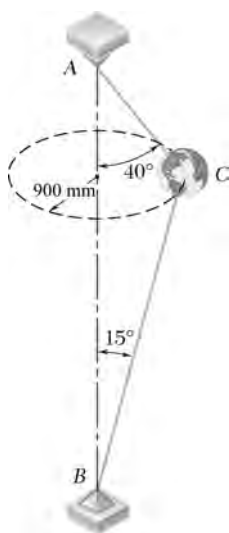
$$N_2 \cos 40^\circ = m_D \frac{v_D^2}{\rho} \sin 60^\circ - m_D g \cos 60^\circ$$

$(v_D)_{\max}$ occurs when $N_2 = (N_2)_{\max}$

$$(v_D)_{\max} = 1.663 \text{ m/s}$$

For $N_1, N_2 \leq 1.1 \text{ N}$

$$1.121 \text{ m/s} \leq v_D \leq 1.663 \text{ m/s} \quad \blacktriangleleft$$



PROBLEM 12.42*

As part of an outdoor display, a 6 kg model C of the earth is attached to wires AC and BC and revolves at a constant speed v in the horizontal circle shown. Determine the range of the allowable values of v if both wires are to remain taut and if the tension in either of the wires is not to exceed 125 N.

SOLUTION

First note

$$a_C = a_n = \frac{v_C^2}{\rho}$$

where

$$\rho = 0.9 \text{ m}$$

$$\leftarrow^+ \Sigma F_x = m_C a_C: T_{CA} \sin 40^\circ + T_{CB} \sin 15^\circ = m_C \frac{v_C^2}{\rho} \quad (1)$$

$$^+ \uparrow \Sigma F_y = 0: T_{CA} \cos 40^\circ - T_{CB} \cos 15^\circ - m_C g = 0 \quad (2)$$

Note that Eq. (2) implies that

$$(a) \quad \text{when} \quad T_{CB} = (T_{CB})_{\max}, \quad T_{CA} = (T_{CA})_{\max}$$

$$(b) \quad \text{when} \quad T_{CB} = (T_{CB})_{\min}, \quad T_{CA} = (T_{CA})_{\min}$$

Case 1: T_{CA} is maximum.

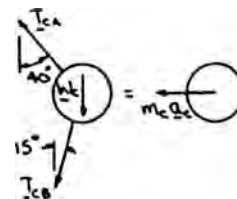
$$\text{Let} \quad T_{CA} = 125 \text{ N}$$

$$\text{Eq. (2)} \quad (125 \text{ N}) \cos 40^\circ - T_{CB} \cos 15^\circ - (6 \times 9.81) = 0$$

$$\text{or} \quad T_{CB} = 68.4732 \text{ N}$$

$$(T_{CB})_{(T_{CA})_{\max}} < 125 \text{ N} \quad \text{O.K.}$$

$$[(T_{CB})_{\max} = 68.4732 \text{ N}]$$



PROBLEM 12.42* (Continued)

Eq. (1)

$$(v_C^2)_{(T_{CA})_{\max}} = \frac{0.9}{6}(125 \sin 40^\circ + 68.4732 \sin 15^\circ) \text{ N}$$

or

$$(v_C)_{(T_{CA})_{\max}} = 3.8354 \text{ m/s}$$

Now we form

$$(\cos 15^\circ) (1) + (\sin 15^\circ) (2)$$

$$T_{CA} \sin 40^\circ \cos 15^\circ + T_{CA} \cos 40^\circ \sin 15^\circ = m_C \frac{v_C^2}{\rho} \cos 15^\circ + m_C g \sin 15^\circ$$

or

$$T_{CA} \sin 55^\circ = m_C \frac{v_C^2}{\rho} \cos 15^\circ + m_C g \sin 15^\circ \quad (3)$$

$(v_C)_{\max}$ occurs when $T_{CA} = (T_{CA})_{\max}$

$$(v_C)_{\max} = 3.84 \text{ m/s}$$

Case 2: T_{CA} is minimum.

Because $(T_{CA})_{\min}$ occurs when $T_{CB} = (T_{CB})_{\min}$,

let $T_{CB} = 0$ (note that wire BC will not be taut).

Eq. (2)

$$T_{CA} \cos 40^\circ - (6g) = 0$$

or

$$T_{CA} = 76.8363 \text{ N} < 125 \text{ N} \quad \text{O.K.}$$

Note: Eq. (3) implies that when $T_{CA} = (T_{CA})_{\min}$, $v_C = (v_C)_{\min}$. Then

Eq. (1)

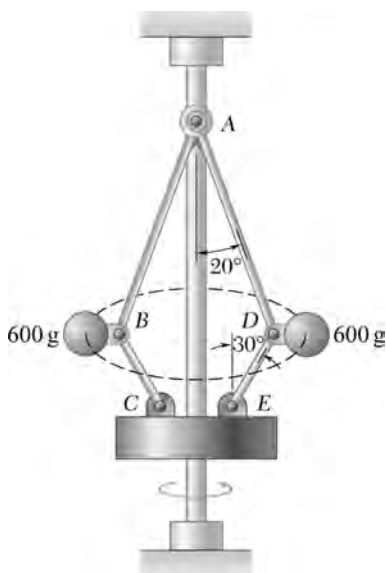
$$(v_C^2)_{\min} = \frac{0.9}{6}(76.8363) \sin 40^\circ$$

or

$$(v_C)_{\min} = 2.7218 \text{ m/s}$$

$0 < T_{CA}, \quad T_{CB} \leq 125 \text{ N}$ when

$$2.72 \text{ m/s} < v_C \leq 3.84 \text{ m/s} \quad \blacktriangleleft$$



PROBLEM 12.43*

The 600 g flyballs of a centrifugal governor revolve at a constant speed v in the horizontal circle of 150 mm radius shown. Neglecting the weights of links AB , BC , AD , and DE and requiring that the links support only tensile forces, determine the range of the allowable values of v so that the magnitudes of the forces in the links do not exceed 80 N.

SOLUTION

First note

$$a = a_n = \frac{v^2}{\rho}$$

where

$$\rho = 0.15 \text{ m}$$

$$\leftarrow^+ \Sigma F_x = ma: T_{DA} \sin 20^\circ + T_{DE} \sin 30^\circ = m \frac{v^2}{\rho}$$

$$^+ \uparrow \Sigma F_y = 0: T_{DA} \cos 20^\circ - T_{DE} \cos 30^\circ - mg = 0 \quad (2)$$

Note that Eq. (2) implies that

$$(a) \quad \text{when} \quad T_{DE} = (T_{DE})_{\max}, \quad T_{DA} = (T_{DA})_{\max}$$

$$(b) \quad \text{when} \quad T_{DE} = (T_{DE})_{\min}, \quad T_{DA} = (T_{DA})_{\min}$$

Case 1: T_{DA} is maximum.

$$\text{Let} \quad T_{DA} = 80 \text{ N}$$

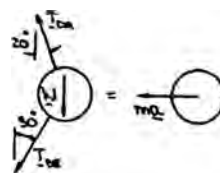
$$\text{Eq. (2)} \quad (80 \text{ N}) \cos 20^\circ - T_{DE} \cos 30^\circ - (0.6 \times 9.81) = 0$$

$$\text{or} \quad T_{DE} = 80.01 \text{ N} \quad \text{unacceptable} (> 80 \text{ N})$$

$$\text{Now let} \quad T_{DE} = 80 \text{ N}$$

$$\text{Eq. (2)} \quad T_{DA} \cos 20^\circ - (80 \text{ N}) \cos 30^\circ - (0.6 \times 9.81) = 0$$

$$\text{or} \quad T_{DA} = 79.9921 \text{ N} \quad \text{O.K.} (< 80 \text{ N})$$



FBD of flyball (1)

PROBLEM 12.43* (Continued)

$$(T_{DA})_{\max} = 79.9921 \text{ N}$$

$$(T_{DE})_{\max} = 80 \text{ N}$$

$$\text{Eq. (1)} \quad (v^2)_{(T_{DA})_{\max}} = \frac{0.15 \text{ m}}{0.6 \text{ kg}} (79.9921 \sin 20^\circ + 80 \sin 30^\circ) \text{ N}$$

$$\text{or} \quad v_{(T_{DA})_{\max}} = 4.10362 \text{ m/s}$$

$$\text{Now form} \quad (\cos 30^\circ)(1) + (\sin 30^\circ)(2)$$

$$T_{DA} \sin 20^\circ \cos 30^\circ + T_{DA} \cos 20^\circ \sin 30^\circ = m \frac{v^2}{\rho} \cos 30^\circ + mg \sin 30^\circ$$

$$\text{or} \quad T_{DA} \sin 50^\circ = m \frac{v^2}{\rho} \cos 30^\circ + mg \sin 30^\circ \quad (3)$$

$$v_{\max} \text{ occurs when } T_{DA} = (T_{DA})_{\max}$$

$$v_{\max} = 4.10362 \text{ m/s}$$

Case 2: T_{DA} is minimum.

Because $(T_{DA})_{\min}$ occurs when $T_{DE} = (T_{DE})_{\min}$,

let $T_{DE} = 0$.

$$\text{Eq. (2)} \quad T_{DA} \cos 20^\circ - (0.6 \times 9.81) = 0$$

$$\text{or} \quad T_{DA} = 6.26375 \text{ N} < 80 \text{ N} \quad \text{O.K.}$$

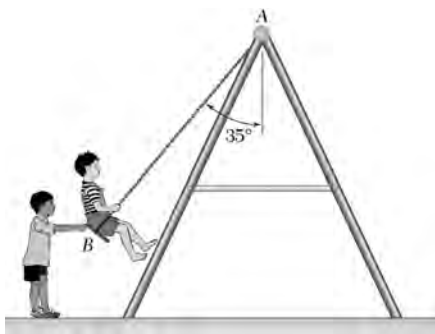
Note: Eq. (3) implies that when $T_{DA} = (T_{DA})_{\min}$, $v = v_{\min}$. Then

$$\text{Eq. (1)} \quad (v^2)_{\min} = \frac{(0.15 \text{ m})}{(0.6 \text{ kg})} (6.26375 \text{ N}) \sin 20^\circ$$

$$\text{or} \quad v_{\min} = 0.53558 \text{ m/s}$$

$$0 \leq T_{AB}, T_{BC}, T_{AD}, T_{DE} \leq 80 \text{ N}$$

$$\text{when} \quad 0.536 \text{ m/s} \leq v \leq 4.10 \text{ m/s} \quad \blacktriangleleft$$



PROBLEM 12.44

A child having a mass of 22 kg sits on a swing and is held in the position shown by a second child. Neglecting the mass of the swing, determine the tension in rope AB (a) while the second child holds the swing with his arms outstretched horizontally, (b) immediately after the swing is released.

SOLUTION

Note: The factors of " $\frac{1}{2}$ " are included in the following free-body diagrams because there are two ropes and only one is considered.

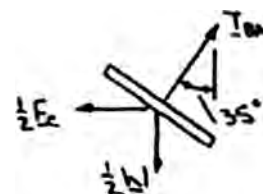
(a) For the swing at rest

$$\Sigma F_y = 0: T_{BA} \cos 35^\circ - \frac{1}{2}W = 0$$

or

$$T_{BA} = \frac{22 \text{ kg} \times 9.81 \text{ m/s}^2}{2 \cos 35^\circ}$$

or



$$T_{BA} = 131.7 \text{ N} \quad \blacktriangleleft$$

(b) At $t = 0$, $v = 0$, so that

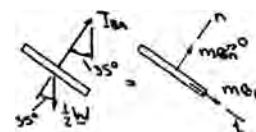
$$a_n = \frac{v^2}{\rho} = 0$$

$$\nearrow \Sigma F_n = 0: T_{BA} - \frac{1}{2}W \cos 35^\circ = 0$$

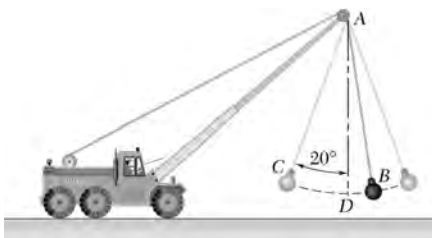
or

$$T_{BA} = \frac{1}{2}(22 \text{ kg})(9.81 \text{ m/s}^2) \cos 35^\circ$$

or



$$T_{BA} = 88.4 \text{ N} \quad \blacktriangleleft$$



PROBLEM 12.45

A 60-kg wrecking ball B is attached to a 15-m-long steel cable AB and swings in the vertical arc shown. Determine the tension in the cable (a) at the top C of the swing, (b) at the bottom D of the swing, where the speed of B is 4.2 m/s.

SOLUTION

- (a) At C , the top of the swing, $v_B = 0$; thus

$$a_n = \frac{v_B^2}{L_{AB}} = 0$$

$$\nearrow \Sigma F_n = 0: T_{BA} - W_B \cos 20^\circ = 0$$

or

$$T_{BA} = (60 \text{ kg})(9.81 \text{ m/s}^2) \times \cos 20^\circ$$

or

$$T_{BA} = 553 \text{ N} \quad \blacktriangleleft$$

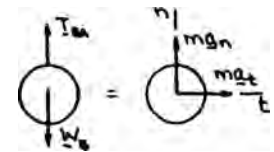
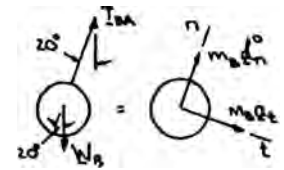
- (b) $\nearrow \Sigma F_n = ma_n: T_{BA} - W_B = m_B \frac{(v_B)_D^2}{L_{AB}}$

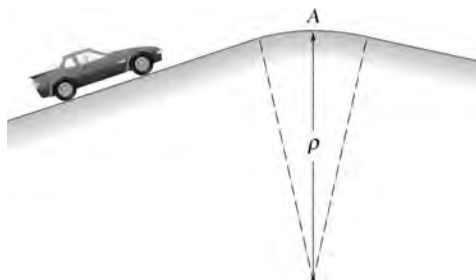
or

$$T_{BA} = (60 \text{ kg}) \left[9.81 \text{ m/s}^2 + \frac{(4.2 \text{ m/s})^2}{15 \text{ m}} \right]$$

or

$$T_{BA} = 659 \text{ N} \quad \blacktriangleleft$$





PROBLEM 12.46

During a high-speed chase, a 1000 kg sports car traveling at a speed of 160 km/h just loses contact with the road as it reaches the crest A of a hill. (a) Determine the radius of curvature ρ of the vertical profile of the road at A. (b) Using the value of ρ found in part a, determine the force exerted on a 80 kg driver by the seat of his 1200 kg car as the car, traveling at a constant speed of 75 km/h, passes through A.

SOLUTION

(a) Note:

$$160 \text{ km/h} = \frac{400}{9} \text{ m/s}$$

$$+\downarrow \Sigma F_n = ma_n: W_{\text{car}} = \frac{W_{\text{car}}}{g} \frac{v_A^2}{\rho}$$

or

$$\rho = \frac{\left(\frac{400}{9} \text{ m/s}\right)^2}{9.81 \text{ m/s}^2}$$

$$= 201.357 \text{ m}$$

or

(b) Note: v is constant $\Rightarrow a_t = 0$; $75 \text{ km/h} = \frac{125}{6} \text{ m/s}$

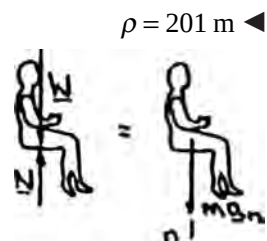
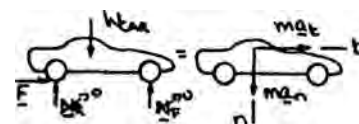
$$+\downarrow \Sigma F_n = ma_n: W - N = \frac{W}{g} \frac{v_A^2}{\rho}$$

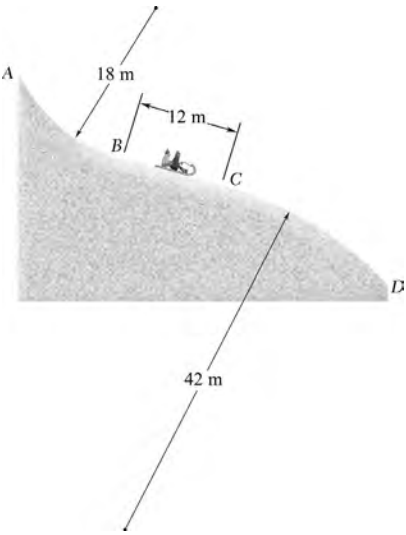
or

$$N = W - \frac{mv_A^2}{S} = 80 \times 9.81 - \frac{80 \times \left(\frac{125}{6}\right)^2}{201.357} = 612.359 \text{ N}$$

or

$$N = 612 \text{ N} \uparrow \blacktriangleleft$$





PROBLEM 12.47

The portion of a toboggan run shown is contained in a vertical plane. Sections AB and CD have radii of curvature as indicated, and section BC is straight and forms an angle of 20° with the horizontal. Knowing that the coefficient of kinetic friction between a sled and the run is 0.10 and that the speed of the sled is 7.5 m/s at B, determine the tangential component of the acceleration of the sled (a) just before it reaches B, (b) just after it passes C.

SOLUTION

(a) Note: Just before B, $\rho_B = 18$ m

$$\nearrow \Sigma F_n = ma_n: N - W \cos 20^\circ = \frac{W}{g} \frac{v_B^2}{\rho_B}$$

or

$$N = W \left(\cos 20^\circ + \frac{v_B^2}{g \rho_B} \right)$$

Sliding:

$$\begin{aligned} F &= \mu_k N \\ &= \mu_k W \left(\cos 20^\circ + \frac{v_B^2}{g \rho_B} \right) \end{aligned}$$

$$\searrow \Sigma F_t = ma_t: W \sin 20^\circ - F = \frac{W}{g} a_t$$

or

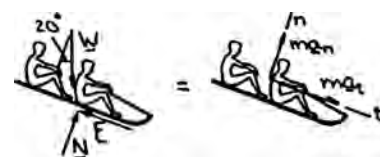
$$a_t = g(\sin 20^\circ - \mu_t \cos 20^\circ) - \mu_k \frac{v_B^2}{\rho_B}$$

Then

$$a_t = 9.81(\sin 20^\circ - 0.1 \cos 20^\circ) - \frac{0.1(7.5 \text{ m/s})^2}{18}$$

or

$$\mathbf{a}_t = 2.12 \text{ m/s}^2 \searrow 20^\circ \blacktriangleleft$$



PROBLEM 12.47 (Continued)

- (b) It is first necessary to determine v_C .

For section BC

$$+\nearrow \Sigma F_y = 0: N_{BC} - W \cos 20^\circ = 0$$

or

$$N_{BC} = W \cos 20^\circ$$

Sliding:

$$F_{BC} = \mu_k N_{BC} = \mu_k W \cos 20^\circ$$

$$\searrow \Sigma F_x = ma_{BC}: W \sin 20^\circ - F_{BC} = \frac{W}{g} a_{BC}$$

or

$$\begin{aligned} a_{BC} &= g(\sin 20^\circ - \mu_k \cos 20^\circ) \\ &= (9.81 \text{ m/s}^2)(\sin 20^\circ - 0.1 \cos 20^\circ) \\ &= 2.43338 \text{ m/s}^2 \end{aligned}$$

For this uniformly accelerated motion we have

$$\begin{aligned} v_C^2 &= v_B^2 + 2a_{BC} \Delta x_{BC} \\ &= (7.5 \text{ m/s})^2 + 2(2.43338 \text{ m/s}^2)(12 \text{ m}) \end{aligned}$$

or

$$v_C = 10.7075 \text{ m/s}$$

Now, just after C, $\rho_C = 42 \text{ m}$

$$\nwarrow \Sigma F_n = ma_n: W \cos 20^\circ - N = \frac{W}{g} \frac{v_C^2}{\rho_C}$$

or

$$N = W \left(\cos 20^\circ - \frac{v_C^2}{g \rho_C} \right)$$

Sliding:

$$\begin{aligned} F &= \mu_k N \\ &= \mu_k W \left(\cos 20^\circ - \frac{v_C^2}{g \rho_C} \right) \end{aligned}$$

$$\nwarrow \Sigma F_t = ma_t: W \sin 20^\circ - F = \frac{W}{g} a_t$$

or

$$a_t = g(\sin 20^\circ - \mu_k \cos 20^\circ) + \mu_k \frac{v_C^2}{\rho_C}$$

Note:

$$g(\sin 20^\circ - \mu_k \cos 20^\circ) = a_{BC}$$

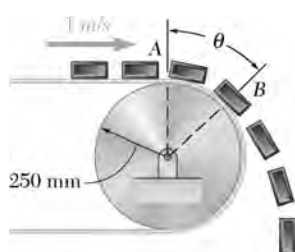
Then

$$a_t = 2.43338 \text{ m/s}^2 + 0.1 \frac{(10.7075 \text{ m/s})^2}{42 \text{ m}} = 2.7068 \text{ m/s}^2$$

or

$$\mathbf{a}_t = 2.71 \text{ m/s}^2 \searrow 20^\circ \blacktriangleleft$$





PROBLEM 12.48

A series of small packages, each with a mass of 0.5 kg are discharged from a conveyor belt as shown. Knowing that the coefficient of static friction between each package and the conveyor belt is 0.40, determine (a) the force exerted by the belt on a package just after it has passed Point A, (b) the angle θ defining Point B where the packages first slip relative to the belt.

SOLUTION

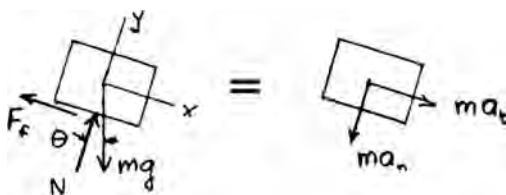
Assume package does not slip.

$$a_t = 0, \quad F_f \leq \mu_s N$$

On the curved portion of the belt

$$a_n = \frac{v^2}{\rho} = \frac{(1 \text{ m/s})^2}{0.250 \text{ m}} = 4 \text{ m/s}^2$$

For any angle θ



$$\nearrow \Sigma F_y = ma_y: \quad N - mg \cos \theta = -ma_n = -\frac{mv^2}{\rho}$$

$$N = mg \cos \theta - \frac{mv^2}{\rho} \quad (1)$$

$$\searrow \Sigma F_x = ma_x: \quad -F_f + mg \sin \theta = ma_t = 0$$

$$F_f = mg \sin \theta \quad (2)$$

(a) At Point A,

$$\theta = 0^\circ$$

$$N = (0.5)(9.81)(1.000) - (0.5)(4) \quad N = 2.905 \text{ N} \quad \blacktriangleleft$$

(b) At Point B,

$$F_f = \mu_s N$$

$$mg \sin \theta = \mu_s (mg \cos \theta - ma_n)$$

$$\sin \theta = \mu_s \left(\cos \theta - \frac{a_n}{g} \right) = 0.40 \left[\cos \theta - \frac{4}{9.81} \right]$$

PROBLEM 12.48 (Continued)

Squaring and using trigonometric identities,

$$1 - \cos^2 \theta = 0.16 \cos^2 \theta - 0.130479 \cos \theta + 0.026601$$

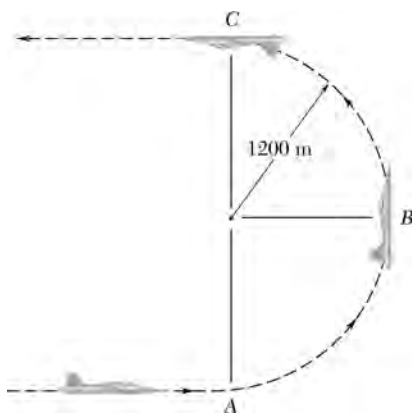
$$1.16 \cos^2 \theta - 0.130479 \cos \theta - 0.97340 = 0$$

$$\cos \theta = 0.97402$$

$$\theta = 13.09^\circ \blacktriangleleft$$

Check that package does not separate from the belt.

$$N = \frac{F_f}{\mu_s} = \frac{mg \sin \theta}{\mu_s} \quad N > 0.$$



PROBLEM 12.49

A 54-kg pilot flies a jet trainer in a half vertical loop of 1200-m radius so that the speed of the trainer decreases at a constant rate. Knowing that the pilot's apparent weights at Points A and C are 1680 N and 350 N, respectively, determine the force exerted on her by the seat of the trainer when the trainer is at Point B.

SOLUTION

First we note that the pilot's apparent weight is equal to the vertical force that she exerts on the seat of the jet trainer.

$$\text{At A: } +\uparrow \Sigma F_n = ma_n: N_A - W = m \frac{v_A^2}{\rho}$$

$$\text{or } v_A^2 = (1200 \text{ m}) \left(\frac{1680 \text{ N}}{54 \text{ kg}} - 9.81 \text{ m/s}^2 \right)$$

$$= 25,561.3 \text{ m}^2/\text{s}^2$$

$$\text{At C: } +\downarrow \Sigma F_n = ma_n: N_C + W = m \frac{v_C^2}{\rho}$$

$$\text{or } v_C^2 = (1200 \text{ m}) \left(\frac{350 \text{ N}}{54 \text{ kg}} + 9.81 \text{ m/s}^2 \right)$$

$$= 19,549.8 \text{ m}^2/\text{s}^2$$

Since a_t = constant, we have from A to C

$$v_C^2 = v_A^2 + 2a_t \Delta s_{AC}$$

$$\text{or } 19,549.8 \text{ m}^2/\text{s}^2 = 25,561.3 \text{ m}^2/\text{s}^2 + 2a_t(\pi \times 1200 \text{ m})$$

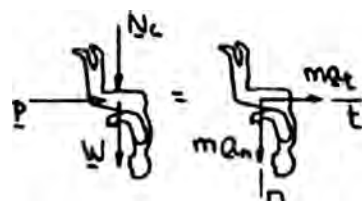
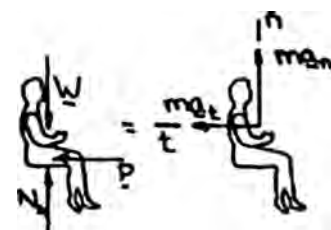
$$\text{or } a_t = -0.79730 \text{ m/s}^2$$

Then from A to B

$$v_B^2 = v_A^2 + 2a_t \Delta s_{AB}$$

$$= 25,561.3 \text{ m}^2/\text{s}^2 + 2(-0.79730 \text{ m/s}^2) \left(\frac{\pi}{2} \times 1200 \text{ m} \right)$$

$$= 22,555 \text{ m}^2/\text{s}^2$$



PROBLEM 12.49 (Continued)

At B: $\leftarrow^+ \Sigma F_n = ma_n: N_B = m \frac{v_B^2}{\rho}$

or $N_B = 54 \text{ kg} \frac{22,555 \text{ m}^2/\text{s}^2}{1200 \text{ m}}$

or $N_B = 1014.98 \text{ N} \leftarrow$

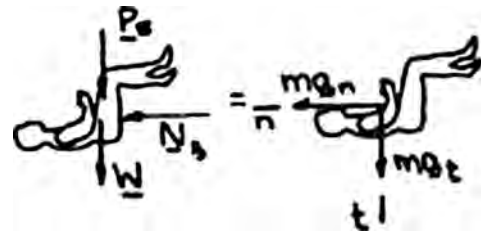
$+\downarrow \Sigma F_t = ma_t: W + P_B = m|a_t|$

or $P_B = (54 \text{ kg})(0.79730 - 9.81) \text{ m/s}^2$

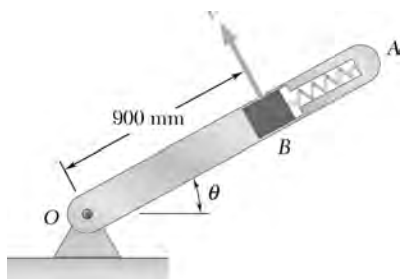
or $P_B = 486.69 \text{ N} \uparrow$

Finally, $(F_{\text{pilot}})_B = \sqrt{N_B^2 + P_B^2} = \sqrt{(1014.98)^2 + (486.69)^2}$
 $= 1126 \text{ N}$

or



$(F_{\text{pilot}})_B = 1126 \text{ N} \nearrow 25.6^\circ \blacktriangleleft$



PROBLEM 12.50

A 250-g block B fits inside a small cavity cut in arm OA , which rotates in the vertical plane at a constant rate such that $v = 3$ m/s. Knowing that the spring exerts on block B a force of magnitude $P = 1.5$ N and neglecting the effect of friction, determine the range of values of θ for which block B is in contact with the face of the cavity closest to the axis of rotation O .

SOLUTION

$$\uparrow \Sigma F_n = ma_n: \quad P + mg \sin \theta - Q = m \frac{v^2}{\rho}$$

To have contact with the specified surface, we need $Q \geq 0$,

or

$$Q = P + mg \sin \theta - \frac{mv^2}{\rho} \geq 0$$

$$\sin \theta \geq \frac{1}{g} \left(\frac{v^2}{\rho} - \frac{P}{m} \right) \quad (1)$$

Data:

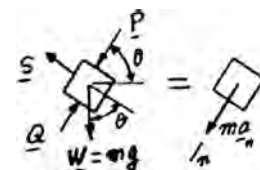
$$m = 0.250 \text{ kg}, \quad v = 3 \text{ m/s}, \quad P = 1.5 \text{ N}, \quad \rho = 0.9 \text{ m}$$

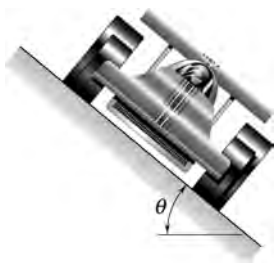
Substituting into (1):

$$\sin \theta \geq \frac{1}{9.81} \left[\frac{(3)^2}{0.9} - \frac{1.5}{0.25} \right]$$

$$\sin \theta \geq 0.40775$$

$$24.1^\circ \leq \theta \leq 155.9^\circ \quad \blacktriangleleft$$





PROBLEM 12.51

A curve in a speed track has a radius of 300 m and a rated speed of 190 km/h. (See sample Problem 12.6 for the definition of rated speed). Knowing that a racing car starts skidding on the curve when traveling at a speed of 290 km/h, determine (a) the banking angle θ , (b) the coefficient of static friction between the tires and the track under the prevailing conditions, (c) the minimum speed at which the same car could negotiate that curve.

SOLUTION

Weight

$$W = mg$$

Acceleration

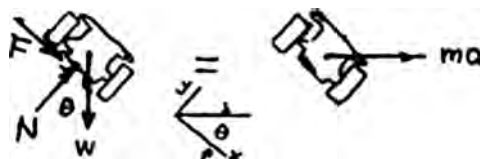
$$a = \frac{v^2}{\rho}$$

$$\Sigma F_x = ma_x: F + W \sin \theta = ma \cos \theta$$

$$F = \frac{mv^2}{\rho} \cos \theta - mg \sin \theta \quad (1)$$

$$\Sigma F_y = ma_y: N - W \cos \theta = ma \sin \theta$$

$$N = \frac{mv^2}{\rho} \sin \theta + mg \cos \theta \quad (2)$$



- (a) Banking angle. Rated speed $v = 190 \text{ km/h} = \frac{475}{9} \text{ m/s}$. $F = 0$ at rated speed.

$$0 = \frac{mv^2}{\rho} \cos \theta - mg \sin \theta$$

$$\tan \theta = \frac{v^2}{\rho g} = \frac{\left(\frac{475}{9}\right)^2}{(300)(9.81)}$$

$$\theta = 43.425^\circ$$

$$\theta = 43.5^\circ \blacktriangleleft$$

- (b) Slipping outward.

$$v = 290 \text{ km/h} = \left(\frac{725}{9}\right) \text{ m/s}$$

$$F = \mu N \quad \mu = \frac{F}{N} = \frac{v^2 \cos \theta - \rho g \sin \theta}{v^2 \sin \theta + \rho g \cos \theta}$$

$$\mu = \frac{\left(\frac{725}{9}\right)^2 \cos (43.425^\circ) - (300)(9.81) \sin 43.425^\circ}{\left(\frac{725}{9}\right)^2 \sin (43.425^\circ) + (300)(9.81) \cos 43.425^\circ}$$

$$= 0.40768$$

$$\mu = 0.408 \blacktriangleleft$$

PROBLEM 12.51 (Continued)

(c) Minimum speed.

$$F = -\mu N$$

$$-\mu = \frac{v^2 \cos \theta - \rho g \sin \theta}{v^2 \sin \theta + \rho g \cos \theta}$$

$$v^2 = \frac{\rho g (\sin \theta - \mu \cos \theta)}{\cos \theta + \mu \sin \theta}$$

$$= \frac{(300)(9.81)(\sin 43.425^\circ - 0.40768 \cos 43.425^\circ)}{(\cos 43.425^\circ + 0.40768 \sin 43.425^\circ)}$$

$$= 1144 \text{ m}^2/\text{s}^2$$

$$v = 33.8258 \text{ m/s}$$

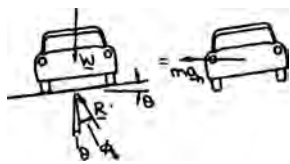
$$v = 121.8 \text{ km/h} \quad \blacktriangleleft$$

PROBLEM 12.52

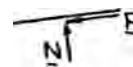
A car is traveling on a banked road at a constant speed v . Determine the range of values of v for which the car does not skid. Express your answer in terms of the radius r of the curve, the banking angle θ , and the angle of static friction ϕ_s between the tires and the pavement.

SOLUTION

Case 1: $v = v_{\max}$



Note: $\mathbf{R} = \mathbf{F} + \mathbf{N}$



$$\leftarrow \Sigma F_n = ma_n: \quad R \sin(\theta + \phi_s) = m \frac{v_{\max}^2}{r} \quad (1)$$

$$+\uparrow \Sigma F_y = 0: \quad R \cos(\theta + \phi_s) - W = 0$$

$$R \cos(\theta + \phi_s) = mg \quad (2)$$

or

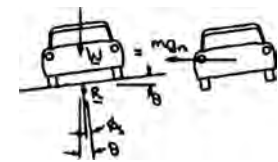
Forming $\frac{(1)}{(2)}$

$$\frac{R \sin(\theta + \phi_s)}{R \cos(\theta + \phi_s)} = \frac{m \frac{v_{\max}^2}{r}}{mg}$$

or

$$v_{\max} = \sqrt{gr \tan(\theta + \phi_s)}$$

Case 2: $v = v_{\min}$



Note: $\mathbf{R} = \mathbf{F} + \mathbf{N}$



$$\leftarrow \Sigma F_n = ma_n: \quad R \sin(\theta - \phi_s) = m \frac{v_{\min}^2}{r} \quad (3)$$

$$+\uparrow \Sigma F_y = 0: \quad R \cos(\theta - \phi_s) - W = 0$$

$$R \cos(\theta - \phi_s) = mg \quad (4)$$

or

Forming $\frac{(3)}{(4)}$

$$\frac{R \sin(\theta - \phi_s)}{R \cos(\theta - \phi_s)} = \frac{m \frac{v_{\min}^2}{r}}{mg}$$

or

$$v_{\min} = \sqrt{gr \tan(\theta - \phi_s)}$$

For the car not to skid

$$\sqrt{gr \tan(\theta - \phi_s)} \leq v \leq \sqrt{gr \tan(\theta + \phi_s)} \quad \blacktriangleleft$$



PROBLEM 12.53

Tilting trains, such as the *American Flyer*, which will run from Washington to New York and Boston, are designed to travel safely at high speeds on curved sections of track, which were built for slower, conventional trains. As it enters a curve, each car is tilted by hydraulic actuators mounted on its trucks. The tilting feature of the cars also increases passenger comfort by eliminating or greatly reducing the side force F_s (parallel to the floor of the car) to which passengers feel subjected. For a train traveling at 160 km/h on a curved section of track banked through an angle $\theta = 6^\circ$ and with a rated speed of 100 km/h, determine (a) the magnitude of the side force felt by a passenger of weight W in a standard car with no tilt ($\phi = 0$), (b) the required angle of tilt ϕ if the passenger is to feel no side force. (See Sample Problem 12.6 for the definition of rated speed.)

SOLUTION

Rated speed:

$$v_R = 100 \text{ km/h} = \frac{250}{9} \text{ m/s}, \quad 160 \text{ km/h} = \frac{400}{9} \text{ m/s}$$

From Sample Problem 12.6,

$$v_R^2 = g \rho \tan \theta$$

or

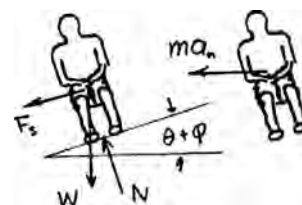
$$\rho = \frac{v_R^2}{g \tan \theta} = \frac{\left(\frac{250}{9}\right)^2}{9.81 \tan 6^\circ} = 748.352 \text{ m}$$

Let the x-axis be parallel to the floor of the car.

$$\begin{aligned} \sum F_x = ma_x: \quad F_s + W \sin(\theta + \phi) &= ma_n \cos(\theta + \phi) \\ &= \frac{mv^2}{\rho} \cos(\theta + \phi) \end{aligned}$$

(a) $\phi = 0$.

$$\begin{aligned} F_s &= W \left[\frac{v^2}{g\rho} \cos(\theta + \phi) - \sin(\theta + \phi) \right] \\ &= W \left[\frac{\left(\frac{400}{9}\right)^2}{(9.81)(748.352)} \cos 6^\circ - \sin 6^\circ \right] \\ &= 0.16306W \end{aligned}$$



$$F_s = 0.1631W \quad \blacktriangleleft$$

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PROBLEM 12.53 (Continued)

(b) For $F_s = 0$,

$$\frac{v^2}{g\rho} \cos(\theta + \phi) - \sin(\theta + \phi) = 0$$

$$\tan(\theta + \phi) = \frac{v^2}{g\rho} = \frac{\left(\frac{400}{9}\right)^2}{(9.81)(748.352)} = 0.26907$$

$$\theta + \phi = 15.060^\circ$$

$$\phi = 15.06^\circ - 6^\circ$$

$$\phi = 9.06^\circ \blacktriangleleft$$



PROBLEM 12.54

Tests carried out with the tilting trains described in Problem 12.53 revealed that passengers feel queasy when they see through the car windows that the train is rounding a curve at high speed, yet do not feel any side force. Designers, therefore, prefer to reduce, but not eliminate that force. For the train of Problem 12.53, determine the required angle of tilt ϕ if passengers are to feel side forces equal to 10% of their weights.

PROBLEM 12.53 Tilting trains, such as the *American Flyer*, which will run from Washington to New York and Boston, are designed to travel safely at high speeds on curved sections of track, which were built for slower, conventional trains. As it enters a curve, each car is tilted by hydraulic actuators mounted on its trucks. The tilting feature of the cars also increases passenger comfort by eliminating or greatly reducing the side force F_s (parallel to the floor of the car) to which passengers feel subjected. For a train traveling at 160 km/h on a curved section of track banked through an angle $\theta = 6^\circ$ and with a rated speed of 100 km/h, determine (a) the magnitude of the side force felt by a passenger of weight W in a standard car with no tilt ($\phi = 0$), (b) the required angle of tilt ϕ if the passenger is to feel no side force. (See Sample Problem 12.6 for the definition of rated speed.)

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From Sample Problem 12.6,

$$v_R^2 = g \rho \tan \theta$$

or

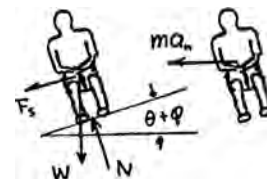
$$\rho = \frac{v_R^2}{g \tan \theta} = \frac{\left(\frac{250}{9}\right)^2}{9.81 \tan 6^\circ} = 748.352 \text{ m}$$

Let the x -axis be parallel to the floor of the car.

$$\begin{aligned} \sum F_x = ma_x: \quad F_s + W \sin(\theta + \phi) &= ma_n \cos(\theta + \phi) \\ &= \frac{mv^2}{\rho} \cos(\theta + \phi) \end{aligned}$$

Solving for F_s ,

$$F_s = W \left[\frac{v^2}{g\rho} \cos(\theta + \phi) - \sin(\theta + \phi) \right]$$



PROBLEM 12.54 (Continued)

Now
$$\frac{v^2}{g\rho} = \frac{\left(\frac{400}{9}\right)^2}{(9.81)(748.352)} = 0.26907 \quad \text{and} \quad F_s = 0.10W$$

So that
$$0.10W = W[0.26907 \cos(\theta + \phi) - \sin(\theta + \phi)]$$

Let
$$u = \sin(\theta + \phi)$$

Then
$$\cos(\theta + \phi) = \sqrt{1 - u^2}$$

$$0.10 = 0.26907\sqrt{1 - u^2} - u \quad \text{or} \quad 0.26907\sqrt{1 - u^2} = 0.10 + u$$

Squaring both sides,
$$0.072399(1 - u^2) = 0.01 + 0.2u + u^2$$

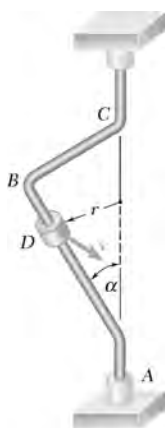
or
$$1.072399u^2 + 0.2u - 0.062399 = 0$$

The positive root of the quadratic equation is $u = 0.16537$

Then,
$$\theta + \phi = \sin^{-1} u = 9.5187^\circ$$

$$\phi = 9.5187^\circ - \theta$$

$$\phi = 3.52^\circ \blacktriangleleft$$



PROBLEM 12.55

A small, 300-g collar D can slide on portion AB of a rod which is bent as shown. Knowing that $\alpha = 40^\circ$ and that the rod rotates about the vertical AC at a constant rate of 5 rad/s, determine the value of r for which the collar will not slide on the rod if the effect of friction between the rod and the collar is neglected.

SOLUTION

First note

$$v_D = r\dot{\theta}_{ABC}$$

$$+\uparrow \Sigma F_y = 0: N \sin 40^\circ - W = 0$$

or

$$N = \frac{mg}{\sin 40^\circ}$$

$$+\rightarrow \Sigma F_n = ma_n: N \cos 40^\circ = m \frac{v_D^2}{r}$$

or

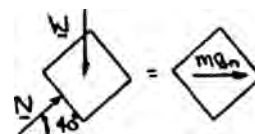
$$\frac{mg}{\sin 40^\circ} \cos 40^\circ = m \frac{(r\dot{\theta}_{ABC})^2}{r}$$

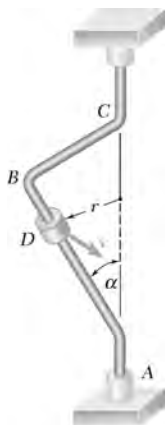
or

$$\begin{aligned} r &= \frac{g}{\dot{\theta}_{ABC}^2 \tan 40^\circ} \\ &= \frac{9.81 \text{ m/s}^2}{(5 \text{ rad/s})^2 \tan 40^\circ} \\ &= 0.468 \text{ m} \end{aligned}$$

or

$$r = 468 \text{ mm} \quad \blacktriangleleft$$





PROBLEM 12.56

A small, 200-g collar D can slide on portion AB of a rod which is bent as shown. Knowing that the rod rotates about the vertical AC at a constant rate and that $\alpha = 30^\circ$ and $r = 600$ mm, determine the range of values of the speed v for which the collar will not slide on the rod if the coefficient of static friction between the rod and the collar is 0.30.

SOLUTION

Case 1: $v = v_{\min}$, impending motion downward

$$+\nearrow \Sigma F_x = ma_x: N - W \sin 30^\circ = m \frac{v^2}{r} \cos 30^\circ$$

$$\text{or} \quad N = m \left(g \sin 30^\circ + \frac{v^2}{r} \cos 30^\circ \right)$$

$$+\searrow \Sigma F_y = ma_y: F - W \cos 30^\circ = -m \frac{v^2}{r} \sin 30^\circ$$

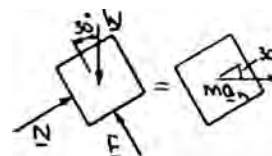
$$\text{or} \quad F = m \left(g \cos 30^\circ - \frac{v^2}{r} \sin 30^\circ \right)$$

$$\text{Now} \quad F = \mu_s N$$

$$\text{Then} \quad m \left(g \cos 30^\circ - \frac{v^2}{r} \sin 30^\circ \right) = \mu_s \times m \left(g \sin 30^\circ + \frac{v^2}{r} \cos 30^\circ \right)$$

$$\begin{aligned} \text{or} \quad v^2 &= gr \frac{1 - \mu_s \tan 30^\circ}{\mu_s + \tan 30^\circ} \\ &= (9.81 \text{ m/s}^2)(0.6 \text{ m}) \frac{1 - 0.3 \tan 30^\circ}{0.3 + \tan 30^\circ} \end{aligned}$$

$$\text{or} \quad v_{\min} = 2.36 \text{ m/s}$$



PROBLEM 12.56 (Continued)

Case 2: $v = v_{\max}$, impending motion upward

$$+\nearrow \Sigma F_x = ma_x: N - W \sin 30^\circ = m \frac{v^2}{r} \cos 30^\circ$$

or
$$N = m \left(g \sin 30^\circ + \frac{v^2}{r} \cos 30^\circ \right)$$

$$+\searrow \Sigma F_y = ma_y: F + W \cos 30^\circ = m \frac{v^2}{r} \sin 30^\circ$$

or
$$F = m \left(-g \cos 30^\circ + \frac{v^2}{r} \sin 30^\circ \right)$$

Now
$$F = \mu_s N$$

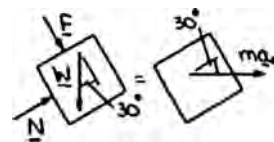
Then
$$m \left(-g \cos 30^\circ + \frac{v^2}{r} \sin 30^\circ \right) = \mu_s \times m \left(g \sin 30^\circ + \frac{v^2}{r} \cos 30^\circ \right)$$

or
$$v^2 = gr \frac{1 + \mu_s \tan 30^\circ}{\tan 30^\circ - \mu_s}$$

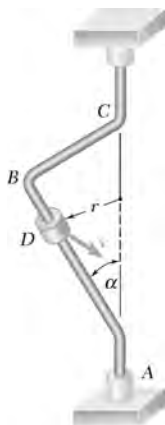
$$= (9.81 \text{ m/s}^2)(0.6 \text{ m}) \frac{1 + 0.3 \tan 30^\circ}{\tan 30^\circ - 0.3}$$

or
$$v_{\max} = 4.99 \text{ m/s}$$

For the collar not to slide



$$2.36 \text{ m/s} \leq v \leq 4.99 \text{ m/s} \quad \blacktriangleleft$$



PROBLEM 12.57

A small, 250 g collar D can slide on portion AB of a rod which is bent as shown. Knowing that $r = 200$ mm and that the rod rotates about the vertical AC at a constant rate of 10 rad/s, determine the smallest allowable value of the coefficient of static friction between the collar and the rod if the collar is not to slide when (a) $\alpha = 15^\circ$, (b) $\alpha = 45^\circ$. Indicate for each case the direction of the impending motion.

SOLUTION

First we note that $v = r\dot{\theta}_{ABC} = (0.2 \text{ m})(10 \text{ rad/s}) = 2 \text{ m/s}$, and that requiring $\mu_s = (\mu_s)_{\min}$ implies that sliding of collar D is impending.

Also, $\mu_s = \tan \phi_s$

Now we consider the two possible cases of impending motion.

Case 1: Impending motion downward.

$$+\nearrow \Sigma F_x = ma_x: \quad N - W \sin \alpha = \frac{W}{g} \frac{v^2}{r} \cos \alpha$$

or

$$N = W \left(\sin \alpha + \frac{v^2}{gr} \cos \alpha \right)$$

$$+\searrow \Sigma F_y = ma_y: \quad F - W \cos \alpha = -\frac{W}{g} \frac{v^2}{r} \sin \alpha$$

or

$$F = W \left(\cos \alpha - \frac{v^2}{gr} \sin \alpha \right)$$

Now

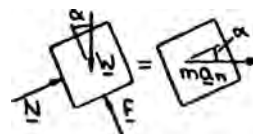
$$F = \mu_s N$$

Then

$$W \left(\cos \alpha - \frac{v^2}{gr} \sin \alpha \right) = \mu_s \times W \left(\sin \alpha + \frac{v^2}{gr} \cos \alpha \right)$$

or

$$\begin{aligned} \frac{v^2}{gr} &= \frac{1 - \mu_s \tan \alpha}{\tan \alpha + \mu_s} = \frac{1 - \tan \phi_s \tan \alpha}{\tan \alpha + \tan \phi_s} \\ &= \frac{1}{\tan(\alpha + \phi_s)} \end{aligned}$$



PROBLEM 12.57 (Continued)

Case 2: Impending motion upward.

$$\nearrow \Sigma F_x = ma_x: N - W \sin \alpha = \frac{W}{g} \frac{v^2}{r} \cos \alpha$$

or
$$N = W \left(\sin \alpha + \frac{v^2}{gr} \cos \alpha \right)$$

$$\nwarrow \Sigma F_y = ma_y: F + W \cos \alpha = \frac{W}{g} \frac{v^2}{r} \sin \alpha$$

or
$$F = W \left(-\cos \alpha + \frac{v^2}{gr} \sin \alpha \right)$$

Now
$$F = \mu_s N$$

Then
$$W \left(-\cos \alpha + \frac{v^2}{gr} \sin \alpha \right) = \mu_s \times W \left(\sin \alpha + \frac{v^2}{gr} \cos \alpha \right)$$

or
$$\begin{aligned} \frac{v^2}{gr} &= \frac{1 + \mu_s \tan \alpha}{\tan \alpha - \mu_s} = \frac{1 + \tan \alpha \tan \phi_s}{\tan \alpha - \tan \phi_s} \\ &= \frac{1}{\tan(\alpha - \phi_s)} \end{aligned}$$

Now
$$\frac{gr}{v^2} = \frac{(9.81 \text{ m/s}^2)(0.2 \text{ m})}{(2 \text{ m/s})^2} = 0.4905$$

Then
$$\tan(\alpha \pm \phi_s) = 0.4905$$

or
$$\alpha \pm \phi_s = 26.128^\circ, \quad \phi_s \geq 0$$

and where the “+” corresponds to impending motion downward and the “−” to impending motion upward.

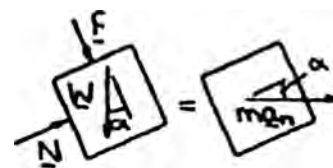
(a) $\alpha = 15^\circ$:

We have
$$15^\circ \pm \phi_s = 26.128^\circ$$

$\phi_s \geq 0 \Rightarrow$ “+” so that
$$\phi_s = 11.128^\circ$$

Then
$$(\mu_s)_{\min} = \tan 11.128^\circ$$

or
$$(\mu_s)_{\min} = 0.1967, \text{ motion impending downward}$$



PROBLEM 12.57 (Continued)

(b) $\alpha = 45^\circ$:

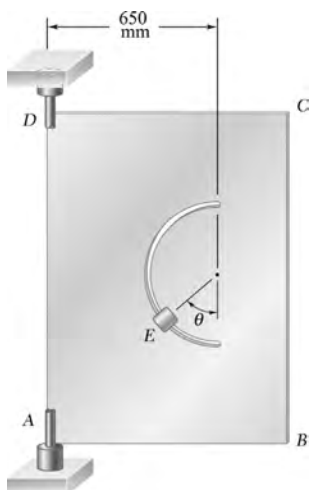
We have $45^\circ \pm \phi_s = 26.128^\circ$

$\phi_s \geq 0 \Rightarrow \text{“-”}$ so that $\phi_s = 18.872^\circ$

Then $(\mu_s)_{\min} = \tan 18.872^\circ$

or $\mu_s = 0.342$ motion impending upward





PROBLEM 12.58

A semicircular slot of 250-mm radius is cut in a flat plate which rotates about the vertical AD at a constant rate of 14 rad/s. A small, 400-g block E is designed to slide in the slot as the plate rotates. Knowing that the coefficients of friction are $\mu_s = 0.35$ and $\mu_k = 0.25$, determine whether the block will slide in the slot if it is released in the position corresponding to (a) $\theta = 80^\circ$, (b) $\theta = 40^\circ$. Also determine the magnitude and the direction of the friction force exerted on the block immediately after it is released.

SOLUTION

First note

$$\begin{aligned}\rho &= (650 - 250 \sin \theta) \text{ mm} \\ &= (0.65 - 0.25 \sin \theta) \text{ m}\end{aligned}$$

$$v_E = \rho \dot{\phi}_{ABCD}$$

Then

$$\begin{aligned}a_n &= \frac{v_E^2}{\rho} = \rho (\dot{\phi}_{ABCD})^2 \\ &= (0.65 - 0.25 \sin \theta) \text{ m} (14 \text{ rad/s})^2 \\ &= (127.4 - 49 \sin \theta) \text{ m/s}^2\end{aligned}$$

Assume that the block is at rest with respect to the plate.

$$\nearrow \Sigma F_x = ma_x: \quad N + W \cos \theta = m \frac{v_E^2}{\rho} \sin \theta$$

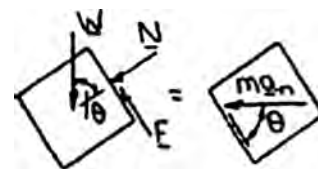
or

$$N = W \left(-\cos \theta + \frac{v_E^2}{g\rho} \sin \theta \right)$$

$$\nwarrow \Sigma F_y = ma_y: \quad -F + W \sin \theta = -m \frac{v_E^2}{\rho} \cos \theta$$

or

$$F = W \left(\sin \theta + \frac{v_E^2}{g\rho} \cos \theta \right)$$



PROBLEM 12.58 (Continued)

(a) We have $\theta = 80^\circ$

Then

$$N = (0.4 \times 9.81) \left[-\cos 80^\circ + \frac{1}{9.81 \text{ m/s}^2} \times (127.4 - 49 \sin 80^\circ) \times \sin 80^\circ \right]$$

$$= 30.4954 \text{ N}$$

$$F = (0.4 \times 9.81) \left[\sin 80^\circ + \frac{1}{9.81 \text{ m/s}^2} \times (127.4 - 49 \sin 80^\circ) \times \cos 80^\circ \right]$$

$$= 9.3617 \text{ N}$$

Now $F_{\max} = \mu_s N = 0.35(30.4954) = 10.6734 \text{ N}$

The block does not slide in the slot, and

$$F = 9.3617 \text{ N} \quad \nearrow 80^\circ \quad \blacktriangleleft$$

(b) We have $\theta = 40^\circ$

Then

$$N = (0.4 \times 9.81) \left[-\cos 40^\circ + \frac{1}{9.81} \times (127.4 - 49 \sin 40^\circ) \times \sin 40^\circ \right]$$

$$= 21.652 \text{ N}$$

$$F = (0.4 \times 9.81) \left[\sin 40^\circ + \frac{1}{9.81} \times (127.4 - 49 \sin 40^\circ) \times \cos 40^\circ \right]$$

$$= 31.9088 \text{ N}$$

Now $F_{\max} = \mu_s N$, from which it follows that

$$F > F_{\max}$$

Block E will slide in the slot

and

$$\mathbf{a}_E = \mathbf{a}_n + \mathbf{a}_{E/\text{plate}}$$

$$= \mathbf{a}_n + (\mathbf{a}_{E/\text{plate}})_t + (\mathbf{a}_{E/\text{plate}})_n$$

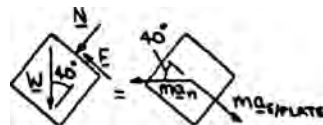
At $t = 0$, the block is at rest relative to the plate, thus $(\mathbf{a}_{E/\text{plate}})_n = 0$ at $t = 0$, so that $\mathbf{a}_{E/\text{plate}}$ must be directed tangentially to the slot.

$$\nearrow \Sigma F_x = ma_x: \quad N + W \cos 40^\circ = m \frac{v_E^2}{\rho} \sin 40^\circ$$

or

$$N = W \left(-\cos 40^\circ + \frac{v_E^2}{g\rho} \sin 40^\circ \right) \quad (\text{as above})$$

$$= 21.652 \text{ N}$$



PROBLEM 12.58 (Continued)

Sliding:

$$\begin{aligned} F &= \mu_k N \\ &= 0.25(21.652 \text{ N}) \\ &= 5.413 \text{ N} \end{aligned}$$

Noting that $\mathbf{F}$ and $\mathbf{a}_{E/\text{plane}}$ must be directed as shown (if their directions are reversed, then $\Sigma \mathbf{F}_x$ is $\nearrow$ while $m\mathbf{a}_x$ is $\nwarrow$), we have
the block slides downward in the slot and

$$\mathbf{F} = 5.413 \text{ N} \nearrow 40^\circ \blacktriangleleft$$



PROBLEM 12.59

Three seconds after a polisher is started from rest, small tufts of fleece from along the circumference of the 225-mm-diameter polishing pad are observed to fly free of the pad. If the polisher is started so that the fleece along the circumference undergoes a constant tangential acceleration of 4 m/s^2 , determine (a) the speed v of a tuft as it leaves the pad, (b) the magnitude of the force required to free a tuft if the average mass of a tuft is 1.6 mg .

SOLUTION

- (a) $a_t = \text{constant} \Rightarrow$ uniformly acceleration motion

Then $v = 0 + a_t t$

At $t = 3 \text{ s}$: $v = (4 \text{ m/s}^2)(3 \text{ s})$

or

$v = 12.00 \text{ m/s} \blacktriangleleft$

- (b) $\Sigma F_t = ma_t$; $F_t = ma_t$

or

$$F_t = (1.6 \times 10^{-6} \text{ kg})(4 \text{ m/s}^2) \\ = 6.4 \times 10^{-6} \text{ N}$$

$$\Sigma F_n = ma_n; \quad F_n = m \frac{v^2}{\rho}$$

At $t = 3 \text{ s}$: $F_n = (1.6 \times 10^{-6} \text{ kg}) \frac{(12 \text{ m/s})^2}{\left(\frac{0.225}{2} \text{ m}\right)}$

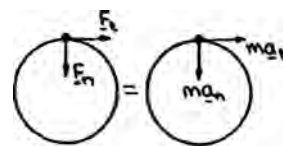
$$= 2.048 \times 10^{-3} \text{ N}$$

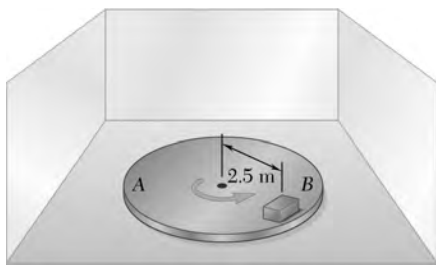
Finally,

$$F_{\text{tuft}} = \sqrt{F_t^2 + F_n^2} \\ = \sqrt{(6.4 \times 10^{-6} \text{ N})^2 + (2.048 \times 10^{-3} \text{ N})^2}$$

or

$F_{\text{tuft}} = 2.05 \times 10^{-3} \text{ N} \blacktriangleleft$





PROBLEM 12.60

A turntable A is built into a stage for use in a theatrical production. It is observed during a rehearsal that a trunk B starts to slide on the turntable 10 s after the turntable begins to rotate. Knowing that the trunk undergoes a constant tangential acceleration of 0.24 m/s^2 , determine the coefficient of static friction between the trunk and the turntable.

SOLUTION

First we note that $(a_B)_t = \text{constant}$ implies uniformly accelerated motion.

$$v_B = 0 + (a_B)_t t$$

At $t = 10 \text{ s}$:

$$v_B = (0.24 \text{ m/s}^2)(10 \text{ s}) = 2.4 \text{ m/s}$$

In the plane of the turntable

$$\Sigma \mathbf{F} = m_B \mathbf{a}_B: \quad \mathbf{F} = m_B (\mathbf{a}_B)_t + m_B (\mathbf{a}_B)_n$$

Then

$$\begin{aligned} F &= m_B \sqrt{(a_B)_t^2 + (a_B)_n^2} \\ &= m_B \sqrt{(a_B)_t^2 + \left(\frac{v_B^2}{\rho} \right)^2} \end{aligned}$$

$$+\uparrow \Sigma F_y = 0: \quad N - W = 0$$

or

$$N = m_B g$$

At $t = 10 \text{ s}$:

$$F = \mu_s N = \mu_s m_B g$$

Then

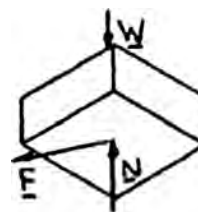
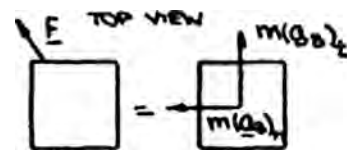
$$\mu_s m_B g = m_B \sqrt{(a_B)_t^2 + \left(\frac{v_B^2}{\rho} \right)^2}$$

or

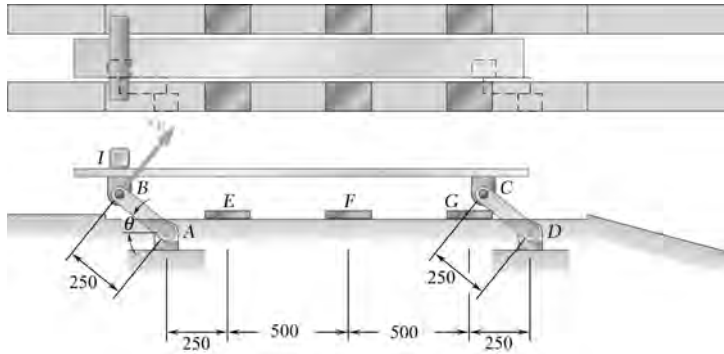
$$\mu_s = \frac{1}{9.81 \text{ m/s}^2} \left\{ (0.24 \text{ m/s}^2)^2 + \left[\frac{(2.4 \text{ m/s})^2}{2.5 \text{ m}} \right]^2 \right\}^{1/2}$$

or

$$\mu_s = 0.236 \quad \blacktriangleleft$$



PROBLEM 12.61

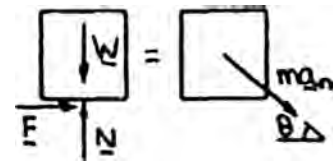


The parallel-link mechanism $ABCD$ is used to transport a component I between manufacturing processes at stations E , F , and G by picking it up at a station when $\theta = 0$ and depositing it at the next station when $\theta = 180^\circ$. Knowing that member BC remains horizontal throughout its motion and that links AB and CD rotate at a constant rate in a vertical plane in such a way that $v_B = 0.7$ m/s, determine (a) the minimum value of the coefficient of static friction between the component and BC if the component is not to slide on BC while being transferred, (b) the values of θ for which sliding is impending.

SOLUTION

$$\rightarrow \Sigma F_x = ma_x: \quad F = \frac{W}{g} \frac{v_B^2}{\rho} \cos \theta$$

$$+\uparrow \Sigma F_y = ma_y: \quad N - W = -\frac{W}{g} \frac{v_B^2}{\rho} \sin \theta$$



or

$$N = W \left(1 - \frac{v_B^2}{g\rho} \sin \theta \right)$$

Now

$$F_{\max} = \mu_s N = \mu_s W \left(1 - \frac{v_B^2}{g\rho} \sin \theta \right)$$

and for the component not to slide

$$F \leq F_{\max}$$

or

$$\frac{W}{g} \frac{v_B^2}{\rho} \cos \theta \leq \mu_s W \left(1 - \frac{v_B^2}{g\rho} \sin \theta \right)$$

or

$$\mu_s \geq \frac{\cos \theta}{\frac{g\rho}{v_B^2} - \sin \theta}$$

PROBLEM 12.61 (Continued)

We must determine the values of θ which maximize the above expression. Thus

$$\frac{d}{d\theta} \left(\frac{\cos \theta}{\frac{g\rho}{v_B^2} - \sin \theta} \right) = \frac{-\sin \theta \left(\frac{g\rho}{v_B^2} - \sin \theta \right) - (\cos \theta)(-\cos \theta)}{\left(\frac{g\rho}{v_B^2} - \sin \theta \right)^2} = 0$$

or $\sin \theta = \frac{v_B^2}{g\rho}$ for $\mu_s = (\mu_s)_{\min}$

Now $\sin \theta = \frac{(0.7 \text{ m/s})^2}{(9.81 \text{ m/s}^2)(0.25 \text{ m})} = 0.199796$

or $\theta = 11.525^\circ$ and $\theta = 168.475^\circ$

(a) From above,

$$(\mu_s)_{\min} = \frac{\cos \theta}{\frac{g\rho}{v_B^2} - \sin \theta} \quad \text{where} \quad \sin \theta = \frac{v_B^2}{g\rho}$$

$$(\mu_s)_{\min} = \frac{\cos \theta}{\frac{1}{\sin \theta} - \sin \theta} = \frac{\cos \theta \sin \theta}{1 - \sin^2 \theta} = \tan \theta$$

$$= \tan 11.525^\circ$$

or $(\mu_s)_{\min} = 0.204 \blacktriangleleft$

(b) We have impending motion

to the left for $\theta = 11.53^\circ \blacktriangleleft$

to the right for $\theta = 168.5^\circ \blacktriangleleft$

PROBLEM 12.62

Knowing that the coefficients of friction between the component I and member BC of the mechanism of Problem 12.61 are $\mu_s = 0.35$ and $\mu_k = 0.25$, determine (a) the maximum allowable constant speed v_B if the component is not to slide on BC while being transferred, (b) the values of θ for which sliding is impending.

SOLUTION

$$+\rightarrow \Sigma F_x = ma_x: \quad F = \frac{W}{g} \frac{v_B^2}{\rho} \cos \theta$$

$$+\uparrow \Sigma F_y = ma_y: \quad N - W = -\frac{W}{g} \frac{v_B^2}{\rho} \sin \theta$$

or

$$N = W \left(1 - \frac{v_B^2}{g\rho} \sin \theta \right)$$

Now

$$\begin{aligned} F_{\max} &= \mu_s N \\ &= \mu_s W \left(1 - \frac{v_B^2}{g\rho} \sin \theta \right) \end{aligned}$$

and for the component not to slide

$$F \leq F_{\max}$$

or

$$\frac{W}{g} \frac{v_B^2}{\rho} \cos \theta \leq \mu_s W \left(1 - \frac{v_B^2}{g\rho} \sin \theta \right)$$

or

$$v_B^2 \leq \mu_s \frac{g\rho}{\cos \theta + \mu_s \sin \theta} \quad (1)$$

To ensure that this inequality is satisfied, $(v_B^2)_{\max}$ must be less than or equal to the minimum value of $\mu_s g\rho / (\cos \theta + \mu_s \sin \theta)$, which occurs when $(\cos \theta + \mu_s \sin \theta)$ is maximum. Thus

$$\frac{d}{d\theta} (\cos \theta + \mu_s \sin \theta) = -\sin \theta + \mu_s \cos \theta = 0$$

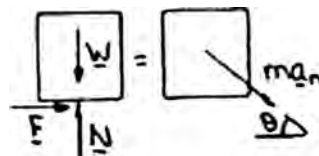
or

$$\tan \theta = \mu_s$$

$$\mu_s = 0.35$$

or

$$\theta = 19.2900^\circ$$



PROBLEM 12.62 (Continued)

- (a) The maximum allowed value of v_B is then

$$\begin{aligned} (v_B^2)_{\max} &= \mu_s \frac{g\rho}{\cos \theta + \mu_s \sin \theta} \quad \text{where} \quad \tan \theta = \mu_s \\ &= g\rho \frac{\tan \theta}{\cos \theta + (\tan \theta) \sin \theta} = g\rho \sin \theta \\ &= (9.81)(0.25\text{m}) \sin 19.29^\circ \end{aligned}$$

or

$$(v_B)_{\max} = 0.900 \text{ m/s} \quad \blacktriangleleft$$

- (b) First note that for $90^\circ < \theta \leq 180^\circ$, Eq. (1) becomes

$$v_B^2 \leq \mu_s \frac{g\rho}{\cos \alpha + \mu_s \sin \alpha}$$

where $\alpha = 180^\circ - \theta$. It then follows that the second value of θ for which motion is impending is

$$\begin{aligned} \theta &= 180^\circ - 19.2900^\circ \\ &= 160.7100^\circ \end{aligned}$$

we have impending motion

to the left for

$$\theta = 19.29^\circ \quad \blacktriangleleft$$

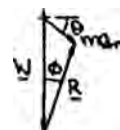
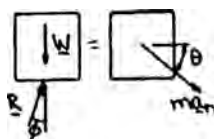
to the right for

$$\theta = 160.7^\circ \quad \blacktriangleleft$$

Alternative solution.

$$\Sigma \mathbf{F} = m\mathbf{a}: \quad \mathbf{W} + \mathbf{R} = m\mathbf{a}_n$$

Then



For impending motion, $\phi = \phi_s$. Also, as shown above, the values of θ for which motion is impending minimize the value of v_B , and thus the value of a_n is $\left(a_n = \frac{v_B^2}{\rho}\right)$. From the above diagram, it can be concluded that a_n is minimum when $m\mathbf{a}_n$ and $\mathbf{R}$ are perpendicular.

PROBLEM 12.62 (Continued)

Therefore,

from the diagram

$$\theta = \phi_s = \tan^{-1} \mu_s \quad (\text{as above})$$

and

$$ma_n = W \sin \phi_s$$

or

$$m \frac{v_B^2}{\rho} = mg \sin \theta$$

or

$$v_B^2 = g\rho \sin \theta \quad (\text{as above})$$

For $90^\circ < \theta \leq 180^\circ$, we have

from the diagram

$$\alpha = 180^\circ - \theta \quad (\text{as above})$$

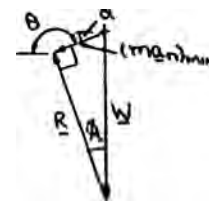
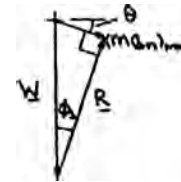
$$\alpha = \phi_s$$

and

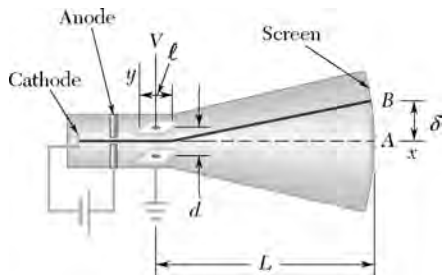
$$ma_n = W \sin \phi_s$$

or

$$v_B^2 = g\rho \sin \theta \quad (\text{as above})$$



PROBLEM 12.63



In the cathode-ray tube shown, electrons emitted by the cathode and attracted by the anode pass through a small hole in the anode and then travel in a straight line with a speed v_0 until they strike the screen at A. However, if a difference of potential V is established between the two parallel plates, the electrons will be subjected to a force $\mathbf{F}$ perpendicular to the plates while they travel between the plates and will strike the screen at Point B, which is at a distance δ from A. The magnitude of the force $\mathbf{F}$ is $F = eV/d$, where $-e$ is the charge of an electron and d is the distance between the plates. Derive an expression for the deflection d in terms of V , v_0 , the charge $-e$ and the mass m of an electron, and the dimensions d , ℓ , and L .

SOLUTION

Consider the motion of one electron. For the horizontal motion, let $x = 0$ at the left edge of the plate and $x = \ell$ at the right edge of the plate. At the screen,

$$x = \frac{\ell}{2} + L$$

Horizontal motion: There are no horizontal forces acting on the electron so that $a_x = 0$.

Let $t_1 = 0$ when the electron passes the left edge of the plate, $t = t_1$ when it passes the right edge, and $t = t_2$ when it impacts on the screen. For uniform horizontal motion,

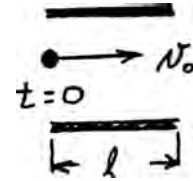
$$x = v_0 t,$$

so that

$$t_1 = \frac{\ell}{v_0}$$

and

$$t_2 = \frac{\ell}{2v_0} + \frac{L}{v_0}.$$



Vertical motion: The gravity force acting on the electron is neglected since we are interested in the deflection produced by the electric force. While the electron is between plates ($0 \leq t \leq t_1$), the vertical force on the electron is $F_y = eV/d$. After it passes the plates ($t_1 \leq t \leq t_2$), it is zero.

$$\text{For } 0 \leq t \leq t_1, \quad \Sigma F_y = ma_y: \quad a_y = \frac{F_y}{m} = \frac{eV}{md}$$

$$v_y = (v_y)_0 + a_y t = 0 + \frac{eVt}{md}$$

$$y = y_0 + (v_y)_0 t + \frac{1}{2} a_y t^2 = 0 + 0 + \frac{eVt^2}{2md}$$

PROBLEM 12.63 (Continued)

At $t = t_1$, $(v_y)_1 = \frac{eVt_1}{md}$ and $y_1 = \frac{eVt_1^2}{2md}$

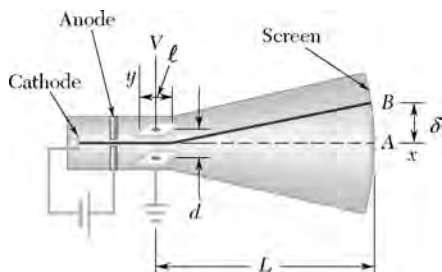
For $t_1 \leq t \leq t_2$, $a_y = 0$

$$y = y_1 + (v_y)_1(t - t_1)$$

At $t = t_2$, $y_2 = \delta = y_1 + (v_y)_1(t_2 - t_1)$

$$\delta = \frac{eVt_1^2}{2md} + \frac{eVt_1}{md}(t_2 - t_1) = \frac{eVt_1}{md} \left(t_2 - \frac{1}{2}t_1 \right)$$

$$= \frac{eV\ell}{mdv_0} \left(\frac{\ell}{2v_0} + \frac{L}{v_0} - \frac{1}{2} \frac{\ell}{v_0} \right) \quad \text{or} \quad \delta = \frac{eV\ell L}{mdv_0^2} \blacktriangleleft$$



PROBLEM 12.64

In Problem 12.63, determine the smallest allowable value of the ratio d/ℓ in terms of e , m , v_0 , and V if at $x = \ell$, the minimum permissible distance between the path of the electrons and the positive plate is $0.05d$.

PROBLEM 12.63 In the cathode-ray tube shown, electrons emitted by the cathode and attracted by the anode pass through a small hole in the anode and then travel in a straight line with a speed v_0 until they strike the screen at A. However, if a difference of potential V is established between the two parallel plates, the electrons will be subjected to a force $\mathbf{F}$ perpendicular to the plates while they travel between the plates and will strike the screen at Point B, which is at a distance δ from A. The magnitude of the force $\mathbf{F}$ is $F = eV/d$, where $-e$ is the charge of an electron and d is the distance between the plates. Neglecting the effects of gravity, derive an expression for the deflection δ in terms of V , v_0 , the charge $-e$ and the mass m of an electron, and the dimensions d , ℓ , and L .

SOLUTION

Consider the motion of one electron. For the horizontal motion, let $x = 0$ at the left edge of the plate and $x = \ell$ at the right edge of the plate. At the screen,

$$x = \frac{\ell}{2} + L$$

Horizontal motion: There are no horizontal forces acting on the electron so that $a_x = 0$.

Let $t_1 = 0$ when the electron passes the left edge of the plate, $t = t_1$ when it passes the right edge, and $t = t_2$ when it impacts on the screen. For uniform horizontal motion,

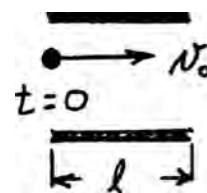
$$x = v_0 t,$$

so that

$$t_1 = \frac{\ell}{v_0}$$

and

$$t_2 = \frac{\ell}{2v_0} + \frac{L}{v_0}.$$



Vertical motion: The gravity force acting on the electron is neglected since we are interested in the deflection produced by the electric force. While the electron is between the plates ($0 \leq t \leq t_1$), the vertical force on the electron is $F_y = eV/d$. After it passes the plates ($t_1 \leq t \leq t_2$), it is zero.

PROBLEM 12.64 (Continued)

For $0 \leq t \leq t_1$,

$$\Sigma F_y = ma_y: \quad a_y = \frac{F_y}{m} = \frac{eV}{md}$$

$$v_y = (v_y)_0 + a_y t = 0 + \frac{eVt}{md}$$

$$y = y_0 + (v_y)_0 t + \frac{1}{2} a_y t^2 = 0 + 0 + \frac{eVt^2}{2md}$$

At $t = t_1$,

$$\frac{\ell}{v_0}, \quad y = \frac{eV\ell^2}{2mdv_0^2}$$

But

$$y < \frac{d}{2} - 0.05d = 0.450d$$

so that

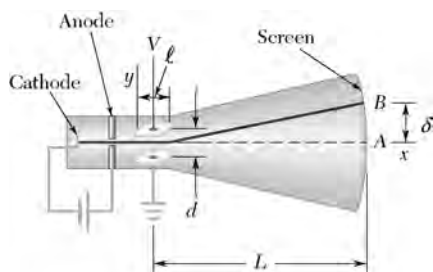
$$\frac{eV\ell^2}{2mdv_0^2} < 0.450d$$

$$\frac{d^2}{\ell^2} > \frac{1}{0.450} \frac{eV}{2mv_0^2} = 1.111 \frac{eV}{mv_0^2}$$

$$\frac{d}{\ell} > 1.054 \sqrt{\frac{eV}{mv_0^2}} \quad \blacktriangleleft$$

PROBLEM 12.65

The current model of a cathode-ray tube is to be modified so that the length of the tube and the spacing between the plates are reduced by 40 percent and 20 percent, respectively. If the size of the screen is to remain the same, determine the new length ℓ' of the plates, assuming that all of the other characteristics of the tube are to remain unchanged. (See Problem 12.63 for a description of a cathode-ray tube.)



PROBLEM 12.63 In the cathode-ray tube shown, electrons emitted by the cathode and attracted by the anode pass through a small hole in the anode and then travel in a straight line with a speed v_0 until they strike the screen at A. However, if a difference of potential V is established between the two parallel plates, the electrons will be subjected to a force F perpendicular to the plates while they travel between the plates and will strike the screen at Point B, which is at a distance δ from A. The magnitude of the force F is $F = eV/d$, where $-e$ is the charge of an electron and d is the distance between the plates. Neglecting the effects of gravity, derive an expression for the deflection δ in terms of V , v_0 , the charge $-e$ and the mass m of an electron, and the dimensions d , ℓ , and L .

SOLUTION

Consider the motion of one electron. For the horizontal motion, let $x = 0$ at the left edge of the plate and $x = \ell$ at the right edge of the plate. At the screen,

$$x = \frac{\ell}{2} + L$$

Horizontal motion: There are no horizontal forces acting on the electron so that $a_x = 0$.

Let $t_1 = 0$ when the electron passes the left edge of the plate, $t = t_1$ when it passes the right edge, and $t = t_2$ when it impacts on the screen. For uniform horizontal motion,

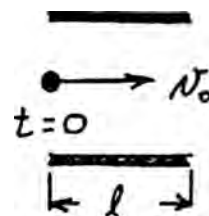
$$x = v_0 t,$$

so that

$$t_1 = \frac{\ell}{v_0}$$

and

$$t_2 = \frac{\ell}{2v_0} + \frac{L}{v_0}.$$



Vertical motion: The gravity force acting on the electron is neglected since we are interested in the deflection produced by the electric force. While the electron is between the plates ($0 \leq t \leq t_1$), the vertical force on the electron is $F_y = eV/d$. After it passes the plates ($t_1 \leq t \leq t_2$), it is zero.

PROBLEM 12.65 (Continued)

For $0 \leq t \leq t_1$,

$$\Sigma F_y = ma_y: \quad a_y = \frac{F_y}{m} = \frac{eV}{md}$$

$$v_y = (v_y)_0 + a_y t = 0 + \frac{eVt}{md}$$

$$y = y_0 + (v_y)_0 t + \frac{1}{2} a_y t^2 = 0 + 0 + \frac{eVt^2}{2md}$$

Let δ , ℓ , L , and d be dimensions taken from the current model and δ' , ℓ' , L' , and d' be those of the modified model.

$$\delta = \frac{eV \ell L}{mdv_0^2}$$

$$\delta' = \frac{eV \ell' L'}{md'v_0^2}$$

Since e , V , m , and v_0 are constants

$$\frac{\delta'}{\delta} = \left(\frac{\ell'}{\ell} \right) \left(\frac{L'}{L} \right) \left(\frac{d}{d'} \right)$$

But

$$\delta' = \delta, \quad L' = L - 0.40L = 0.6L,$$

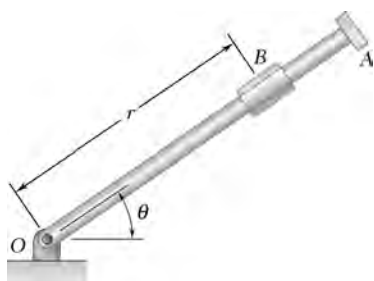
$$d' = d - 0.20L = 0.8d$$

Then

$$\frac{\delta'}{\delta} = \left(\frac{\ell'}{\ell} \right) \left(\frac{0.6L}{L} \right) \left(\frac{d}{0.8d} \right)$$

$$L = 0.75 \frac{\ell'}{\ell}$$

$$\ell' = 1.333\ell \quad \blacktriangleleft$$



PROBLEM 12.66

Rod OA rotates about O in a horizontal plane. The motion of the 300 g collar B is defined by the relations $r = 300 + 100 \cos(0.5\pi t)$ and $\theta = \pi(t^2 - 3t)$, where r is expressed in millimeters, t in seconds, and θ in radians. Determine the radial and transverse components of the force exerted on the collar when (a) $t = 0$, (b) $t = 0.5$ s.

SOLUTION

Polar coordinates and their derivatives.

$$r = 0.3 + 0.1 \cos(0.5\pi t) \text{ m} \quad \theta = \pi(t^2 - 3t) \text{ rad}$$

$$\dot{r} = -0.05\pi \sin(0.5\pi t) \text{ m/s} \quad \dot{\theta} = \pi(2t - 3) \text{ rad/s}$$

$$\ddot{r} = -0.025\pi^2 \cos(0.5\pi t) \text{ m/s}^2 \quad \ddot{\theta} = 2\pi \text{ rad/s}^2$$

(a) For $t = 0$.

$$r = 0.4 \text{ m} \quad \theta = 0$$

$$\dot{r} = 0 \quad \dot{\theta} = -3\pi \text{ rad/s}$$

$$\ddot{r} = -0.025\pi^2 \text{ m/s}^2 \quad \ddot{\theta} = 2\pi \text{ rad/s}^2$$

Components of acceleration.

$$\begin{aligned} a_r &= \ddot{r} - r\dot{\theta}^2 \\ &= -0.025\pi^2 - 0.4(-3\pi)^2 \\ &= -35.777 \text{ m/s}^2 \end{aligned}$$

$$\begin{aligned} a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} \\ &= (0.4)(2\pi) + 2(-3\pi)(0) \\ &= 2.513 \text{ m/s}^2 \end{aligned}$$

Components of force.

$$F_r = ma_r = (0.3)(-35.777) \quad F_r = -10.73 \text{ N} \blacktriangleleft$$

$$F_\theta = ma_\theta = (0.3)(2.513) \quad F_\theta = 0.754 \text{ N} \blacktriangleleft$$

(b) For $t = 0.5$ s.

$$r = 0.37071 \text{ m} \quad \theta = -3.927 \text{ rad}$$

$$\dot{r} = -0.11107 \text{ m/s} \quad \dot{\theta} = -2\pi \text{ rad/s}$$

$$\ddot{r} = -0.17447 \text{ m/s}^2 \quad \ddot{\theta} = 2\pi \text{ rad/s}^2$$

PROBLEM 12.66 (Continued)

Components of acceleration.

$$\begin{aligned}a_r &= \ddot{r} - r\dot{\theta}^2 \\ &= -14.809 \text{ m/s}^2\end{aligned}$$

$$\begin{aligned}a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} \\ &= 3.725 \text{ m/s}^2\end{aligned}$$

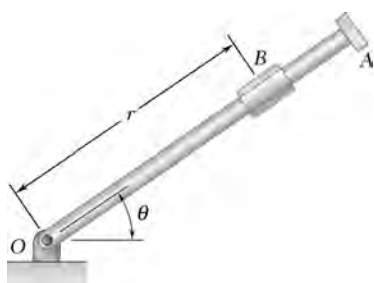
Components of force.

$$F_r = ma_r = (0.3)(-14.809)$$

$$F_r = -4.44 \text{ N} \quad \blacktriangleleft$$

$$F_\theta = ma_\theta = (0.3)(3.725)$$

$$F_\theta = 1.118 \text{ N} \quad \blacktriangleleft$$



PROBLEM 12.67

For the motion defined in Problem 12.66, determine the radial and transverse components of the force exerted on the collar when $t = 1.5$ s.

PROBLEM 12.66 Rod OA rotates about O in a horizontal plane. The motion of the 300 g collar B is defined by the relations $r = 300 + 100 \cos(0.5\pi t)$ and $\theta = \pi(t^2 - 3t)$, where r is expressed in millimeters, t in seconds, and θ in radians. Determine the radial and transverse components of the force exerted on the collar when (a) $t = 0$, (b) $t = 0.5$ s.

SOLUTION

Polar coordinates and their derivatives.

$$\begin{aligned} r &= 0.3 + 0.1 \cos(0.5\pi t) \text{ m} & \theta &= \pi(t^2 - 3t) \text{ rad} \\ \dot{r} &= -0.05\pi \sin(0.5\pi t) \text{ m/s} & \dot{\theta} &= \pi(2t - 3) \text{ rad/s} \\ \ddot{r} &= -0.025\pi^2 \cos(0.5\pi t) \text{ m/s}^2 & \ddot{\theta} &= 2\pi \text{ rad/s}^2 \end{aligned}$$

For $t = 1.5$ s.

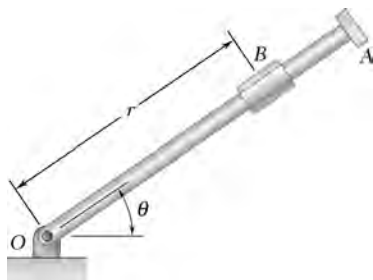
$$\begin{aligned} r &= 0.22929 \text{ m} & \theta &= -7.068 \text{ rad} \\ \dot{r} &= -0.11107 \text{ m/s} & \dot{\theta} &= 0 \\ \ddot{r} &= 0.17447 \text{ m/s}^2 & \ddot{\theta} &= 2\pi \text{ rad/s}^2 \end{aligned}$$

Components of acceleration.

$$\begin{aligned} a_r &= \ddot{r} - r\dot{\theta}^2 = 0.17447 \text{ m/s}^2 \\ a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} = 1.441 \text{ m/s}^2 \end{aligned}$$

Components of force.

$$\begin{aligned} F_r &= ma_r = (0.3)(0.17447) & F_r &= 0.0523 \text{ N} \blacktriangleleft \\ F_\theta &= ma_\theta = (0.3)(1.44107) & F_\theta &= 0.432 \text{ N} \blacktriangleleft \end{aligned}$$



PROBLEM 12.68

Rod OA oscillates about O in a horizontal plane. The motion of the 2.5 kg collar B is defined by the relations $r = 250/(t + 4)$ and $\theta = (2/\pi) \sin \pi t$, where r is expressed in mm, t in seconds, and θ in radians. Determine the radial and transverse components of the force exerted on the collar when (a) $t = 1$ s, (b) $t = 6$ s.

SOLUTION

We have $r = \frac{250}{t+4} \text{ mm} = \frac{0.25}{t+4} \text{ m}$ $\theta = \left(\frac{2}{\pi} \sin \pi t \right) \text{ rad}$

Then $\dot{r} = -\frac{0.25}{(t+4)^2} \text{ m/s}$ $\dot{\theta} = (2 \cos \pi t) \text{ rad/s}$

and $\ddot{r} = \frac{0.5}{(t+4)^3} \text{ m/s}^2$ $\ddot{\theta} = -(2\pi \sin \pi t) \text{ rad/s}^2$

(a) At $t = 1$ s: $r = 0.05 \text{ m}$
 $\dot{r} = -0.01 \text{ m/s}$ $\dot{\theta} = -2 \text{ rad/s}$
 $\ddot{r} = 0.004 \text{ m/s}^2$ $\ddot{\theta} = 0$

Now $a_r = \ddot{r} - r\dot{\theta}^2$
 $= (0.004 \text{ m/s}^2) - (0.05 \text{ m})(-2 \text{ rad/s})^2$
 $= -0.196 \text{ m/s}^2$

and $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$
 $= 0 + 2(-0.01)(-2)$
 $= 0.04 \text{ m/s}^2$

Finally $F_r = m_B a_r$
 $= 2.5 \text{ kg}(-0.196 \text{ m/s}^2)$

or $F_r = -0.49 \text{ N} \blacktriangleleft$

$F_\theta = m_B a_\theta$
 $= (2.5 \text{ kg})(0.04 \text{ m/s}^2)$

or $F_\theta = 0.1 \text{ N} \blacktriangleleft$

PROBLEM 12.68 (Continued)

(b) At $t = 6$ s:

$$r = 0.025 \text{ m}$$

$$\dot{r} = -0.0025 \text{ m/s} \quad \dot{\theta} = 2 \text{ rad/s}$$

$$\ddot{r} = 5 \times 10^{-4} \text{ m/s}^2 \quad \ddot{\theta} = 0$$

Now $a_r = \ddot{r} - r\dot{\theta}^2 = 5 \times 10^{-4} (0.025^{\text{m}})(2 \text{ rad/s}^2) = -0.0995 \text{ m/s}^2$

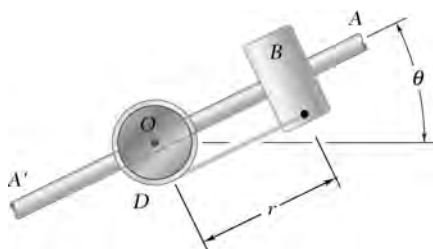
and $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 + 2(-0.0025 \text{ m/s})(2 \text{ rad/s}) = -0.01 \text{ m/s}^2$

Finally $F_r = m_B a_r$
 $= (2.5 \text{ kg})(-0.0995 \text{ m/s}^2)$

or $F_r = -0.249 \text{ N} \blacktriangleleft$

$$F_\theta = m_B a_\theta$$
$$= (2.5 \text{ kg})(-0.01 \text{ m/s}^2)$$

or $F_\theta = -0.0250 \text{ N} \blacktriangleleft$



PROBLEM 12.69

A collar B of mass m slides on the frictionless arm AA' . The arm is attached to drum D and rotates about O in a horizontal plane at the rate $\dot{\theta} = ct$, where c is a constant. As the arm-drum assembly rotates, a mechanism within the drum winds in the cord so that the collar moves toward O with a constant speed k . Knowing that at $t = 0$, $r = r_0$, express as a function of m , c , k , r_0 , and t , (a) the tension T in the cord, (b) the magnitude of the horizontal force Q exerted on B by arm AA' .

SOLUTION

Kinematics

We have

$$\frac{dr}{dt} = \dot{r} = -k$$

At $t = 0$, $r = r_0$:

$$\int_{r_0}^r dr = \int_0^t -k dt$$

or

$$r = r_0 - kt$$

Also,

$$\ddot{r} = 0 \quad \dot{\theta} = ct$$

$$\ddot{\theta} = c$$

Now

$$\begin{aligned} a_r &= \ddot{r} - r\dot{\theta}^2 \\ &= 0 - (r_0 - kt)(ct)^2 \\ &= -c^2(r_0 - kt)t^2 \end{aligned}$$

and

$$\begin{aligned} a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} \\ &= (r_0 - kt)(c) + 2(-k)(ct) \\ &= c(r_0 - 3kt) \end{aligned}$$

Kinetics

$$(a) \quad \nearrow \Sigma F_r = ma_r:$$

$$-T = m[-c^2(r_0 - kt)t^2]$$

or

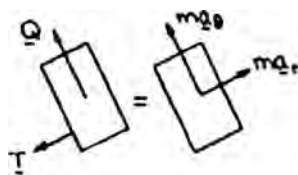
$$T = mc^2(r_0 - kt)t^2 \quad \blacktriangleleft$$

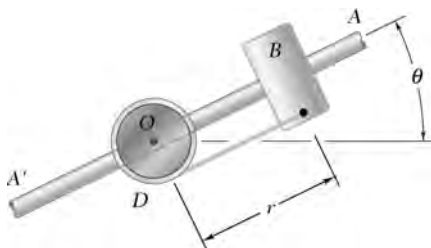
$$(b) \quad \nwarrow \Sigma F_\theta = ma_\theta:$$

$$Q = m[c(r_0 - 3kt)]$$

or

$$Q = mc(r_0 - 3kt) \quad \blacktriangleleft$$





PROBLEM 12.70

The 3-kg collar B slides on the frictionless arm AA' . The arm is attached to drum D and rotates about O in a horizontal plane at the rate $\dot{\theta} = 0.75t$, where $\dot{\theta}$ and t are expressed in rad/s and seconds, respectively. As the arm-drum assembly rotates, a mechanism within the drum releases cord so that the collar moves outward from O with a constant speed of 0.5 m/s. Knowing that at $t = 0$, $r = 0$, determine the time at which the tension in the cord is equal to the magnitude of the horizontal force exerted on B by arm AA' .

SOLUTION

Kinematics

We have

$$\frac{dr}{dt} = \dot{r} = 0.5 \text{ m/s}$$

At $t = 0$, $r = 0$:

$$\int_0^r dr = \int_0^t 0.5 dt$$

or

$$r = (0.5t) \text{ m}$$

Also,

$$\ddot{r} = 0$$

$$\dot{\theta} = (0.75t) \text{ rad/s}$$

$$\ddot{\theta} = 0.75 \text{ rad/s}^2$$

Now

$$a_r = \ddot{r} - r\dot{\theta}^2 = 0 - [(0.5t) \text{ m}][(0.75t) \text{ rad/s}]^2 = -(0.28125t^3) \text{ m/s}^2$$

and

$$\begin{aligned} a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} \\ &= [(0.5t) \text{ m}][0.75 \text{ rad/s}^2] + 2(0.5 \text{ m/s})[(0.75t) \text{ rad/s}] \\ &= (1.125t) \text{ m/s}^2 \end{aligned}$$

Kinetics

$$\nearrow \Sigma F_r = ma_r: -T = (3 \text{ kg})(-0.28125t^3) \text{ m/s}^2$$

or

$$T = (0.84375t^3) \text{ N}$$

$$\nearrow \Sigma F_\theta = m_B a_\theta: Q = (3 \text{ kg})(1.125t) \text{ m/s}^2$$

or

$$Q = (3.375t) \text{ N}$$

Now require that

$$T = Q$$

or

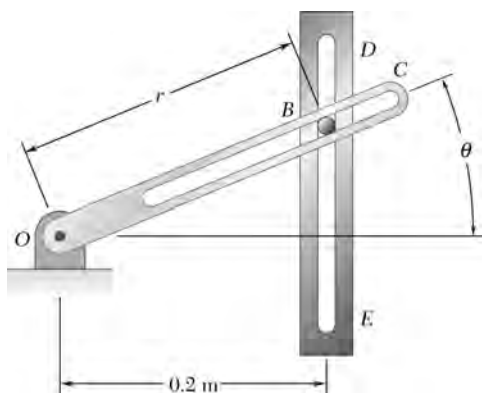
$$(0.84375t^3) \text{ N} = (3.375t) \text{ N}$$

or

$$t^2 = 2.00 \text{ s}$$

or

$$t^2 = 2.00 \text{ s} \quad \blacktriangleleft$$



PROBLEM 12.71

The 100-g pin B slides along the slot in the rotating arm OC and along the slot DE which is cut in a fixed horizontal plate. Neglecting friction and knowing that rod OC rotates at the constant rate $\dot{\theta}_0 = 12 \text{ rad/s}$, determine for any given value of θ (a) the radial and transverse components of the resultant force $\mathbf{F}$ exerted on pin B , (b) the forces $\mathbf{P}$ and $\mathbf{Q}$ exerted on pin B by rod OC and the wall of slot DE , respectively.

SOLUTION

Kinematics

From the drawing of the system, we have

$$r = \frac{0.2}{\cos \theta} \text{ m}$$

Then

$$\dot{r} = \left(0.2 \frac{\sin \theta}{\cos^2 \theta} \dot{\theta} \right) \text{ m/s} \quad \dot{\theta} = 12 \text{ rad/s}$$

and

$$\ddot{\theta} = 0$$

$$\begin{aligned} \ddot{r} &= 0.2 \frac{\cos \theta (\cos^2 \theta) - \sin \theta (-2 \cos \theta \sin \theta)}{\cos^4 \theta} \dot{\theta}^2 \\ &= \left(0.2 \frac{1 + \sin^2 \theta}{\cos^3 \theta} \dot{\theta}^2 \right) \text{ m/s}^2 \end{aligned}$$

Substituting for $\dot{\theta}$

$$\begin{aligned} \dot{r} &= 0.2 \frac{\sin \theta}{\cos^2 \theta} (12) = \left(2.4 \frac{\sin \theta}{\cos^2 \theta} \right) \text{ m/s} \\ \ddot{r} &= 0.2 \frac{1 + \sin^2 \theta}{\cos^3 \theta} (12)^2 = \left(28.8 \frac{1 + \sin^2 \theta}{\cos^3 \theta} \right) \text{ m/s}^2 \end{aligned}$$

Now

$$\begin{aligned} a_r &= \ddot{r} - r \dot{\theta}^2 \\ &= \left(28.8 \frac{1 + \sin^2 \theta}{\cos^3 \theta} \right) - \left(\frac{0.2}{\cos \theta} \right) (12)^2 \\ &= \left(57.6 \frac{\sin^2 \theta}{\cos^3 \theta} \right) \text{ m/s}^2 \end{aligned}$$

PROBLEM 12.71 (Continued)

and

$$\begin{aligned} a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} \\ &= 0 + 2\left(2.4 \frac{\sin \theta}{\cos^2 \theta}\right)(12) \\ &= \left(57.6 \frac{\sin \theta}{\cos^2 \theta}\right) \text{ m/s}^2 \end{aligned}$$

Kinetics

(a) We have

$$F_r = m_B a_r = (0.1 \text{ kg}) \left[\left(57.6 \frac{\sin^2 \theta}{\cos^3 \theta} \right) \text{ m/s}^2 \right]$$

or

$$F_r = (5.76 \text{ N}) \tan^2 \theta \sec \theta \quad \blacktriangleleft$$

and

$$F_\theta = m_B a_\theta = (0.1 \text{ kg}) \left[\left(57.6 \frac{\sin \theta}{\cos^2 \theta} \right) \text{ m/s}^2 \right]$$

or

$$F_\theta = (5.76 \text{ N}) \tan \theta \sec \theta \quad \blacktriangleleft$$

(b) Now

$$+\uparrow \Sigma F_y: F_\theta \cos \theta + F_r \sin \theta = P \cos \theta$$

or

$$P = 5.76 \tan \theta \sec \theta + (5.76 \tan^2 \theta \sec \theta) \tan \theta$$

or

$$P = (5.76 \text{ N}) \tan \theta \sec^2 \theta \quad \swarrow \theta \quad \blacktriangleleft$$

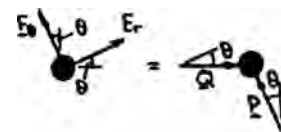
$$+\nearrow \Sigma F_r: F_r = Q \cos \theta$$

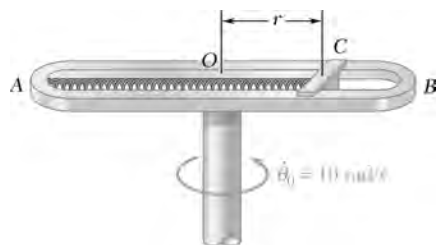
or

$$Q = (5.76 \tan^2 \theta \sec \theta) \frac{1}{\cos \theta}$$

or

$$Q = (5.76 \text{ N}) \tan^2 \theta \sec^2 \theta \rightarrow \quad \blacktriangleleft$$





PROBLEM 12.72*

Slider C has a weight of 250 g and may move in a slot cut in arm AB, which rotates at the constant rate $\dot{\theta}_0 = 10$ rad/s in a horizontal plane. The slider is attached to a spring of constant $k = 50$ N/m, which is unstretched when $r = 0$. Knowing that the slider is released from rest with no radial velocity in the position $r = 500$ mm and neglecting friction, determine for the position $r = 300$ mm (a) the radial and transverse components of the velocity of the slider, (b) the radial and transverse components of its acceleration, (c) the horizontal force exerted on the slider by arm AB.

SOLUTION

Let l_0 be the radial coordinate when the spring is unstretched. Force exerted by the spring.

$$F_r = -k(r - l_0)$$

$$\Sigma F_r = ma_r: -k(r - l_0) = m(\ddot{r} - r\dot{\theta}^2)$$

$$\ddot{r} = \left(\dot{\theta}^2 - \frac{k}{m} \right) r + \frac{kl_0}{m} \quad (1)$$

But

$$\ddot{r} = \frac{d}{dt}(\dot{r}) = \frac{d\dot{r}}{dr} \frac{dr}{dt} = \dot{r} \frac{d\dot{r}}{dr}$$

$$\dot{r} d\dot{r} = \ddot{r} dr = \left[\left(\dot{\theta}^2 - \frac{k}{m} \right) r + \frac{kl_0}{m} \right] dr$$

Integrate using the condition $\dot{r} = \dot{r}_0$ when $r = r_0$.

$$\frac{1}{2} \dot{r}^2 \Big|_{\dot{r}_0}^{\dot{r}} = \left[\frac{1}{2} \left(\dot{\theta}^2 - \frac{k}{m} \right) r^2 + \frac{kl_0}{m} r \right] \Big|_{r_0}^r$$

$$\frac{1}{2} \dot{r}^2 - \frac{1}{2} \dot{r}_0^2 = \frac{1}{2} \left(\dot{\theta}^2 - \frac{k}{m} \right) (r^2 - r_0^2) + \frac{kl_0}{m} (r - r_0)$$

$$\dot{r}^2 = \dot{r}_0^2 + \left(\dot{\theta}^2 - \frac{k}{m} \right) (r^2 - r_0^2) + \frac{2kl_0}{m} (r - r_0)$$

Data:

$$m = 0.25 \text{ kg}$$

$$\dot{\theta} = 10 \text{ rad/s}, \quad k = 50 \text{ N/m}, \quad l_0 = 0$$

$$\dot{r}_0 = (v_r)_0 = 0, \quad r_0 = 500 \text{ mm} = 0.5 \text{ m}, \quad r = 300 \text{ mm} = 0.3 \text{ m}$$

PROBLEM 12.72* (Continued)

- (a) Components of velocity when $r = 0.3$ m

$$\begin{aligned}\dot{r}^2 &= 0 + \left(10^2 - \frac{50}{0.25}\right)(0.3^2 - 0.5^2) + 0 \\ &= 16 \text{ m}^2/\text{s}^2 \\ v_r &= \dot{r} = \pm 4.00 \text{ m/s}\end{aligned}$$

Since r is decreasing, v_r is negative

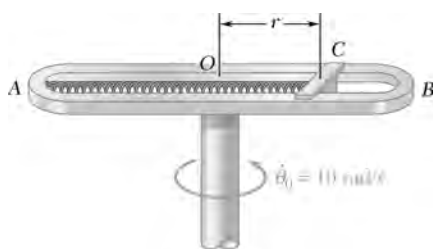
$$\begin{aligned}\dot{r} &= -4 \text{ m/s} & v_r &= -4.00 \text{ m/s} \quad \blacktriangleleft \\ v_\theta &= r\dot{\theta} = (0.3)(10) & v_\theta &= 3.00 \text{ m/s} \quad \blacktriangleleft\end{aligned}$$

- (b) Components of acceleration.

$$\begin{aligned}F_r &= -kr + kl_0 = -(50)(0.3) + 0 = -15 \text{ N} \\ a_r &= \frac{F_r}{m} = -\frac{15}{0.25} & a_r &= -60.0 \text{ m/s}^2 \quad \blacktriangleleft \\ a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 + (2)(-4)(10) & a_\theta &= -80.0 \text{ m/s}^2 \quad \blacktriangleleft\end{aligned}$$

- (c) Transverse component of force.

$$F_\theta = ma_\theta = (0.25)(-80) \quad F_\theta = -20.0 \text{ N} \quad \blacktriangleleft$$



PROBLEM 12.73*

Solve Problem 12.72, assuming that the spring is unstretched when slider C is located 50 mm to the left of the midpoint O of arm AB ($r = 50$ mm).

PROBLEM 12.72 Slider C has a weight of 250 g and may move in a slot cut in arm AB , which rotates at the constant rate $\dot{\theta}_0 = 10$ rad/s in a horizontal plane. The slider is attached to a spring of constant $k = 50$ N/m, which is unstretched when $r = 0$. Knowing that the slider is released with no radial velocity in the position $r = 500$ mm and neglecting friction, determine for the position $r = 300$ mm (a) the radial and transverse components of the velocity of the slider, (b) the radial and transverse components of its acceleration, (c) the horizontal force exerted on the slider by arm AB .

SOLUTION

Let l_0 be the radial coordinate when the spring is unstretched. Force exerted by the spring.

$$F_r = -k(r - l_0)$$

$$\Sigma F_r = ma_r: -k(r - l_0) = m(\ddot{r} - r\dot{\theta}^2)$$

$$\ddot{r} = \left(\dot{\theta}^2 - \frac{k}{m} \right) r + \frac{kl_0}{m} \quad (1)$$

But

$$\ddot{r} = \frac{d}{dt}(\dot{r}) = \frac{d\dot{r}}{dr} \frac{dr}{dt} = \dot{r} \frac{d\dot{r}}{dr}$$

$$\dot{r} d\dot{r} = \ddot{r} dr = \left[\left(\dot{\theta}^2 - \frac{k}{m} \right) r + \frac{kl_0}{m} \right] dr$$

Integrate using the condition $\dot{r} = \dot{r}_0$ when $r = r_0$.

$$\frac{1}{2} \dot{r}^2 \Big|_{\dot{r}_0}^{\dot{r}} = \left[\frac{1}{2} \left(\dot{\theta}^2 - \frac{k}{m} \right) r^2 + \frac{kl_0}{m} r \right] \Big|_{r_0}^r$$

$$\frac{1}{2} \dot{r}^2 - \frac{1}{2} \dot{r}_0^2 = \frac{1}{2} \left(\dot{\theta}^2 - \frac{k}{m} \right) (r^2 - r_0^2) + \frac{kl_0}{m} (r - r_0)$$

$$\dot{r}^2 = \dot{r}_0^2 + \left(\dot{\theta}^2 - \frac{k}{m} \right) (r^2 - r_0^2) + \frac{2kl_0}{m} (r - r_0)$$

PROBLEM 12.73* (Continued)

Data:

$$m = 0.25 \text{ kg}$$

$$\dot{\theta} = 10 \text{ rad/s}, \quad k = 50 \text{ N/m}, \quad l_0 = 0 - 50 \text{ mm} = -50 \text{ mm} = -0.05 \text{ m}$$

$$\dot{r}_0 = (v_r)_0 = 0, \quad r_0 = 0.5 \text{ m}, \quad r = 0.3 \text{ m}$$

(a) Components of velocity when $r = 0.3 \text{ m}$

$$\begin{aligned} \dot{r}^2 &= 0 + \left(10^2 - \frac{50}{0.25} \right) (0.3^2 - 0.5^2) + \frac{(2)(50)(-0.05)}{0.25} (0.3 - 0.5) \\ &= 20 \text{ m}^2/\text{s}^2 \\ v_r = \dot{r} &= \pm 4.47214 \text{ m/s} \end{aligned}$$

Since r is decreasing, v_r is negative.

$$\dot{r} = -4.47214 \text{ m/s}$$

$$v_r = -4.47 \text{ m/s} \quad \blacktriangleleft$$

$$v_\theta = r\dot{\theta} = (0.3)(10)$$

$$v_\theta = 3.00 \text{ m/s} \quad \blacktriangleleft$$

(b) Components of acceleration.

$$F_r = -kr + kl_0 = -(50)(0.3) + (50)(-0.05) = -17.5 \text{ N}$$

$$a_r = \frac{F_r}{m} = -\frac{17.5}{0.25}$$

$$a_r = -70.0 \text{ m/s}^2 \quad \blacktriangleleft$$

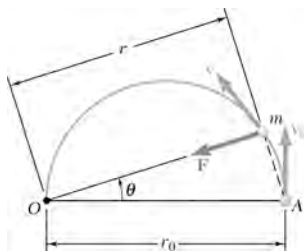
$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 + (2)(-4.47214)(10)$$

$$a_\theta = -89.4 \text{ m/s}^2 \quad \blacktriangleleft$$

(c) Transverse component of force.

$$F_\theta = ma_\theta = (0.25)(-89.4428)$$

$$F_\theta = -22.4 \text{ N} \quad \blacktriangleleft$$



PROBLEM 12.74

A particle of mass m is projected from Point A with an initial velocity $\mathbf{v}_0$ perpendicular to line OA and moves under a central force $\mathbf{F}$ along a semicircular path of diameter OA . Observing that $r = r_0 \cos \theta$ and using Eq. (12.27), show that the speed of the particle is $v = v_0 / \cos^2 \theta$.

SOLUTION

Since the particle moves under a central force, $h = \text{constant}$.

Using Eq. (12.27),

$$h = r^2 \dot{\theta} = h_0 = r_0 v_0$$

or

$$\dot{\theta} = \frac{r_0 v_0}{r^2} = \frac{r_0 v_0}{r_0^2 \cos^2 \theta} = \frac{v_0}{r_0 \cos^2 \theta}$$

Radial component of velocity.

$$v_r = \dot{r} = \frac{d}{dt}(r_0 \cos \theta) = -(r_0 \sin \theta) \dot{\theta}$$

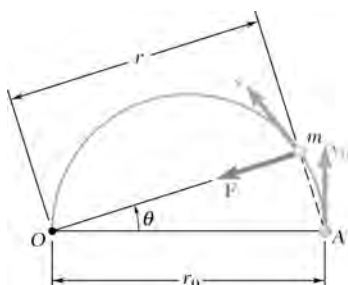
Transverse component of velocity.

$$v_\theta = r \dot{\theta} = (r_0 \cos \theta) \dot{\theta}$$

Speed.

$$v = \sqrt{v_r^2 + v_\theta^2} = r_0 \dot{\theta} = \frac{r_0 v_0}{r_0 \cos^2 \theta}$$

$$v = \frac{v_0}{\cos^2 \theta} \quad \blacktriangleleft$$



PROBLEM 12.75

For the particle of Problem 12.74, determine the tangential component F_t of the central force $\mathbf{F}$ along the tangent to the path of the particle for (a) $\theta = 0$, (b) $\theta = 45^\circ$.

SOLUTION

Since the particle moves under a central force, $h = \text{constant}$

Using Eq. (12.27),

$$h = r^2 \dot{\theta} = h_0 = r_0 v_0$$

$$\dot{\theta} = \frac{r_0 v_0}{r^2} = \frac{r_0 v_0}{r_0^2 \cos^2 \theta} = \frac{v_0}{r_0 \cos^2 \theta}$$

Radial component of velocity.

$$v_r = \dot{r} = \frac{d}{dt}(r_0 \cos \theta) = -(r_0 \sin \theta) \dot{\theta}$$

Transverse component of velocity.

$$v_\theta = r \dot{\theta} = (r_0 \cos \theta) \dot{\theta}$$

Speed.

$$v = \sqrt{v_r^2 + v_\theta^2} = r_0 \dot{\theta} = \frac{r_0 v_0}{r_0 \cos^2 \theta} = \frac{v_0}{\cos^2 \theta}$$

Tangential component of acceleration.

$$a_t = \frac{dv}{dt} = v_0 \frac{(-2)(-\sin \theta) \dot{\theta}}{\cos^3 \theta} = \frac{2v_0 \sin \theta}{\cos^3 \theta} \cdot \frac{v_0}{r_0 \cos^2 \theta}$$

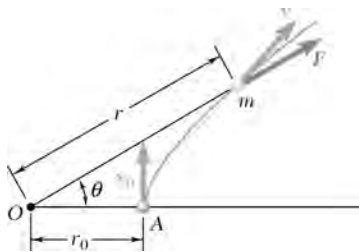
$$= \frac{2v_0^2 \sin \theta}{r_0 \cos^5 \theta}$$

Tangential component of force.

$$F_t = ma_t: \quad F_t = \frac{2mv_0^2 \sin \theta}{r_0 \cos^5 \theta}$$

$$(a) \quad \theta = 0, \quad F_t = 0 \quad F_t = 0 \quad \blacktriangleleft$$

$$(b) \quad \theta = 45^\circ, \quad F_t = \frac{2mv_0^2 \sin 45^\circ}{\cos^5 45^\circ} \quad F_t = \frac{8mv_0^2}{r_0} \quad \blacktriangleleft$$



PROBLEM 12.76

A particle of mass m is projected from point A with an initial velocity $\mathbf{v}_0$ perpendicular to line OA and moves under a central force $\mathbf{F}$ directed away from the center of force O . Knowing that the particle follows a path defined by the equation $r = r_0/\sqrt{\cos 2\theta}$ and using Eq. (12.27), express the radial and transverse components of the velocity $\mathbf{v}$ of the particle as functions of θ .

SOLUTION

Since the particle moves under a central force, $h = \text{constant}$.

Using Eq. (12.27),

$$h = r^2 \dot{\theta} = h_0 = r_0 v_0$$

or

$$\dot{\theta} = \frac{r_0 v_0}{r^2} = \frac{r_0 v_0 \cos 2\theta}{r_0^2} = \frac{v_0}{r_0} \cos 2\theta$$

Radial component of velocity.

$$v_r = \dot{r} = \frac{dr}{d\theta} \dot{\theta} = \frac{d}{d\theta} \left(\frac{r_0}{\sqrt{\cos 2\theta}} \right) \dot{\theta} = r_0 \frac{\sin 2\theta}{(\cos 2\theta)^{3/2}} \dot{\theta}$$

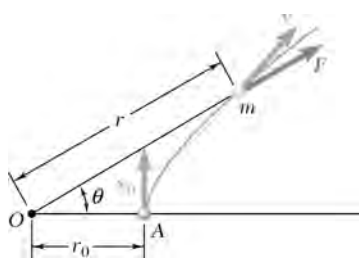
$$= r_0 \frac{\sin 2\theta}{(\cos 2\theta)^{3/2}} \frac{v_0}{r} \cos 2\theta$$

$$v_r = v_0 \frac{\sin 2\theta}{\sqrt{\cos 2\theta}} \quad \blacktriangleleft$$

Transverse component of velocity.

$$v_\theta = \frac{h}{r} = \frac{r_0 v_0}{r_0} \sqrt{\cos 2\theta}$$

$$v_\theta = v_0 \sqrt{\cos 2\theta} \quad \blacktriangleleft$$



PROBLEM 12.77

For the particle of Problem 12.76, show (a) that the velocity of the particle and the central force $\mathbf{F}$ are proportional to the distance r from the particle to the center of force O , (b) that the radius of curvature of the path is proportional to r^3 .

PROBLEM 12.76 A particle of mass m is projected from Point A with an initial velocity $\mathbf{v}_0$ perpendicular to line OA and moves under a central force $\mathbf{F}$ directed away from the center of force O . Knowing that the particle follows a path defined by the equation $r = r_0 / \sqrt{\cos 2\theta}$ and using Eq. (12.27), express the radial and transverse components of the velocity $\mathbf{v}$ of the particle as functions of θ .

SOLUTION

Since the particle moves under a central force, $h = \text{constant}$.

Using Eq. (12.27),

$$h = r^2 \dot{\theta} = h_0 = r_0 v_0 \quad \text{or} \quad \dot{\theta} = \frac{r_0 v_n}{r^2} = \frac{r_0 v_0 \cos 2\theta}{r_0^2} = \frac{v_0}{r_0} \cos 2\theta$$

Differentiating the expression for r with respect to time,

$$\dot{r} = \frac{dr}{d\theta} \dot{\theta} = \frac{d}{d\theta} \left(\frac{r_0}{\sqrt{\cos 2\theta}} \right) \dot{\theta} = r_0 \frac{\sin 2\theta}{(\cos 2\theta)^{3/2}} \dot{\theta} = r_0 \frac{\sin 2\theta}{(\cos 2\theta)^{3/2}} \frac{v_0}{r_0} \cos 2\theta = v_0 \frac{\sin 2\theta}{\sqrt{\cos 2\theta}}$$

Differentiating again,

$$\ddot{r} = \frac{d\dot{r}}{d\theta} \dot{\theta} = \frac{d}{d\theta} \left(v_0 \frac{\sin 2\theta}{\sqrt{\cos 2\theta}} \right) \dot{\theta} = v_0 \frac{2 \cos^2 2\theta + \sin^2 2\theta}{(\cos 2\theta)^{3/2}} \dot{\theta} = \frac{v_0^2}{r_0} \frac{2 \cos^2 2\theta + \sin^2 2\theta}{\sqrt{\cos 2\theta}}$$

$$(a) \quad v_r = \dot{r} = v_0 \frac{\sin 2\theta}{\sqrt{\cos 2\theta}} = \frac{v_0 r}{r_0} \sin 2\theta \quad v_\theta = r \dot{\theta} = \frac{v_0 r}{r_0} \cos 2\theta$$

$$v = \sqrt{(v_r)^2 + (v_\theta)^2} = \frac{v_0 r}{r_0} \sqrt{\sin^2 2\theta + \cos^2 2\theta} \quad v = \frac{v_0 r}{r_0} \quad \blacktriangleleft$$

$$\begin{aligned} a_r = \ddot{r} - r \dot{\theta}^2 &= \frac{v_0^2}{r_0} \frac{2 \cos^2 2\theta + \sin^2 2\theta}{\sqrt{\cos 2\theta}} - \frac{r_0}{\sqrt{\cos 2\theta}} \frac{v_0^2}{r_0^2} \cos^2 2\theta \\ &= \frac{v_0^2}{r_0} \frac{\cos^2 2\theta + \sin^2 2\theta}{\sqrt{\cos 2\theta}} = \frac{v_0}{r_0 \sqrt{\cos 2\theta}} = \frac{v_0^2 r}{r_0^2} \end{aligned}$$

$$F_r = m a_r = \frac{m v_0^2 r}{r_0^2} : \quad F_r = \frac{m v_0^2 r}{r_0^2} \quad \blacktriangleleft$$

PROBLEM 12.77 (Continued)

Since the particle moves under a central force, $a_\theta = 0$.

Magnitude of acceleration.

$$a = \sqrt{a_r^2 + a_\theta^2} = \frac{v_0^2 r}{r_0^2}$$

Tangential component of acceleration.

$$a_t = \frac{dv}{dt} = \frac{d}{dt} \left(\frac{v_0 r}{r_0} \right) = \frac{v_0}{r_0} \dot{r} = \frac{v_0^2 r}{r_0^2} \sin 2\theta$$

Normal component of acceleration.

$$a_n = \sqrt{a^2 - a_t^2} = \frac{v_0^2 r}{r_0^2} \sqrt{1 - \sin^2 2\theta} = \frac{v_0^2 r \cos 2\theta}{r_0^2}$$

But $\cos 2\theta = \left(\frac{r_0}{r} \right)^2$

Hence, $a_n = \frac{v_0^2}{r}$

(b) But $a_n = \frac{v^2}{\rho}$ or $\rho = \frac{v^2}{a_n} = \frac{v_0^2 r^2}{r_0^2} \cdot \frac{r}{v_0^2} \quad \rho = \frac{r^3}{r_0^2} \blacktriangleleft$

PROBLEM 12.78

The radius of the orbit of a moon of a given planet is equal to twice the radius of that planet. Denoting by ρ the mean density of the planet, show that the time required by the moon to complete one full revolution about the planet is $(24 \pi / G \rho)^{1/2}$, where G is the constant of gravitation.

SOLUTION

For gravitational force and a circular orbit,

$$|F_r| = \frac{GMm}{r^2} = \frac{mv^2}{r} \quad \text{or} \quad v = \sqrt{\frac{GM}{r}}$$

Let τ be the periodic time to complete one orbit.

$$v\tau = 2\pi r \quad \text{or} \quad \tau \sqrt{\frac{GM}{r}} = 2\pi r$$

Solving for τ ,

$$\tau = \frac{2\pi r^{3/2}}{\sqrt{GM}}$$

But $M = \frac{4}{3}\pi R^3 \rho$, hence, $\sqrt{GM} = 2\sqrt{\frac{\pi}{3} G \rho} R^{3/2}$

Then
$$\tau = \sqrt{\frac{3\pi}{G\rho}} \left(\frac{r}{R}\right)^{3/2}$$

Using $r = 2R$ as a given leads to

$$\tau = 2^{3/2} \sqrt{\frac{3\pi}{G\rho}} = \sqrt{\frac{24\pi}{G\rho}} \quad \tau = (24\pi/G\rho)^{1/2} \blacktriangleleft$$

PROBLEM 12.79

Show that the radius r of the orbit of a moon of a given planet can be determined from the radius R of the planet, the acceleration of gravity at the surface of the planet, and the time τ required by the moon to complete one full revolution about the planet. Determine the acceleration of gravity at the surface of the planet Jupiter knowing that $R = 71,492$ km and that $\tau = 3.551$ days and $r = 670.9 \times 10^3$ km for its moon Europa.

SOLUTION

We have
$$F = G \frac{Mm}{r^2} \quad [\text{Eq. (12.28)}]$$

and
$$F = F_n = ma_n = m \frac{v^2}{r}$$

Then
$$G \frac{Mm}{r^2} = m \frac{v^2}{r}$$

or
$$v^2 = \frac{GM}{r}$$

Now
$$GM = gR^2 \quad [\text{Eq. (12.30)}]$$

so that
$$v^2 = \frac{gR^2}{r} \quad \text{or} \quad v = R\sqrt{\frac{g}{r}}$$

For one orbit,
$$\tau = \frac{2\pi r}{v} = \frac{2\pi r}{R\sqrt{\frac{g}{r}}}$$

or
$$r = \left(\frac{g\tau^2 R^2}{4\pi^2} \right)^{1/3} \quad \text{Q.E.D.}$$

Solving for g ,
$$g = 4\pi^2 \frac{r^3}{\tau^2 R^2}$$

and noting that $\tau = 3.551$ days $= 306,806$ s, then

$$\begin{aligned} g_{\text{Jupiter}} &= 4\pi^2 \frac{r_{\text{Eur}}^3}{\tau_{\text{Eur}}^2 R_{\text{Jup}}^2} \\ &= 4\pi^2 \frac{(670.9 \times 10^6 \text{ m})^3}{(306,806 \text{ s})^2 (71.492 \times 10^6 \text{ m})^2} \end{aligned}$$

or
$$g_{\text{Jupiter}} = 24.8 \text{ m/s}^2 \quad \blacktriangleleft$$

Note:
$$g_{\text{Jupiter}} \approx 2.53 g_{\text{Earth}}$$



PROBLEM 12.80

Communication satellites are placed in a geosynchronous orbit, i.e., in a circular orbit such that they complete one full revolution about the earth in one sidereal day (23.934 h), and thus appear stationary with respect to the ground. Determine (a) the altitude of these satellites above the surface of the earth, (b) the velocity with which they describe their orbit. Give the answers in SI units.

SOLUTION

For gravitational force and a circular orbit,

$$|F_r| = \frac{GMm}{r^2} = \frac{mv^2}{r} \quad \text{or} \quad v = \sqrt{\frac{GM}{r}}$$

Let τ be the period time to complete one orbit.

$$\text{But} \quad v\tau = 2\pi r \quad \text{or} \quad v^2\tau^2 = \frac{GM\tau^2}{r} = 4\pi^2 r^2$$

$$\text{Then} \quad r^3 = \frac{GM\tau^2}{4\pi^2} \quad \text{or} \quad r = \left(\frac{GM\tau^2}{4\pi^2} \right)^{1/3}$$

$$\text{Data:} \quad \tau = 23.934 \text{ h} = 86.1624 \times 10^3 \text{ s}$$

$$(a) \quad g = 9.81 \text{ m/s}^2, \quad R = 6.37 \times 10^6 \text{ m}$$

$$GM = gR^2 = (9.81)(6.37 \times 10^6)^2 = 398.06 \times 10^{12} \text{ m}^3/\text{s}^2$$

$$r = \left[\frac{(398.06 \times 10^{12})(86.1624 \times 10^3)^2}{4\pi^2} \right]^{1/3} = 42.145 \times 10^6 \text{ m}$$

$$\text{altitude } h = r - R = 35.775 \times 10^6 \text{ m}$$

$$h = 35,800 \text{ km} \quad \blacktriangleleft$$

(b)

$$v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{398.06 \times 10^{12}}{42.145 \times 10^6}} = 3.07 \times 10^3 \text{ m/s}$$

$$v = 3.07 \text{ km/s} \quad \blacktriangleleft$$

PROBLEM 12.81

Determine the mass of the earth, knowing that the mean radius of the moon's orbit about the earth is 384,000 km and that the moon requires 27.32 days to complete one full revolution about the earth.

SOLUTION

We have

$$F = G \frac{Mm}{r^2} \quad [\text{Eq. (12.28)}]$$

and

$$F = F_n = ma_n = m \frac{v^2}{r}$$

Then

$$G \frac{Mm}{r^2} = m \frac{v^2}{r}$$

or

$$M = \frac{r}{G} v^2$$

Now

$$v = \frac{2\pi r}{\tau}$$

so that

$$M = \frac{r}{G} \left(\frac{2\pi r}{\tau} \right)^2 = \frac{1}{G} \left(\frac{2\pi}{\tau} \right)^2 r^3$$

Noting that

$$\tau = 27.32 \text{ days} = 2.3604 \times 10^6 \text{ s}$$

and

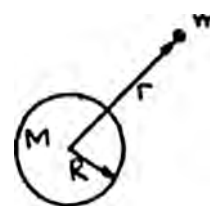
$$r = 384,000 \text{ km} = 3.84 \times 10^8 \text{ m}$$

we have

$$M = \frac{1}{66.73 \times 10^{-12} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2}} \left(\frac{2\pi}{2.3604 \times 10^6 \text{ s}} \right)^2 (3.84 \times 10^8)^3$$

or

$$M = 6.01 \times 10^{24} \text{ kg} \quad \blacktriangleleft$$



PROBLEM 12.82

A spacecraft is placed into a polar orbit about the planet Mars at an altitude of 380 km. Knowing that the mean density of Mars is 3.94 Mg/m^3 and that the radius of Mars is 3397 km, determine (a) the time τ required for the spacecraft to complete one full revolution about Mars, (b) the velocity with which the spacecraft describes its orbit.

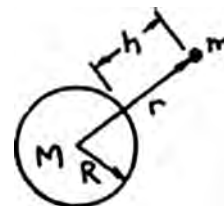
SOLUTION

(a) From the solution to Problem 12.78, we have

$$\tau = \sqrt{\frac{3\pi}{G\rho}} \left(\frac{r}{R} \right)^{3/2}$$

where

$$\begin{aligned} r &= R + h = (3397 + 380) \text{ km} \\ &= 3777 \text{ km} \end{aligned}$$



Then

$$\begin{aligned} \tau &= \left[\frac{3\pi}{(66.73 \times 10^{-12} \text{ m}^3/\text{kg} \cdot \text{s}^2)(3.94 \times 10^3 \text{ kg/m}^3)} \right]^{1/2} \left(\frac{3777 \text{ km}}{3397 \text{ km}} \right)^{3/2} \\ &= 7019.5 \text{ s} \end{aligned}$$

or

$$\tau = 1 \text{ h } 57 \text{ min} \blacktriangleleft$$

(b) We have

$$\begin{aligned} v &= \frac{2\pi r}{\tau} \\ &= \frac{2\pi(3777 \times 10^3 \text{ m})}{7019.5 \text{ s}} \end{aligned}$$

or

$$v = 3380 \text{ m/s} \blacktriangleleft$$

PROBLEM 12.83

A satellite is placed into a circular orbit about the planet Saturn at an altitude of 3400 km. The satellite describes its orbit with a velocity of 87,500 km/h. Knowing that the radius of the orbit about Saturn and the periodic time of Atlas, one of Saturn's moons, are 136.9×10^3 and 0.6017, respectively, determine (a) the radius of Saturn, (b) the mass of Saturn. (The *periodic time* of a satellite is the time it requires to complete one full revolution about the planet.)

SOLUTION

Velocity of Atlas.

$$v_A = \frac{2\pi r_A}{\tau_A}$$

where

$$r_A = 136.9 \times 10^3 \text{ km} = 136.9 \times 10^6 \text{ m}$$

and

$$\tau_A = 0.6017 \text{ days} = 51,987 \text{ s}$$

$$v_A = \frac{(2\pi)(136.9 \times 10^6)}{51,987} = 16.54583 \times 10^3 \text{ m/s}$$

Gravitational force.

$$F = \frac{GMm}{r^2} = \frac{mv^2}{r}$$

from which

$$GM = rv^2 = \text{constant}$$

For the satellite,

$$r_s v_s^2 = r_A v_A^2$$

$$r_s = \frac{r_A v_A^2}{v_s^2}$$

where

$$v_s = 87,500 \text{ km/h} = 24.30556 \times 10^3 \text{ m/s}$$

$$r_s = \frac{(136.9 \times 10^6)(16.54583 \times 10^3)^2}{(24.30556 \times 10^3)^2} = 63.4409 \times 10^6 \text{ m}$$

$$r_s = 63,400 \text{ km}$$

(a) Radius of Saturn.

$$R = r_s - (\text{altitude}) = 63,400 - 3400$$

$$R = 60,000 \text{ km} \quad \blacktriangleleft$$

(b) Mass of Saturn.

$$M = \frac{r_A v_A^2}{G} = \frac{(136.9 \times 10^6)(16.54583 \times 10^3)^2}{66.73 \times 10^{-12}}$$

$$M = 5.62 \times 10^{24} \text{ kg} \quad \blacktriangleleft$$

PROBLEM 12.84

The periodic times (see Problem 12.83) of the planet Uranus' moons Juliet and Titania have been observed to be 0.4931 days and 8.706 days, respectively. Knowing that the radius of Juliet's orbit is 64,360 km, determine (a) the mass of Uranus, (b) the radius of Titania's orbit.

SOLUTION

Velocity of Juliet.

$$v_J = \frac{2\pi r_J}{\tau_J}$$

where

$$r_J = 64,360 \text{ km} = 64.36 \times 10^6 \text{ m}$$

and

$$\tau_J = 0.4931 \text{ days} = 42,604 \text{ s}$$

$$v_J = \frac{(2\pi)(64.36 \times 10^6)}{42,604} = 9.4917 \times 10^3 \text{ m/s}$$

Gravitational force.

$$F = \frac{GMm}{r^2} = \frac{mv^2}{r}$$

from which

$$GM = rv^2 = \text{constant}$$

(a) Mass of Uranus.

$$M = \frac{r_J v_J^2}{G}$$

$$M = \frac{(64.36 \times 10^6)(9.4917 \times 10^3)^2}{66.73 \times 10^{-12}} \\ = 86.893 \times 10^{24} \text{ kg}$$

$$M = 86.9 \times 10^{24} \text{ kg} \blacktriangleleft$$

(b) Radius of Titania's orbit.

$$GM = r_T v_T^2 = \frac{4\pi^2 r_T^3}{\tau_T^2} = \frac{4\pi^2 r_J^3}{\tau_J^2}$$

$$r_T^3 = r_J^3 \left(\frac{\tau_T}{\tau_J} \right)^2 = (64.36 \times 10^6)^3 \left(\frac{8.706}{0.4931} \right)^2 = 81.103 \times 10^{24} \text{ m}^3$$

$$r_T = 436.39 \times 10^6 \text{ m}$$

$$r_T = 436,000 \text{ km} \blacktriangleleft$$

PROBLEM 12.85

A 600-kg spacecraft first is placed into a circular orbit about the earth at an altitude of 4500 km and then is transferred to a circular orbit about the moon. Knowing that the mass of the moon is 0.01230 times the mass of the earth and that the radius of the moon is 1740 km, determine (a) the gravitational force exerted on the spacecraft as it was orbiting the earth, (b) the required radius of the orbit of the spacecraft about the moon if the periodic times (see Problem 12.83) of the two orbits are to be equal, (c) the acceleration of gravity at the surface of the moon.

SOLUTION

First note that

$$R_E = 6370 \text{ km}$$

Then

$$r_E = R_E + h_E = (6370 + 4500) \text{ km} \\ = 10870 \text{ km}$$

(a) We have

$$F = G \frac{M_M}{r^2} \quad [\text{Eq. (12.28)}]$$

and

$$GM = gR^2 \quad [\text{Eq. (12.29)}]$$

Then

$$F = gR^2 \frac{m}{r^2} = mg \left(\frac{R}{r} \right)^2$$

For the earth orbit,

$$F = 600 \times 9.81 \times \left(\frac{6370 \text{ km}}{10870 \text{ km}} \right)^2$$

or

$$F = 2021.343 \text{ N}$$

$$F = 2020 \text{ N} \quad \blacktriangleleft$$

(b) From the solution to Problem 12.81, we have

$$M = \frac{1}{G} \left(\frac{2\pi}{\tau} \right)^2 r^3$$

Then

$$\tau = \frac{2\pi r^{3/2}}{\sqrt{GM}}$$

Now

$$\tau_E = \tau_M \Rightarrow \frac{2\pi r_E^{3/2}}{\sqrt{GM_E}} = \frac{2\pi r_M^{3/2}}{\sqrt{GM_M}} \quad (1)$$

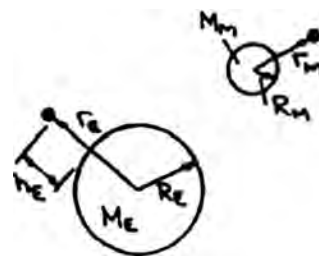
or

$$r_M = \left(\frac{M_M}{M_E} \right)^{1/3} r_E = (0.01230)^{1/3} (10870 \text{ km})$$

or

$$= 2509.176 \text{ km}$$

$$r_M = 2510 \text{ km} \quad \blacktriangleleft$$



PROBLEM 12.85 (Continued)

(c) We have $GM = gR^2$ [Eq.(12.29)]

Substituting into Eq. (1)

$$\frac{2\pi r_E^{3/2}}{R_E \sqrt{g_E}} = \frac{2\pi r_M^{3/2}}{R_M \sqrt{g_M}}$$

or
$$g_M = \left(\frac{R_E}{R_M}\right)^2 \left(\frac{r_M}{r_E}\right)^3 g_E = \left(\frac{R_E}{R_M}\right)^2 \left(\frac{M_M}{M_E}\right) g_E$$

using the results of Part (b). Then

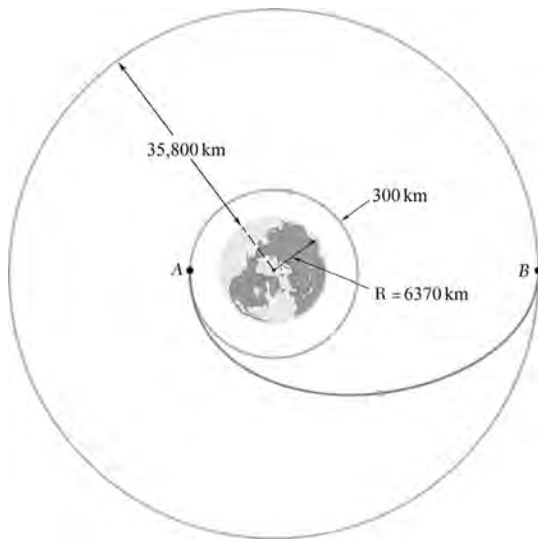
$$g_M = \left(\frac{6370 \text{ km}}{1740 \text{ km}}\right)^2 (0.01230)(9.81 \text{ m/s}^2)$$

or

$$g_{\text{moon}} = 1.62 \text{ m/s}^2 \blacktriangleleft$$

Note:
$$g_{\text{moon}} \approx \frac{1}{6} g_{\text{earth}}$$

PROBLEM 12.86



To place a communications satellite into a geosynchronous orbit (see Problem 12.80) at an altitude of 35800 km above the surface of the earth, the satellite first is released from a space shuttle, which is in a circular orbit at an altitude of 300 km, and then is propelled by an upper-stage booster to its final altitude. As the satellite passes through A, the booster's motor is fired to insert the satellite into an elliptic transfer orbit. The booster is again fired at B to insert the satellite into a geosynchronous orbit. Knowing that the second firing increases the speed of the satellite by 1440 m/s, determine (a) the speed of the satellite as it approaches B on the elliptic transfer orbit, (b) the increase in speed resulting from the first firing at A.

SOLUTION

For earth,

$$R = 6370 \text{ km} = 6.37 \times 10^6 \text{ m}$$

$$GM = gR^2 = (9.81 \text{ m/s}^2)(6.37 \times 10^6)^2 = 3.9806 \times 10^{14} \text{ m}^3/\text{s}^2$$

$$r_A = 6370 + 300 = 6670 \text{ km} = 6.67 \times 10^6 \text{ m}$$

$$r_B = 6370 + 35800 = 42,170 \text{ km} = 42.17 \times 10^6 \text{ m}$$

Speed on circular orbit through A.

$$\begin{aligned} (v_A)_{\text{circ}} &= \sqrt{\frac{GM}{r_A}} \\ &= \sqrt{\frac{3.9806 \times 10^{14}}{6.67 \times 10^6}} \\ &= 7.7252 \times 10^3 \text{ m/s} \end{aligned}$$

Speed on circular orbit through B.

$$\begin{aligned} (v_B)_{\text{circ}} &= \sqrt{\frac{GM}{r_B}} \\ &= \sqrt{\frac{3.9806 \times 10^{14}}{42.17 \times 10^6}} \\ &= 3.07236 \times 10^3 \text{ m/s} \end{aligned}$$

PROBLEM 12.86 (Continued)

- (a) Speed on transfer trajectory at B.

$$\begin{aligned}(v_B)_{tr} &= 3.07236 \times 10^3 - 1440 \\ &= 1632.36 \times \text{m/s} \quad \blacktriangleleft\end{aligned}$$

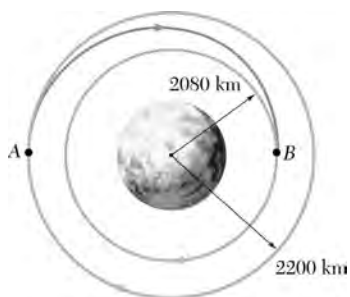
Conservation of angular momentum for transfer trajectory.

$$\begin{aligned}r_A(v_A)_{tr} &= r_B(v_B)_{tr} \\ (v_A)_{tr} &= \frac{r_B(v_B)_{tr}}{r_A} \\ &= \frac{(42.17 \times 10^6)(1632.36)}{(6.67 \times 10^6)} \\ &= 10,320.33 \text{ m/s}\end{aligned}$$

- (b) Change in speed at A.

$$\begin{aligned}\Delta v_A &= (v_A)_{tr} - (v_A)_{\text{circ}} \\ &= 10,320.33 - 7725.2 \\ &= 2595.13 \text{ m/s}\end{aligned}$$

$$\Delta v_A = 2600 \text{ m/s} \quad \blacktriangleleft$$



PROBLEM 12.87

A space vehicle is in a circular orbit of 2200-km radius around the moon. To transfer it to a smaller circular orbit of 2080-km radius, the vehicle is first placed on an elliptic path AB by reducing its speed by 26.3 m/s as it passes through A . Knowing that the mass of the moon is 73.49×10^{21} kg, determine (a) the speed of the vehicle as it approaches B on the elliptic path, (b) the amount by which its speed should be reduced as it approaches B to insert it into the smaller circular orbit.

SOLUTION

For a circular orbit, $\Sigma F_n = ma_n$: $F = m \frac{v^2}{r}$

Eq. (12.28): $F = G \frac{Mm}{r^2}$

Then $G \frac{Mm}{r^2} = m \frac{v^2}{r}$

or $v^2 = \frac{GM}{r}$

Then $(v_A)_{\text{circ}}^2 = \frac{66.73 \times 10^{-12} \text{ m}^3/\text{kg} \cdot \text{s}^2 \times 73.49 \times 10^{21} \text{ kg}}{2200 \times 10^3 \text{ m}}$

or $(v_A)_{\text{circ}} = 1493.0 \text{ m/s}$

and $(v_B)_{\text{circ}}^2 = \frac{66.73 \times 10^{-12} \text{ m}^3/\text{kg} \cdot \text{s}^2 \times 73.49 \times 10^{21} \text{ kg}}{2080 \times 10^3 \text{ m}}$

or $(v_B)_{\text{circ}} = 1535.5 \text{ m/s}$

(a) We have $(v_A)_{\text{TR}} = (v_A)_{\text{circ}} + \Delta v_A$
 $= (1493.0 - 26.3) \text{ m/s}$
 $= 1466.7 \text{ m/s}$

Conservation of angular momentum requires that

$$r_A m (v_A)_{\text{TR}} = r_B m (v_B)_{\text{TR}}$$

or $(v_B)_{\text{TR}} = \frac{2200 \text{ km}}{2080 \text{ km}} \times 1466.7 \text{ m/s}$
 $= 1551.3 \text{ m/s}$

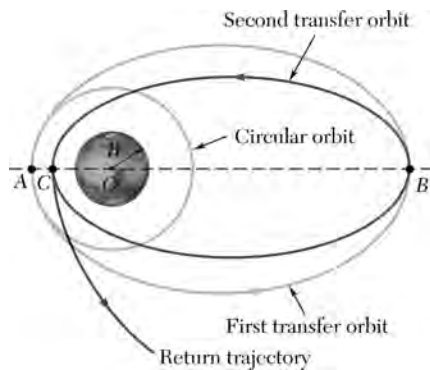
or $(v_B)_{\text{TR}} = 1551 \text{ m/s} \blacktriangleleft$

(b) Now $(v_B)_{\text{circ}} = (v_B)_{\text{TR}} + \Delta v_B$

or $\Delta v_B = (1535.5 - 1551.3) \text{ m/s}$

or $\Delta v_B = -15.8 \text{ m/s} \blacktriangleleft$

PROBLEM 12.88



Plans for an unmanned landing mission on the planet Mars called for the earth-return vehicle to first describe a circular orbit at an altitude $d_A = 2200$ km above the surface of the planet with a velocity of 2771 m/s. As it passed through point A, the vehicle was to be inserted into an elliptic transfer orbit by firing its engine and increasing its speed by $\Delta v_A = 1046$ m/s. As it passed through point B, at an altitude $d_B = 100,000$ km, the vehicle was to be inserted into a second transfer orbit located in a slightly different plane, by changing the direction of its velocity and reducing its speed by $\Delta v_B = -22.0$ m/s. Finally, as the vehicle passed through Point C, at an altitude $d_C = 1000$ km, its speed was to be increased by $\Delta v_C = 660$ m/s to insert it into its return trajectory. Knowing that the radius of the planet Mars is $R = 3400$ km, determine the velocity of the vehicle after completion of the last maneuver.

SOLUTION

$$\tau_A = 3400 + 2200 = 5600 \text{ km} = 5.60 \times 10^6 \text{ m}$$

$$\tau_B = 3400 + 100,000 = 103,400 \text{ km} = 103.4 \times 10^6 \text{ m}$$

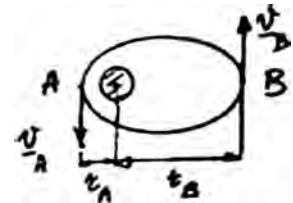
$$\tau_C = 3400 + 1000 = 4400 \text{ km} = 4.40 \times 10^6 \text{ m}$$

First transfer orbit.

$$v_A = 2771 \text{ m/s} + 1046 \text{ m/s} = 3817 \text{ m/s}$$

Conservation of angular momentum:

$$\begin{aligned} \tau_A m v_A &= \tau_B m v_B \\ (5.60 \times 10^6)(3817) &= (103.4 \times 10^6) v_B \\ v_B &= 206.7 \text{ m/s} \end{aligned}$$

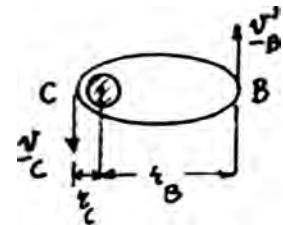


Second transfer orbit.

$$\begin{aligned} v'_B &= v_B + \Delta v_B \\ &= 206.7 - 22.0 = 184.7 \text{ m/s} \end{aligned}$$

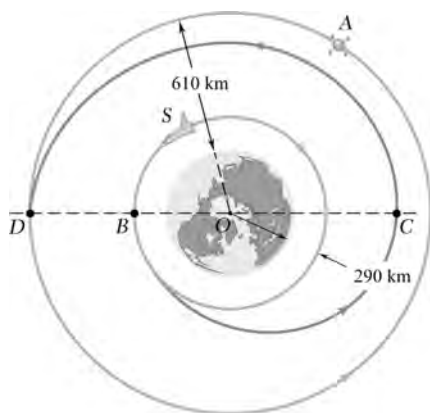
Conservation of angular momentum:

$$\begin{aligned} \tau_B m v'_B &= \tau_C m v_C \\ (103.4 \times 10^6)(184.7) &= (4.40 \times 10^6) v_C \\ v_C &= 4340 \text{ m/s} \end{aligned}$$



After last maneuver.

$$v = v_C + \Delta v_C = 4340 + 660 = 5000 \text{ m/s}$$



PROBLEM 12.89

A space shuttle S and a satellite A are in the circular orbits shown. In order for the shuttle to recover the satellite, the shuttle is first placed in an elliptic path BC by increasing its speed by $\Delta v_B = 84$ m/s as it passes through B . As the shuttle approaches C , its speed is increased by $\Delta v_C = 78$ m/s to insert it into a second elliptic transfer orbit CD . Knowing that the distance from O to C is 6860 km, determine the amount by which the speed of the shuttle should be increased as it approaches D to insert it into the circular orbit of the satellite.

SOLUTION

First note

$$R = 6370 \text{ km} = 6.37 \times 10^6 \text{ m}$$

$$r_A = (6370 + 610) \text{ km} = 6.98 \times 10^6 \text{ km}$$

$$r_B = (6370 + 290) \text{ km} = 6660 \text{ km} = 6.66 \times 10^6 \text{ km}$$

For a circular orbit,

$$\Sigma F_n = ma_n: \quad F = m \frac{v^2}{r}$$

Eq. (12.28):

$$F = G \frac{Mm}{r^2}$$

Then

$$G \frac{Mm}{r^2} = m \frac{v^2}{r}$$

or

$$v^2 = \frac{GM}{r} = \frac{gR^2}{r} \quad \text{using Eq. (12.29).}$$

Then

$$(v_A)_{\text{circ}}^2 = \frac{9.81 \times (6.37 \times 10^6)^2}{6.98 \times 10^6}$$

or

$$(v_A)_{\text{circ}} = 7551.73 \text{ m/s}$$

and

$$(v_B)_{\text{circ}}^2 = \frac{9.81 \times (6.37 \times 10^6)^2}{(6.66 \times 10^6)}$$

or

$$(v_B)_{\text{circ}} = 7731.02 \text{ m/s}$$

We have

$$\begin{aligned} (v_B)_{TR_{BC}} &= (v_B)_{\text{circ}} + \Delta v_B = (7731.02 + 84) \text{ m/s} \\ &= 7815.02 \text{ m/s} \end{aligned}$$

PROBLEM 12.89 (Continued)

Conservation of angular momentum requires that

$$BC: r_B m(v_B)_{TR_{BC}} = r_C m(v_C)_{TR_{BC}} \quad (1)$$

$$CD: r_C m(v_C)_{TR_{CD}} = r_A m(v_D)_{TR_{CD}} \quad (2)$$

$$\begin{aligned} \text{From Eq. (1)} \quad (v_C)_{TR_{BC}} &= \frac{r_B}{r_C} (v_B)_{TR_{BC}} = \frac{6.66 \times 10^6}{6.86 \times 10^6} \times 7815.02 \text{ m/s} \\ &= 7587.18 \text{ m/s} \end{aligned}$$

$$\begin{aligned} \text{Now} \quad (v_C)_{TR_{CD}} &= (v_C)_{TR_{BC}} + \Delta v_C = (7587.18 + 78) \text{ m/s} \\ &= 7665.18 \text{ m/s} \end{aligned}$$

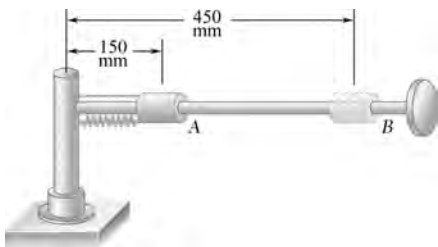
$$\begin{aligned} \text{From Eq. (2)} \quad (v_D)_{TR_{CD}} &= \frac{r_C}{r_A} (v_C)_{TR_{CD}} = \frac{(6.86 \times 10^6)}{(6.98 \times 10^6)} \times 7665.18 \\ &= 7533.40 \text{ m/s} \end{aligned}$$

$$\text{Finally,} \quad (v_A)_{\text{circ}} = (v_D)_{TR_{CD}} + \Delta v_D$$

$$\text{or} \quad \Delta v_D = (7551.73 - 7533.4) \text{ m/s}$$

$$\text{or} \quad \Delta v_D = 18.33 \text{ m/s} \quad \blacktriangleleft$$

PROBLEM 12.90



A 1.5 kg collar can slide on a horizontal rod, which is free to rotate about a vertical shaft. The collar is initially held at A by a cord attached to the shaft. A spring of constant 35 N/m is attached to the collar and to the shaft and is undeformed when the collar is at A. As the rod rotates at the rate $\dot{\theta} = 16 \text{ rad/s}$, the cord is cut and the collar moves out along the rod. Neglecting friction and the mass of the rod, determine (a) the radial and transverse components of the acceleration of the collar at A, (b) the acceleration of the collar relative to the rod at A, (c) the transverse component of the velocity of the collar at B.

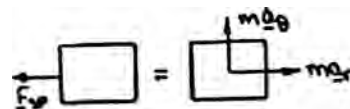
SOLUTION

First note

$$F_{sp} = k(r - r_A)$$

(a) $F_\theta = 0$ and at A,

$$F_r = -F_{sp} = 0$$



$$(a_A)_r = 0 \quad \blacktriangleleft$$

$$(a_A)_\theta = 0 \quad \blacktriangleleft$$

(b) $\sum F_r = ma_r$:

$$-F_{sp} = m(\ddot{r} - r\dot{\theta}^2)$$

Noting that

$$a_{\text{collar/rod}} = \ddot{r}, \text{ we have at A}$$

$$0 = m[a_{\text{collar/rod}} - (0.15 \text{ m})(16 \text{ rad/s})^2]$$

or

$$(a_{\text{collar/rod}})_A = 38.4 \text{ m/s}^2$$

(c) After the cord is cut, the only horizontal force acting on the collar is due to the spring. Thus, angular momentum about the shaft is conserved.

$$r_A m(v_A)_\theta = r_B m(v_B)_\theta \quad \text{where} \quad (v_A)_\theta = r_A \dot{\theta}_0$$

Then

$$(v_B)_\theta = \frac{0.15 \text{ m}}{0.45 \text{ m}} [(0.15 \text{ m})(16 \text{ rad/s})] = 0.8 \text{ m/s}$$

or

$$(v_B)_\theta = 0.800 \text{ m/s} \quad \blacktriangleleft$$

PROBLEM 12.91

For the collar of Problem 12.90, assuming that the rod initially rotates at the rate $\dot{\theta} = 12$ rad/s, determine for position B of the collar (a) the transverse component of the velocity of the collar, (b) the radial and transverse components of its acceleration, (c) the acceleration of the collar relative to the rod.

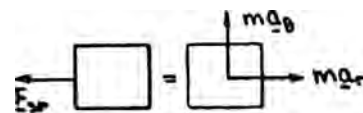
SOLUTION

First we note

$$F_{sp} = k(r - r_A)$$

At B :

$$\begin{aligned}(F_{sp})_B &= 35 \text{ N/m}(0.45 - 0.15) \text{ m} \\ &= 10.5 \text{ N}\end{aligned}$$



- (a) After the cord is cut, the only horizontal force acting on the collar is due to the spring. Thus, angular momentum about the shaft is conserved.

$$r_A m(v_A)_\theta = r_B m(v_B)_\theta$$

where

$$(v_A)_\theta = r_A \dot{\theta}_0$$

Then

$$(v_B)_\theta = \frac{0.15 \text{ m}}{0.45 \text{ m}} [(0.15 \text{ m})(12 \text{ rad/s})] = 0.6 \text{ m/s}$$

or

$$(v_B)_\theta = 0.600 \text{ m/s} \quad \blacktriangleleft$$

- (b)

$$(F_B)_\theta = 0 \Rightarrow$$

$$(a_B)_\theta = 0 \quad \blacktriangleleft$$

Now

$$\sum F_r = m a_r: -(F_{sp})_B = m(a_B)_r$$

or

$$\begin{aligned}(a_B)_r &= -\frac{10.5 \text{ N}}{1.5 \text{ kg}} \\ &= -7 \text{ m/s}^2\end{aligned}$$

or

$$(a_B)_r = -7.00 \text{ m/s}^2 \quad \blacktriangleleft$$

- (c) We have

$$a_r = \ddot{r} - r\dot{\theta}^2$$

Now

$$a_{\text{collar/rod}} = \ddot{r} \quad \text{and} \quad \dot{\theta}_B = \frac{(v_B)_\theta}{r_B}$$

Then at B :

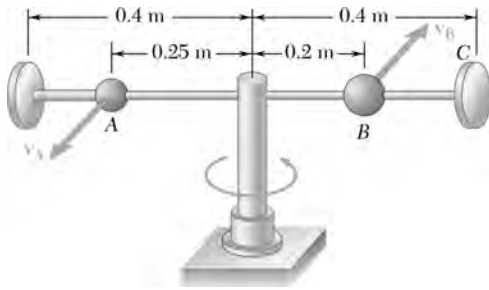
$$(a_{\text{collar/rod}})_B = -7 \text{ m/s}^2 + (0.45 \text{ m}) \left(\frac{0.6}{0.45} \right)^2$$

$$= -6.2 \text{ m/s}^2$$

or

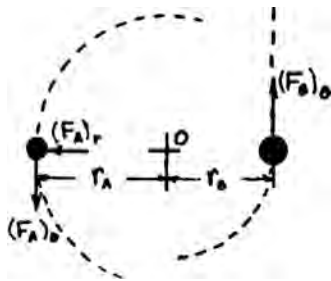
$$(a_{\text{collar/rod}})_\theta = -6.20 \text{ m/s}^2 \quad \blacktriangleleft$$

PROBLEM 12.92



A 200-g ball A and a 400-g ball B are mounted on a horizontal rod which rotates freely about a vertical shaft. The balls are held in the positions shown by pins. The pin holding B is suddenly removed and the ball moves to position C as the rod rotates. Neglecting friction and the mass of the rod and knowing that the initial speed of A is $v_A = 2.5$ m/s, determine (a) the radial and transverse components of the acceleration of ball B immediately after the pin is removed, (b) the acceleration of ball B relative to the rod at that instant, (c) the speed of ball A after ball B has reached the stop at C .

SOLUTION



Let r and θ be polar coordinates with the origin lying at the shaft.

Constraint of rod: $\theta_B = \theta_A + \pi$ radians; $\dot{\theta}_B = \dot{\theta}_A = \dot{\theta}$; $\ddot{\theta}_B = \ddot{\theta}_A = \ddot{\theta}$.

(a) Components of acceleration.

Sketch the free body diagrams of the balls showing the radial and transverse components of the forces acting on them. Owing to frictionless sliding of B along the rod, $(F_B)_r = 0$.

Radial component of acceleration of B .

$$F_r = m_B(a_B)_r : \quad (a_B)_r = 0 \quad \blacktriangleleft$$

Transverse components of acceleration.

$$\begin{aligned} (a_A)_\theta &= r_A \ddot{\theta} + 2\dot{r}_A \dot{\theta} = r_A \ddot{\theta} \\ (a_B)_\theta &= r_B \ddot{\theta} + 2\dot{r}_B \dot{\theta} \end{aligned} \quad (1)$$

Since the rod is massless, it must be in equilibrium. Draw its free-body diagram, applying Newton's Third Law.

$$\sum M_O = 0: r_A(F_A)_\theta + r_B(F_B)_\theta = r_A m_A (a_A)_\theta + r_B m_B (a_B)_\theta = 0$$

$$r_A m_A r_A \ddot{\theta} + r_B m_B (r_B \ddot{\theta} + 2\dot{r}_B \dot{\theta}) = 0$$

$$\ddot{\theta} = \frac{-2r_B \dot{r}_B \dot{\theta}}{m_A r_A^2 + m_B r_B^2}$$

$$\text{At } t = 0, \quad \dot{r}_B = 0 \quad \text{so that} \quad \ddot{\theta} = 0.$$

$$\text{From Eq. (1),} \quad (a_B)_\theta = 0 \quad \blacktriangleleft$$

PROBLEM 12.92 (Continued)

(b) Acceleration of B relative to the rod.

$$\text{At } t = 0, \quad (v_A)_\theta = 2.5 \text{ m/s}, \quad \dot{\theta} = \frac{(v_A)_\theta}{r_A} = \frac{2.5}{0.25} = 10 \text{ rad/s}$$

$$\ddot{r}_B - r_B \dot{\theta}^2 = (a_B)_r = 0$$

$$\ddot{r}_B = r_B \dot{\theta}^2 = (0.2)(10)^2 = 20 \text{ m/s}^2$$

$$\ddot{r}_B = 20.0 \text{ m/s}^2 \quad \blacktriangleleft$$

(c) Speed of A.

Substituting $\frac{d}{dt}(mr^2\dot{\theta})$ for rF_θ in each term of the moment equation gives

$$\frac{d}{dt}(m_A r_A^2 \dot{\theta}) + \frac{d}{dt}(m_B r_B^2 \dot{\theta}) = 0$$

Integrating with respect to time,

$$m_A r_A^2 \dot{\theta} + m_B r_B^2 \dot{\theta} = (m_A r_A^2 \dot{\theta})_0 + (m_B r_B^2 \dot{\theta})_0$$

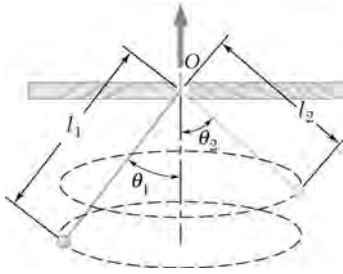
Applying to the final state with ball B moved to the stop at C,

$$(m_A r_A^2 + m_B r_C^2) \dot{\theta}_f = [m_A r_A^2 + m_B (r_B)_0^2] \dot{\theta}_0$$

$$\begin{aligned} \dot{\theta}_f &= \frac{m_A r_A^2 + m_B r_B^2}{m_A r_A^2 + m_B r_C^2} \\ &= \frac{(0.2)(0.25)^2 + (0.4)(0.2)^2}{(0.2)(0.25)^2 + (0.4)(0.4)^2} (10) \\ &= 3.7255 \text{ rad/s} \end{aligned}$$

$$(v_A)_f = r_A \dot{\theta}_f = (0.25)(3.7255) = 0.93137 \text{ m/s}$$

$$(v_A)_f = 0.931 \text{ m/s} \quad \blacktriangleleft$$



PROBLEM 12.93

A small ball swings in a horizontal circle at the end of a cord of length l_1 , which forms an angle θ_1 with the vertical. The cord is then slowly drawn through the support at O until the length of the free end is l_2 . (a) Derive a relation among l_1 , l_2 , θ_1 , and θ_2 . (b) If the ball is set in motion so that initially $l_1 = 0.8$ m and $\theta_1 = 35^\circ$, determine the angle θ_2 when $l_2 = 0.6$ m.

SOLUTION

(a) For State 1 or 2, neglecting the vertical component of acceleration,

$$+\uparrow \Sigma F_y = 0: T \cos \theta - W = 0$$

$$T = W \cos \theta$$

$$+\rightarrow \Sigma F_x = ma_n: T \sin \theta = W \sin \theta \cos \theta = \frac{mv^2}{\rho}$$

But $\rho = \ell \sin \theta$ so that

$$v^2 = \frac{\rho W}{m} \sin^2 \theta \cos \theta = \ell g \sin \theta \tan \theta$$

$$v_1 = \sqrt{\ell_1 g \sin \theta_1 \tan \theta_1}$$

$$v_2 = \sqrt{\ell_2 g \sin \theta_2 \tan \theta_2}$$

and

$$\Sigma M_y = 0: H_y = \text{constant}$$

$$r_1 m v_1 = r_2 m v_2 \quad \text{or} \quad v_1 \ell_1 \sin \theta_1 = v_2 \ell_2 \sin \theta_2$$

$$\ell_1^{3/2} g \sin \theta_1 \sqrt{\sin \theta_1 \tan \theta_1} = \ell_2^{3/2} \sin \theta_2 \sqrt{\sin \theta_2 \tan \theta_2}$$

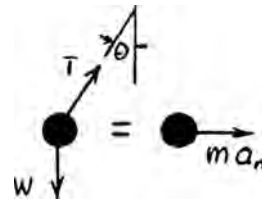
$$\ell_1^3 \sin^3 \theta_1 \tan \theta_1 = \ell_2^3 \sin^3 \theta_2 \tan \theta_2 \quad \blacktriangleleft$$

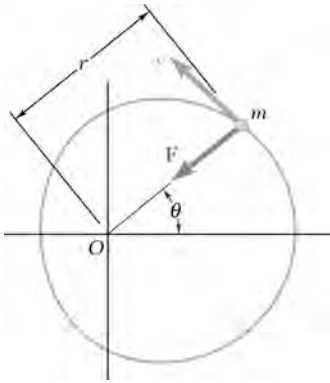
(b) With $\theta_1 = 35^\circ$, $\ell_1 = 0.8$ m, and $\ell_2 = 0.6$ m

$$(0.8)^3 \sin^3 35^\circ \tan 35^\circ = (0.6)^3 \sin^3 \theta_2 \tan \theta_2$$

$$\sin^3 \theta_2 \tan \theta_2 - 0.31320 = 0$$

$$\theta_2 = 43.6^\circ \quad \blacktriangleleft$$





PROBLEM 12.94

A particle of mass m describes the cardioid $r = r_0(1 + \cos \theta)/2$ under a central force $\mathbf{F}$ directed toward the center of force O . Using Eq. (12.37), show that $\mathbf{F}$ is inversely proportional to the fourth power of the distance r from the particle to O .

SOLUTION

We have

$$\frac{d^2u}{d\theta^2} + u = \frac{F}{mh^2u^2} \quad \text{Eq. (12.37)}$$

where

$$u = \frac{1}{r} \quad \text{and} \quad mh^2 = \text{constant}$$

$$F \propto u^2 \left(\frac{d^2u}{d\theta^2} + u \right)$$

Now

$$u = \frac{1}{r} = \frac{2}{r_0} \frac{1}{1 + \cos \theta}$$

Then

$$\frac{du}{d\theta} = \frac{d}{d\theta} \left(\frac{2}{r_0} \frac{1}{1 + \cos \theta} \right) = \frac{2}{r_0} \frac{\sin \theta}{(1 + \cos \theta)^2}$$

and

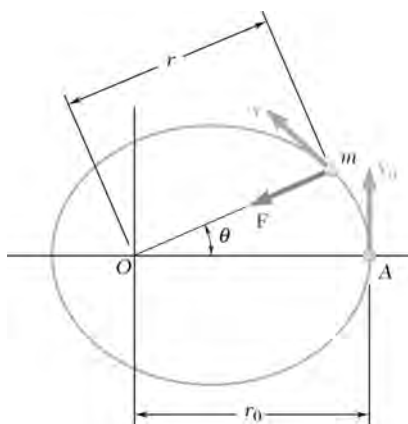
$$\begin{aligned} \frac{d^2u}{d\theta^2} &= \frac{2}{r_0} \frac{\cos \theta (1 + \cos \theta)^2 - \sin \theta [2(1 + \cos \theta)](-\sin \theta)}{(1 + \cos \theta)^4} \\ &= \frac{2}{r_0} \frac{1 + \cos \theta + \sin^2 \theta}{(1 + \cos \theta)^3} = \frac{2}{r_0} \left[\frac{1}{(1 + \cos \theta)^2} + \frac{1 - \cos^2 \theta}{(1 + \cos \theta)^3} \right] \\ &= \frac{2}{r_0} \frac{2 - \cos \theta}{(1 + \cos \theta)^2} = \frac{2}{r_0} \left(\frac{r_0}{2r} \right)^2 \left[2 - \left(\frac{2r}{r_0} - 1 \right) \right] \\ &= \frac{r_0}{2r^2} \left(3 - \frac{2r}{r_0} \right) \end{aligned}$$

Then

$$F \propto \left(\frac{1}{r} \right)^2 \left[\frac{r_0}{2r^2} \left(3 - \frac{2r}{r_0} \right) + \frac{1}{r} \right] = \frac{3}{2} \frac{r_0}{r^4}$$

$$F \propto \frac{1}{r^4} \quad \text{Q.E.D.} \quad \blacktriangleleft$$

Note: $F > 0$ implies that $\mathbf{F}$ is attractive.



PROBLEM 12.95

A particle of mass m is projected from Point A with an initial velocity $\mathbf{v}_0$ perpendicular to OA and moves under a central force $\mathbf{F}$ along an elliptic path defined by the equation $r = r_0 / (2 - \cos \theta)$. Using Eq. (12.37), show that $\mathbf{F}$ is inversely proportional to the square of the distance r from the particle to the center of force O .

SOLUTION

$$u = \frac{1}{r} = \frac{2 - \cos \theta}{r_0}, \quad \frac{du}{d\theta} = \frac{\sin \theta}{r_0}, \quad \frac{d^2u}{d\theta^2} = \frac{\cos \theta}{r_0}$$

$$\frac{d^2u}{d\theta^2} + u = \frac{2}{r_0} = \frac{F}{mh^2u^2} \quad \text{by Eq. (12.37).}$$

Solving for F ,

$$F = \frac{2mh^2u^2}{r_0} = \frac{2mh^2}{r_0r^2}$$

Since m , h , and r_0 are constants, F is proportional to $\frac{1}{r^2}$, or inversely proportional to r^2 .

PROBLEM 12.96

A particle of mass m describes the path defined by the equation $r = r_0 \sin \theta$ under a central force $\mathbf{F}$ directed toward the center of force O . Using Eq. (12.37), show that $\mathbf{F}$ is inversely proportional to the fifth power of the distance r from the particle to O .

SOLUTION

We have

$$\frac{d^2u}{d\theta^2} + u = \frac{F}{mh^2u^2} \quad \text{Eq. (12.37)}$$

where

$$u = \frac{1}{r} \quad \text{and} \quad mh^2 = \text{constant}$$

$$F \propto u^2 \left(\frac{d^2u}{d\theta^2} + u \right)$$

Now

$$u = \frac{1}{r} = \frac{1}{r_0 \sin \theta}$$

Then

$$\frac{du}{d\theta} = \frac{1}{d\theta} \left(\frac{1}{r_0 \sin \theta} \right) = -\frac{1}{r_0} \frac{\cos \theta}{\sin^2 \theta}$$

and

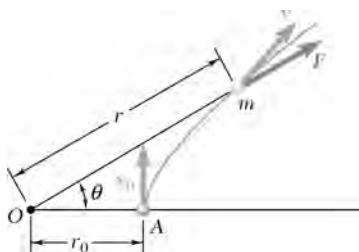
$$\begin{aligned} \frac{d^2u}{d\theta^2} &= -\frac{1}{r_0} \left[\frac{-\sin \theta (\sin^2 \theta) - \cos \theta (2 \sin \theta \cos \theta)}{\sin^4 \theta} \right] \\ &= \frac{1}{r_0} \frac{1 + \cos^2 \theta}{\sin^3 \theta} \end{aligned}$$

Then

$$\begin{aligned} F &\propto \left(\frac{1}{r} \right)^2 \left(\frac{1}{r_0} \frac{1 + \cos^2 \theta}{\sin^3 \theta} + \frac{1}{r_0 \sin \theta} \right) \\ &= \frac{1}{r_0} \frac{1}{r^2} \left(\frac{1 + \cos^2 \theta}{\sin^3 \theta} + \frac{\sin^2 \theta}{\sin^3 \theta} \right) \\ &= \frac{2}{r_0} \frac{1}{r^2} \frac{1}{\sin^3 \theta} \quad \sin^3 \theta = \left(\frac{r}{r_0} \right)^3 \\ &= \frac{2r_0^2}{r^5} \end{aligned}$$

$$F \propto \frac{1}{r^5} \quad \text{Q.E.D.} \quad \blacktriangleleft$$

Note: $F > 0$ implies that $\mathbf{F}$ is attractive.



PROBLEM 12.97

For the particle of Problem 12.76, and using Eq. (12.37), show that the central force $\mathbf{F}$ is proportional to the distance r from the particle to the center of force O .

PROBLEM 12.76 A particle of mass m is projected from Point A with an initial velocity $\mathbf{v}_0$ perpendicular to line OA and moves under a central force $\mathbf{F}$ directed away from the center of force O . Knowing that the particle follows a path defined by the equation $r = r_0/\sqrt{\cos 2\theta}$ and using Eq. (12.27), express the radial and transverse components of the velocity $\mathbf{v}$ of the particle as functions of θ .

SOLUTION

We have

$$\frac{d^2u}{d\theta^2} + u = \frac{F}{mh^2u^2} \quad \text{Eq. (12.37)}$$

where

$$u = \frac{1}{r} \quad \text{and} \quad mh^2 = \text{constant}$$

$$F \propto u^2 \left(\frac{d^2u}{d\theta^2} + u \right)$$

Now

$$u = \frac{1}{r} = \frac{1}{r_0} \sqrt{\cos^2 \theta}$$

Then

$$\frac{du}{d\theta} = \frac{d}{d\theta} \left(\frac{1}{r_0} \sqrt{\cos 2\theta} \right) = -\frac{1}{r_0} \frac{\sin 2\theta}{\sqrt{\cos 2\theta}}$$

and

$$\begin{aligned} \frac{d^2u}{d\theta^2} &= -\frac{1}{r_0} \frac{2 \cos 2\theta \sqrt{\cos 2\theta} - \sin 2\theta (-\sin 2\theta / \sqrt{\cos 2\theta})}{\cos 2\theta} \\ &= -\frac{1}{r_0} \frac{1 + \cos^2 2\theta}{(\cos 2\theta)^{3/2}} = -\frac{1}{r_0} \left(\frac{r}{r_0} \right)^3 \left[1 + \left(\frac{r_0}{r} \right)^4 \right] \\ &= -\frac{r^3}{r_0^4} \left[1 + \left(\frac{r_0}{r} \right)^4 \right] \end{aligned}$$

Then

$$F \propto \left(\frac{1}{r} \right)^2 \left\{ -\frac{r^3}{r_0^4} \left[1 + \left(\frac{r_0}{r} \right)^4 \right] + \frac{1}{r} \right\} = -\frac{r}{r_0^4}$$

$F \propto r$ Q.E.D. ◀

Note: $F < 0$ implies that $\mathbf{F}$ is repulsive.

PROBLEM 12.98

It was observed that during the Galileo spacecraft's first flyby of the earth, its minimum altitude was 960 km above the surface of the earth. Assuming that the trajectory of the spacecraft was parabolic, determine the maximum velocity of Galileo during its first flyby of the earth.

SOLUTION

First we note

$$R = 6.37 \times 10^6 \text{ m}$$

so that

$$\begin{aligned} r_0 &= (6.37 \times 10^6 + 960 \times 10^3) \text{ m} \\ &= 7.33 \times 10^6 \text{ m} \end{aligned}$$

Now

$$v_{\max} = v_0$$

and from page 709 of the text

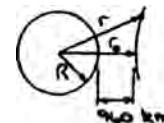
$$v_0 = \sqrt{\frac{2GM}{r_0}} = \sqrt{\frac{2gR^2}{r_0}} \quad \text{using Eq. (12.30).}$$

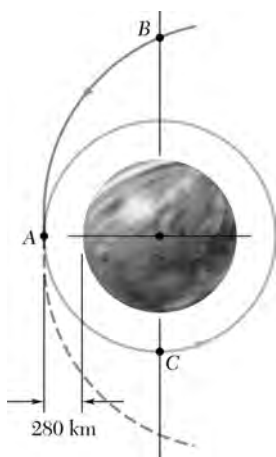
Then

$$\begin{aligned} v_{\max} &= \left[\frac{2 \times 9.81 \text{ m/s}^2 \times (6.37 \times 10^6 \text{ m})^2}{7.33 \times 10^6 \text{ m}} \right]^{1/2} \\ &= 10,421.7 \text{ m/s} \end{aligned}$$

or

$$v_{\max} = 10.42 \text{ km/s} \quad \blacktriangleleft$$





PROBLEM 12.99

As a space probe approaching the planet Venus on a parabolic trajectory reaches Point A closest to the planet, its velocity is decreased to insert it into a circular orbit. Knowing that the mass and the radius of Venus are 4.87×10^{24} kg and 6052 km, respectively, determine (a) the velocity of the probe as it approaches A, (b) the decrease in velocity required to insert it into the circular orbit.

SOLUTION

First note

$$r_A = (6052 + 280) \text{ km} = 6332 \text{ km}$$

- (a) From page 709 of the text, the velocity at the point of closest approach on a parabolic trajectory is given by

$$v_0 = \sqrt{\frac{2GM}{r_0}}$$

Thus,

$$(v_A)_{\text{par}} = \left[\frac{2 \times 66.73 \times 10^{-12} \text{ m}^3/\text{kg} \cdot \text{s}^2 \times 4.87 \times 10^{24} \text{ kg}}{6332 \times 10^3 \text{ m}} \right]^{1/2}$$

$$= 10,131.4 \text{ m/s}$$

or

$$(v_A)_{\text{par}} = 10.13 \text{ km/s} \quad \blacktriangleleft$$

- (b) We have

$$(v_A)_{\text{circ}} = (v_A)_{\text{par}} + \Delta v_A$$

Now

$$(v_A)_{\text{circ}} = \sqrt{\frac{GM}{r_0}} \quad \text{Eq. (12.44)}$$

$$= \frac{1}{\sqrt{2}} (v_A)_{\text{par}}$$

Then

$$\Delta v_A = \frac{1}{\sqrt{2}} (v_A)_{\text{par}} - (v_A)_{\text{par}}$$

$$= \left(\frac{1}{\sqrt{2}} - 1 \right) (10.1314 \text{ km/s})$$

$$= -2.97 \text{ km/s}$$

$$|\Delta v_A| = 2.97 \text{ km/s} \quad \blacktriangleleft$$

PROBLEM 12.100

It was observed that during its second flyby of the earth, the Galileo spacecraft had a velocity of $13.85 \times 10^3 \text{ m/s}$ as it reached its minimum altitude of 300 km above the surface of the earth. Determine the eccentricity of the trajectory of the spacecraft during this portion of its flight.

SOLUTION

First we note

$$R = 6370 \text{ km} = 6.37 \times 10^6 \text{ m}$$

and

$$\begin{aligned} r_0 &= (6370 + 300) \text{ km} = 6670 \text{ km} \\ &= 6.67 \times 10^6 \text{ km} \end{aligned}$$

We have

$$\frac{1}{r} = \frac{GM}{h^2} (1 + \varepsilon \cos \theta) \quad \text{Eq. (12.39)}$$

At Point O,

$$r = r_0, \quad \theta = 0, \quad h = h_0 = r_0 v_0$$

Also,

$$GM = gR^2 \quad \text{Eq. (12.30)}$$

Then

$$\frac{1}{r_0} = \frac{gR^2}{(r_0 v_0)^2} (1 + \varepsilon)$$

or

$$\begin{aligned} \varepsilon &= \frac{r_0 v_0^2}{gR^2} - 1 \\ &= \frac{(6.67 \times 10^6)(13.85 \times 10^3 \text{ m/s})^2}{(9.81 \text{ m/s}^2)(6.37 \times 10^6)^2} - 1 \end{aligned}$$

or

$$\varepsilon = 2.21 \quad \blacktriangleleft$$



PROBLEM 12.101

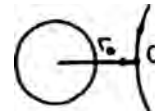
It was observed that as the Galileo spacecraft reached the point on its trajectory closest to Io, a moon of the planet Jupiter, it was at a distance of 2800 km from the center of Io and had a velocity of 15×10^3 m/s. Knowing that the mass of Io is 0.01496 times the mass of the earth, determine the eccentricity of the trajectory of the spacecraft as it approached Io.

SOLUTION

First note

$$r_0 = 2800 \text{ km} = 2.8 \times 10^6 \text{ m}$$

$$R_{\text{earth}} = 6370 \text{ km} = 6.37 \times 10^6 \text{ m}$$



We have

$$\frac{1}{r} = \frac{GM}{h^2} (1 + \varepsilon \cos \theta) \quad \text{Eq. (12.39)}$$

At Point O,

$$r = r_0, \quad \theta = 0, \quad h = h_0 = r_0 v_0$$

Also,

$$GM_{\text{Io}} = G(0.01496 M_{\text{earth}})$$

$$= 0.01496 g R_{\text{earth}}^2 \quad \text{using Eq. (12.30).}$$

Then

$$\frac{1}{r_0} = \frac{0.01496 g R_{\text{earth}}^2}{(r_0 v_0)^2} (1 + \varepsilon)$$

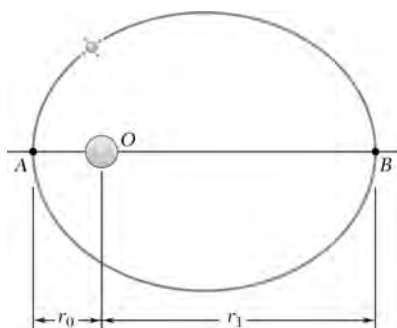
or

$$\varepsilon = \frac{r_0 v_0^2}{0.01496 g R_{\text{earth}}^2} - 1$$

$$= \frac{(2.8 \times 10^6 \text{ m})(15 \times 10^3 \text{ m/s})^2}{0.01496(9.81 \text{ m/s}^2)(6.37 \times 10^6 \text{ m})^2} - 1$$

or

$$\varepsilon = 104.8 \quad \blacktriangleleft$$



PROBLEM 12.102

A satellite describes an elliptic orbit about a planet of mass M . Denoting by r_0 and r_1 , respectively, the minimum and maximum values of the distance r from the satellite to the center of the planet, derive the relation

$$\frac{1}{r_0} + \frac{1}{r_1} = \frac{2GM}{h^2}$$

where h is the angular momentum per unit mass of the satellite.

SOLUTION

Using Eq. (12.39),

$$\frac{1}{r_A} = \frac{GM}{h^2} + C \cos \theta_A$$

and

$$\frac{1}{r_B} = \frac{GM}{h^2} + C \cos \theta_B.$$

But

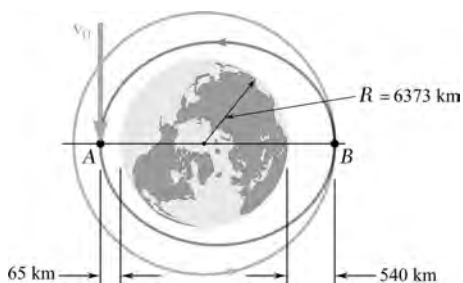
$$\theta_B = \theta_A + 180^\circ,$$

so that

$$\cos \theta_A = -\cos \theta_B.$$

Adding,

$$\frac{1}{r_A} + \frac{1}{r_B} = \frac{1}{r_0} + \frac{1}{r_1} = \frac{2GM}{h^2}$$



PROBLEM 12.103

At main engine cutoff of its thirteenth flight, the space shuttle Discovery was in an elliptic orbit of minimum altitude 65 km and maximum altitude 540 km above the surface of the earth. Knowing that at Point A the shuttle had a velocity $\mathbf{v}_0$ parallel to the surface of the earth and that the shuttle was transferred to a circular orbit as it passed through Point B, determine (a) the speed v_0 of the shuttle at A, (b) the increase in speed required at B to insert the shuttle into the circular orbit.

SOLUTION

For earth, $R = 6373 \text{ km} = 6.373 \times 10^6 \text{ m}$

$$GM = gR^2 = (9.81 \text{ m/s}^2)(6.373 \times 10^6)^2 = 3.9843 \times 10^{14} \text{ m}^3/\text{s}^2$$

$$r_A = 6373 + 65 = 6438 \text{ km} = 6.438 \times 10^6 \text{ m}$$

$$r_B = 6373 + 540 = 6913 \text{ km} = 6.913 \times 10^6 \text{ m}$$

Elliptic trajectory.

Using Eq. (12.39),

$$\frac{1}{r_A} = \frac{GM}{h^2} + C \cos \theta_A \quad \text{and} \quad \frac{1}{r_B} = \frac{GM}{h^2} + C \cos \theta_B.$$

But

$$\theta_B = \theta_A + 180^\circ, \quad \text{so that} \quad \cos \theta_A = -\cos \theta_B$$

Adding,

$$\begin{aligned} \frac{1}{r_A} + \frac{1}{r_B} &= \frac{r_A + r_B}{r_A r_B} = \frac{2GM}{h^2} \\ h &= \sqrt{\frac{2GM r_A r_B}{r_A + r_B}} \\ &= \sqrt{\frac{(2)(3.9843 \times 10^{14})(6.438 \times 10^6)(6.913 \times 10^6)}{(6.438 + 6.913) \times 10^6}} \\ &= 5.15398 \times 10^{10} \text{ m}^2/\text{s} \end{aligned}$$

(a) Speed v_0 at A.

$$v_0 = v_A = \frac{h}{r_A} = \frac{5.15398 \times 10^{10}}{6.438 \times 10^6} \quad v_0 = 8.01 \times 10^3 \text{ m/s} \quad \blacktriangleleft$$

$$\begin{aligned} (v_B)_1 &= \frac{h}{r_B} \\ &= \frac{5.15398 \times 10^{10}}{6.913 \times 10^6} \\ &= 7.4555 \times 10^3 \text{ m/s} \end{aligned}$$

PROBLEM 12.103 (Continued)

For a circular orbit through Point B ,

$$\begin{aligned}(v_B)_{\text{circ}} &= \sqrt{\frac{GM}{r_B}} \\ &= \sqrt{\frac{3.9843 \times 10^{14}}{6.913 \times 10^6}} \\ &= 7.5918 \times 10^3 \text{ m/s}\end{aligned}$$

(b) Increase in speed at Point B .

$$\begin{aligned}\Delta v_B &= (v_B)_{\text{circ}} - (v_B)_1 \\ &= (7.5918 - 7.4555) \times 10^3 = 136.3 \text{ m/s} \qquad \Delta v_B = 136.3 \text{ m/s} \blacktriangleleft\end{aligned}$$

PROBLEM 12.104

A space probe is describing a circular orbit about a planet of radius R . The altitude of the probe above the surface of the planet is αR and its speed is v_0 . To place the probe in an elliptic orbit which will bring it closer to the planet, its speed is reduced from v_0 to βv_0 , where $\beta < 1$, by firing its engine for a short interval of time. Determine the smallest permissible value of β if the probe is not to crash on the surface of the planet.

SOLUTION

For the circular orbit,
$$v_0 = \sqrt{\frac{GM}{r_A}} \quad \text{Eq. (12.44),}$$

where
$$r_A = R + \alpha R = R(1 + \alpha)$$

Then
$$GM = v_0^2 R(1 + \alpha)$$

From the solution to Problem 12.102, we have for the elliptic orbit,

$$\frac{1}{r_A} + \frac{1}{r_B} = \frac{2GM}{h^2}$$

Now
$$h = h_A = r_A(v_A)_{AB} = [R(1 + \alpha)](\beta v_0)$$

Then
$$\frac{1}{R(1 + \alpha)} + \frac{1}{r_B} = \frac{2v_0^2 R(1 + \alpha)}{[R(1 + \alpha)\beta v_0]^2} = \frac{2}{\beta^2 R(1 + \alpha)}$$

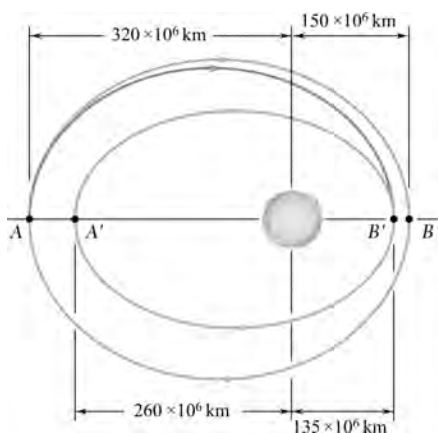
Now $\beta_{\min}$ corresponds to $r_B \rightarrow R$.

Then
$$\frac{1}{R(1 + \alpha)} + \frac{1}{R} = \frac{2}{\beta_{\min}^2 R(1 + \alpha)}$$

or



$$\beta_{\min} = \sqrt{\frac{2}{2 + \alpha}} \quad \blacktriangleleft$$



PROBLEM 12.105

As it describes an elliptic orbit about the sun, a spacecraft reaches a maximum distance of 320×10^6 km from the center of the sun at Point A (called the aphelion) and a minimum distance of 150×10^6 km at Point B (called the perihelion). To place the spacecraft in a smaller elliptic orbit with aphelion at A' and perihelion at B' , where A' and B' are located 260×10^6 and 135×10^6 km, respectively, from the center of the sun, the speed of the spacecraft is first reduced as it passes through A and then is further reduced as it passes through B' . Knowing that the mass of the sun is 332.8×10^3 times the mass of the earth, determine (a) the speed of the spacecraft at A, (b) the amounts by which the speed of the spacecraft should be reduced at A and B' to insert it into the desired elliptic orbit.

SOLUTION

First note

$$R_{\text{earth}} = 6370 \text{ km} = 6.37 \times 10^6 \text{ m}$$

$$r_A = 320 \times 10^6 \text{ km} = 320 \times 10^9 \text{ m}$$

$$r_B = 150 \times 10^6 \text{ km} = 150 \times 10^9 \text{ m}$$

From the solution to Problem 12.102, we have for any elliptic orbit about the sun

$$\frac{1}{r_1} + \frac{1}{r_2} = \frac{2GM_{\text{sun}}}{h^2}$$

(a) For the elliptic orbit AB, we have

$$r_1 = r_A, \quad r_2 = r_B, \quad h = h_A = r_A v_A$$

Also,

$$\begin{aligned} GM_{\text{sun}} &= G[(332.8 \times 10^3)M_{\text{earth}}] \\ &= gR_{\text{earth}}^2 (332.8 \times 10^3) \quad \text{using Eq. (12.30).} \end{aligned}$$

Then

$$\frac{1}{r_A} + \frac{1}{r_B} = \frac{2gR_{\text{earth}}^2 (332.8 \times 10^3)}{(r_A v_A)^2}$$

or

$$\begin{aligned} v_A &= \frac{R_{\text{earth}}}{r_A} \left(\frac{665.6g \times 10^3}{\frac{1}{r_A} + \frac{1}{r_B}} \right)^{1/2} \\ &= \frac{(6.37 \times 10^6 \text{ m})}{(320 \times 10^9 \text{ m})} \left(\frac{665.6 \times 9.81 \times 10^3}{\frac{1}{320 \times 10^9} + \frac{1}{150 \times 10^9}} \right)^{1/2} \\ &= 16,255.6 \text{ m/s} \end{aligned}$$

or

$$v_A = 16.3 \times 10^3 \text{ m/s} \quad \blacktriangleleft$$

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PROBLEM 12.105 (Continued)

(b) From Part (a), we have

$$2GM_{\text{sun}} = (r_A v_A)^2 \left(\frac{1}{r_A} + \frac{1}{r_B} \right)$$

Then, for any other elliptic orbit about the sun, we have

$$\frac{1}{r_1} + \frac{1}{r_2} = \frac{(r_A v_A)^2 \left(\frac{1}{r_A} + \frac{1}{r_B} \right)}{h^2}$$

For the elliptic transfer orbit AB' , we have

$$r_1 = r_A, \quad r_2 = r_{B'}, \quad h = h_{\text{tr}} = r_A (v_A)_{\text{tr}}$$

Then

$$\frac{1}{r_A} + \frac{1}{r_{B'}} = \frac{(r_A v_A)^2 \left(\frac{1}{r_A} + \frac{1}{r_B} \right)}{[r_A (v_A)_{\text{tr}}]^2}$$

or

$$\begin{aligned} (v_A)_{\text{tr}} &= v_A \left(\frac{\frac{1}{r_A} + \frac{1}{r_B}}{\frac{1}{r_A} + \frac{1}{r_{B'}}} \right)^{1/2} = v_A \left(\frac{1 + \frac{r_A}{r_B}}{1 + \frac{r_A}{r_{B'}}} \right)^{1/2} \\ &= (16,255.6 \text{ m/s}) \left(\frac{1 + \frac{320}{150}}{1 + \frac{320}{135}} \right)^{1/2} \\ &= 15,673.6 \text{ m/s} \end{aligned}$$

Now

$$h_{\text{tr}} = (h_A)_{\text{tr}} = (h_{B'})_{\text{tr}}: \quad r_A (v_A)_{\text{tr}} = r_{B'} (v_{B'})_{\text{tr}}$$

Then

$$(v_{B'})_{\text{tr}} = \frac{320 \times 10^9 \text{ km}}{135 \times 10^9 \text{ km}} \times 15,673.6 \text{ m/s} = 37,152.2 \text{ m/s}$$

For the elliptic orbit $A'B'$, we have

$$r_1 = r_{A'}, \quad r_2 = r_{B'}, \quad h = r_{B'} v_{B'}$$

Then

$$\frac{1}{r_{A'}} + \frac{1}{r_{B'}} = \frac{(r_A v_A)^2 \left(\frac{1}{r_A} + \frac{1}{r_B} \right)}{(r_{B'} v_{B'})^2}$$

or

$$\begin{aligned} v_{B'} &= v_A \frac{r_A}{r_{B'}} \left(\frac{\frac{1}{r_A} + \frac{1}{r_B}}{\frac{1}{r_{A'}} + \frac{1}{r_{B'}}} \right)^{1/2} \\ &= (16,255.6 \text{ m/s}) \left(\frac{320 \times 10^9 \text{ m}}{135 \times 10^9 \text{ m}} \right) \left(\frac{\frac{1}{320 \times 10^9} + \frac{1}{150 \times 10^9}}{\frac{1}{260 \times 10^9} + \frac{1}{135 \times 10^9}} \right)^{1/2} \\ &= 35,942.0 \text{ m/s} \end{aligned}$$

PROBLEM 12.105 (Continued)

Finally,

$$(v_A)_{tr} = v_A + \Delta v_A$$

or

$$\Delta v_A = (15673.6 - 16,255.6) \text{ m/s}$$

or

$$|\Delta v_A| = 582 \text{ m/s} \blacktriangleleft$$

and

$$v_{B'} = (v_{B'})_{tr} + \Delta v_B$$

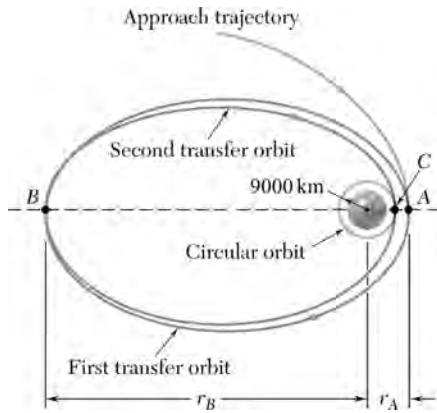
or

$$\begin{aligned} \Delta v_{B'} &= (35,942 - 37,152.2) \text{ m/s} \\ &= -1210.2 \text{ m/s} \end{aligned}$$

or

$$|\Delta v_B| = 1210 \text{ m/s} \blacktriangleleft$$

PROBLEM 12.106



A space probe is to be placed in a circular orbit of 9000 km radius about the planet Venus in a specified plane. As the probe reaches A, the point of its original trajectory closest to Venus, it is inserted in a first elliptic transfer orbit by reducing its speed by Δv_A . This orbit brings it to Point B with a much reduced velocity. There the probe is inserted in a second transfer orbit located in the specified plane by changing the direction of its velocity and further reducing its speed by Δv_B . Finally, as the probe reaches Point C, it is inserted in the desired circular orbit by reducing its speed by Δv_C . Knowing that the mass of Venus is 0.82 times the mass of the earth, that $r_A = 15 \times 10^3$ km and $r_B = 300 \times 10^3$ km, and that the probe approaches A on a parabolic trajectory, determine by how much the velocity of the probe should be reduced (a) at A, (b) at B, (c) at C.

SOLUTION

For Earth,

$$R = 6370 \text{ km} = 6.37 \times 10^6 \text{ m}, \quad g = 9.81 \text{ m/s}^2$$

$$GM_{\text{earth}} = gR^2 = (9.81)(6.37 \times 10^6)^2 = 3.98059 \times 10^{14} \text{ m}^3/\text{s}^2$$

For Venus,

$$GM = 0.82 \quad GM_{\text{earth}} = 3.2641 \times 10^{14} \text{ m}^3/\text{s}^2$$

For a parabolic trajectory with $r_A = 15 \times 10^3 \text{ km} = 15 \times 10^6 \text{ m}$

$$(v_A)_1 = v_{\text{esc}} = \sqrt{\frac{2GM}{r_A}} = \sqrt{\frac{(2)(3.2641 \times 10^{14})}{15 \times 10^6}} = 6.5971 \times 10^3 \text{ m/s}$$

First transfer orbit AB.

$$r_B = 300 \times 10^3 \text{ km} = 300 \times 10^6 \text{ m}$$

At Point A, where $\theta = 180^\circ$

$$\frac{1}{r_A} = \frac{GM}{h_{AB}^2} + C \cos 180^\circ = \frac{GM}{h_{AB}^2} - C \quad (1)$$

At Point B, where $\theta = 0^\circ$

$$\frac{1}{r_B} = \frac{GM}{h_{AB}^2} + C \cos 0 = \frac{GM}{h_{AB}^2} + C \quad (2)$$

Adding,

$$\frac{1}{r_A} + \frac{1}{r_B} = \frac{r_B + r_A}{r_A r_B} = \frac{2GM}{h_{AB}^2}$$

PROBLEM 12.106 (Continued)

Solving for h_{AB} ,

$$h_{AB} = \sqrt{\frac{2GM r_A r_B}{r_B + r_A}} = \sqrt{\frac{(2)(3.2641 \times 10^{14})(15 \times 10^6)(300 \times 10^6)}{(315 \times 10^6)}}$$

$$= 9.65712 \times 10^{10} \text{ m}^2/\text{s}$$

$$(v_A)_2 = \frac{h_{AB}}{r_A} = \frac{9.65712 \times 10^{10}}{15 \times 10^6} = 6.43808 \times 10^3 \text{ m/s}$$

$$(v_B)_1 = \frac{h_{AB}}{r_B} = \frac{9.65712 \times 10^{10}}{300 \times 10^6} = 0.321904 \times 10^3 \text{ m/s}$$

Second transfer orbit BC.

$$r_C = 9000 \text{ km} = 9 \times 10^6 \text{ m}$$

At Point B, where $\theta = 0$

$$\frac{1}{r_B} = \frac{GM}{h_{BC}^2} + C \cos 0 = \frac{GM}{h_{BC}^2} + C$$

At Point C, where $\theta = 180^\circ$

$$\frac{1}{r_C} = \frac{GM}{h_{BC}^2} + C \cos 180^\circ = \frac{GM}{h_{BC}^2} - C$$

Adding,

$$\frac{1}{r_B} + \frac{1}{r_C} = \frac{r_B + r_C}{r_B r_C} = \frac{2GM}{h_{BC}^2}$$

$$h_{BC} = \sqrt{\frac{2GM r_B r_C}{r_B + r_C}}$$

$$= \sqrt{\frac{(2)(3.2641 \times 10^{14})(300 \times 10^6)(9 \times 10^6)}{(309 \times 10^6)}}$$

$$= 7.55265 \times 10^{10} \text{ m}^2/\text{s}$$

$$(v_B)_2 = \frac{h_{BC}}{r_B} = \frac{7.55265 \times 10^{10}}{300 \times 10^6} = 251.755 \text{ m/s}$$

$$(v_C)_1 = \frac{h_{BC}}{r_C} = \frac{7.55265 \times 10^{10}}{9 \times 10^6} = 8391.83 \text{ m/s}$$

Final circular orbit.

$$r_C = 9 \times 10^6 \text{ m}$$

$$(v_C)_2 = \sqrt{\frac{GM}{r_C}}$$

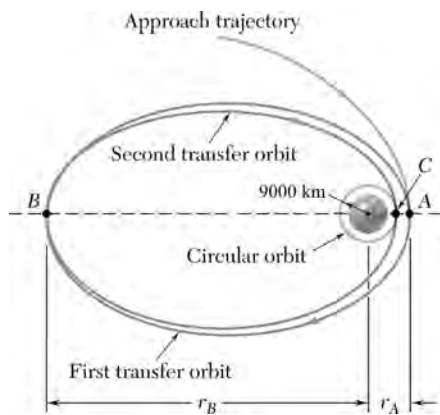
$$= \sqrt{\frac{3.2641 \times 10^{14}}{9 \times 10^6}}$$

$$= 6.02227 \times 10^3 \text{ m/s}$$

PROBLEM 12.106 (Continued)

Speed reductions.

- (a) At A: $(v_A)_1 - (v_A)_2 = 6.5971 \times 10^3 - 6.43808 \times 10^3 = 159.02 \text{ m/s}$
 $\Delta v_A = 159.0 \text{ m/s} \blacktriangleleft$
- (b) At B: $(v_B)_1 - (v_B)_2 = 0.321904 \times 10^3 - 251.755 = 70.149 \text{ m/s}$
 $\Delta v_B = 70.1 \text{ m/s} \blacktriangleleft$
- (c) At C: $(v_C)_1 - (v_C)_2 = (8391.83 - 6.02227 \times 10^3)$
 $= 2369.13 \text{ m/s}$
 $\Delta v_C = 2370 \text{ m/s} \blacktriangleleft$



PROBLEM 12.107

For the space probe of Problem 12.106, it is known that $r_A = 15 \times 10^3$ km and that the velocity of the probe is reduced to 6000 m/s, as it passes through A. Determine (a) the distance from the center of Venus to Point B, (b) the amounts by which the velocity of the probe should be reduced at B and C, respectively.

SOLUTION

For Earth,

$$R = 6370 \text{ km} = 6.37 \times 10^6 \text{ m}, \quad g = 9.81 \text{ m/s}^2$$

$$GM_{\text{earth}} = gR^2 = (9.81)(6.37 \times 10^6)^2 = 3.98059 \times 10^{14} \text{ m}^3/\text{s}^2$$

For Venus,

$$GM = 0.82 \quad GM_{\text{earth}} = 3.2641 \times 10^{14} \text{ m}^3/\text{s}^2$$

Transfer orbit AB:

$$v_A = 6000 \text{ m/s}, \quad r_A = 15 \times 10^3 \text{ km} = 15 \times 10^6 \text{ m}$$

$$h_{AB} = r_A v_A = (15 \times 10^6)(6000) = 9 \times 10^{10} \text{ m}^2/\text{s}$$

At Point A, where $\theta = 180^\circ$

$$\frac{1}{r_A} = \frac{GM}{h_{AB}^2} + C \cos 180^\circ = \frac{GM}{h_{AB}^2} - C$$

At Point B, where $\theta = 0^\circ$

$$\frac{1}{r_B} = \frac{GM}{h_{AB}^2} + C \cos 0 = \frac{GM}{h_{AB}^2} + C$$

Adding,

$$\begin{aligned} \frac{1}{r_A} + \frac{1}{r_B} &= \frac{2GM}{h_{AB}^2} \\ \frac{1}{r_B} &= \frac{2GM}{h_{AB}^2} - \frac{1}{r_A} \\ &= \frac{(2)(3.2641 \times 10^{14})}{(9 \times 10^{10})^2} - \frac{1}{15 \times 10^6} \\ &= 1.39284 \times 10^{-8} \text{ m}^{-1} \\ r_B &= 71.79578 \times 10^6 \text{ m} \end{aligned}$$

$$r_B = 71.8 \times 10^3 \text{ km} \quad \blacktriangleleft$$

PROBLEM 12.107 (Continued)

(a) Radial coordinate r_B .

$$r_B = 71.79578 \times 10^6 \text{ m}$$

$$(v_B)_1 = \frac{h_{AB}}{r_B} = \frac{9 \times 10^{10}}{71.79578 \times 10^6} = 1.25356 \times 10^3 \text{ m/s}$$

Second transfer orbit BC .

$$r_C = 9000 \text{ km} = 9 \times 10^6 \text{ m}$$

At Point B , where $\theta = 0$

$$\frac{1}{r_B} = \frac{GM}{h_{BC}^2} + C \cos 0 = \frac{GM}{h_{BC}^2} + C$$

At Point C , where $\theta = 180^\circ$

$$\frac{1}{r_C} = \frac{GM}{h_{BC}^2} + C \cos 180^\circ = \frac{GM}{h_{BC}^2} - C$$

Adding,

$$\frac{1}{r_B} + \frac{1}{r_C} = \frac{r_B + r_C}{r_B r_C} = \frac{2GM}{h_{BC}^2}$$

$$h_{BC} = \sqrt{\frac{2GM r_B r_C}{r_B + r_C}} = \sqrt{\frac{(2)(3.2641 \times 10^{14})(71.79578 \times 10^6)(9 \times 10^6)}{(80.79578 \times 10^6)}}$$

$$= 7.22555 \times 10^{10} \text{ m}^2/\text{s}$$

$$(v_B)_2 = \frac{h_{BC}}{r_B} = \frac{7.22555 \times 10^{10}}{71.79578 \times 10^6} = 1.0064 \times 10^3 \text{ m/s}$$

$$(v_C)_1 = \frac{h_{BC}}{r_C} = \frac{7.22555 \times 10^{10}}{9 \times 10^6} = 8.0284 \times 10^3 \text{ m/s}$$

Circular orbit with

$$r_C = 9 \times 10^6 \text{ m}$$

$$(v_C)_2 = \sqrt{\frac{GM}{r_C}} = \sqrt{\frac{3.2641 \times 10^{14}}{9 \times 10^6}} = 6.0223 \times 10^3 \text{ m/s}$$

(b) Speed reductions at B and C .

At B : $(v_B)_1 - (v_B)_2 = 1.25356 \times 10^3 - 1.0064 \times 10^3 = 247.16 \text{ m/s}$

$$\Delta v_B = 247 \text{ m/s} \blacktriangleleft$$

At C : $(v_C)_1 - (v_C)_2 = 8.0284 \times 10^3 - 6.0223 \times 10^3$
 $= 2006.1 \text{ m/s}$

$$\Delta v_C = 2010 \text{ m/s} \blacktriangleleft$$

PROBLEM 12.108

Determine the time needed for the probe of 12.106 to travel from A to B on its first transfer orbit.

SOLUTION

From Problem 12.106 for the first transfer orbit

$$r_A = 15 \times 10^6 \text{ m}$$

$$r_B = 300 \times 10^6 \text{ m}$$

$$h_{AB} = 9.65712 \times 10^{10} \text{ m}^2/\text{s}$$

$$\begin{aligned} a &= \frac{1}{2}(r_A + r_B) \\ &= \frac{1}{2}(15 \times 10^6 + 300 \times 10^6) \\ &= 157.5 \times 10^6 \text{ m} \end{aligned}$$

$$\begin{aligned} b &= \sqrt{r_A r_B} \\ &= \sqrt{(15 \times 10^6)(300 \times 10^6)} \\ &= 67.082 \times 10^6 \text{ m} \end{aligned}$$

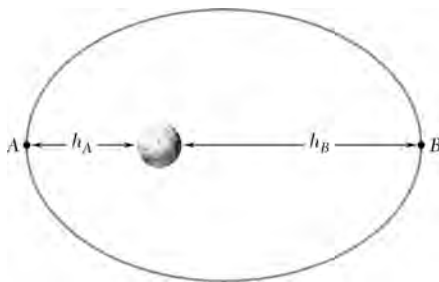
Periodic time for full elliptical orbit.

$$\begin{aligned} \tau &= \frac{2\pi ab}{h_{AB}} \\ &= \frac{(2\pi)(157.5 \times 10^6)(67.082 \times 10^6)}{9.65712 \times 10^{10}} \\ &= 687.415 \times 10^3 \text{ s} \end{aligned}$$

Time to travel from A to B .

$$t_{AB} = \frac{1}{2}\tau = 343.707 \times 10^3 \text{ s}$$

$$\begin{aligned} t_{AB} &= 95 \text{ hrs.} 28 \text{ min } 27 \text{ sec.} \\ &\approx 95 \text{ hrs.} 30 \text{ min.} \end{aligned}$$



PROBLEM 12.109

The Clementine spacecraft described an elliptic orbit of minimum altitude $h_A = 400$ km and a maximum altitude of $h_B = 2940$ km above the surface of the moon. Knowing that the radius of the moon is 1737 km and the mass of the moon is 0.01230 times the mass of the earth, determine the periodic time of the spacecraft.

SOLUTION

For earth, $R = 6370 \text{ km} = 6.370 \times 10^6 \text{ m}$

$$GM = gR^2 = (9.81)(6.370 \times 10^6)^2 = 398.06 \times 10^{12} \text{ m}^3/\text{s}^2$$

For moon,

$$GM = (0.01230)(398.06 \times 10^{12}) = 4.896 \times 10^{12} \text{ m}^3/\text{s}^2$$

$$r_A = 1737 + 400 = 2137 \text{ km} = 2.137 \times 10^6 \text{ m}$$

$$r_B = 1737 + 2940 = 4677 \text{ km} = 4.677 \times 10^6 \text{ m}$$

Using Eq. (12.39),

$$\frac{1}{r_A} = \frac{GM}{h^2} + C \cos \theta_A \quad \text{and} \quad \frac{1}{r_B} = \frac{GM}{h^2} + C \cos \theta_B.$$

But

$$\theta_B = \theta_A + 180^\circ, \quad \text{so that} \quad \cos \theta_A = -\cos \theta_B.$$

Adding,

$$\frac{1}{r_A} + \frac{1}{r_B} = \frac{r_A + r_B}{r_A r_B} = \frac{2GM}{h_{AB}^2}$$

$$h_{AB} = \sqrt{\frac{2GM r_A r_B}{r_A + r_B}} = \sqrt{\frac{(2)(4.896 \times 10^{12})(2.137 \times 10^6)(4.677 \times 10^6)}{6.804 \times 10^6}}$$

$$= 3.78856 \times 10^9 \text{ m}^2/\text{s}$$

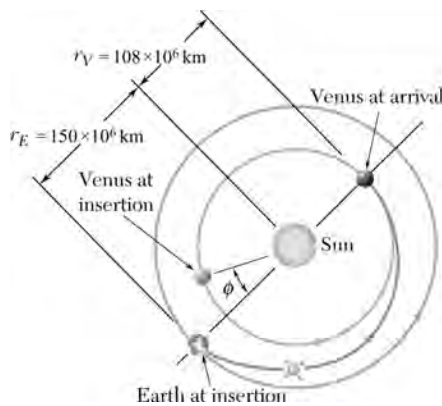
$$a = \frac{1}{2}(r_A + r_B) = 3.402 \times 10^6 \text{ m}$$

$$b = \sqrt{r_A r_B} = 3.16145 \times 10^6 \text{ m}$$

Periodic time.

$$\tau = \frac{2\pi ab}{h_{AB}} = \frac{2\pi(3.402 \times 10^6)(3.16145 \times 10^6)}{3.78856 \times 10^9} = 17.837 \times 10^3 \text{ s}$$

$$\tau = 4.95 \text{ h} \quad \blacktriangleleft$$



PROBLEM 12.110

A space probe in a low earth orbit is inserted into an elliptic transfer orbit to the planet Venus. Knowing that the mass of the sun is 332.8×10^3 times the mass of the earth and assuming that the probe is subjected only to the gravitational attraction of the sun, determine the value of ϕ , which defines the relative position of Venus with respect to the earth at the time the probe is inserted into the transfer orbit.

SOLUTION

First determine the time t_{probe} for the probe to travel from the earth to Venus. Now

$$t_{\text{probe}} = \frac{1}{2} \tau_{\text{tr}}$$

where τ_{tr} is the periodic time of the elliptic transfer orbit. Applying Kepler's Third Law to the orbits about the sun of the earth and the probe, we obtain

$$\frac{\tau_{\text{tr}}^2}{\tau_{\text{earth}}^2} = \frac{a_{\text{tr}}^3}{a_{\text{earth}}^3}$$

where

$$\begin{aligned} a_{\text{tr}} &= \frac{1}{2}(r_E + r_V) \\ &= \frac{1}{2}(150 \times 10^6 + 108 \times 10^6) \text{ km} \\ &= 129 \times 10^6 \text{ km} \end{aligned}$$

and

$$a_{\text{earth}} \approx r_E \quad (\text{Note: } \epsilon_{\text{earth}} = 0.0167)$$

Then

$$\begin{aligned} t_{\text{probe}} &= \frac{1}{2} \left(\frac{a_{\text{tr}}}{r_E} \right)^{3/2} \tau_{\text{earth}} \\ &= \frac{1}{2} \left(\frac{129 \times 10^6 \text{ km}}{150 \times 10^6 \text{ km}} \right)^{3/2} (365.25 \text{ days}) \\ &= 145.649 \text{ days} \\ &= 12.5841 \times 10^6 \text{ s} \end{aligned}$$

In time t_{probe} , Venus travels through the angle θ_V given by

$$\theta_V = \dot{\theta}_V t_{\text{probe}} = \frac{v_V}{r_V} t_{\text{probe}}$$

PROBLEM 12.110 (Continued)

Assuming that the orbit of Venus is circular (*note*: $\epsilon_{\text{venus}} = 0.0068$), then, for a circular orbit

$$v_v = \sqrt{\frac{GM_{\text{sun}}}{r_v}} \quad [\text{Eq. (12.44)}]$$

Now

$$\begin{aligned} GM_{\text{sun}} &= G(332.8 \times 10^3 M_{\text{earth}}) \\ &= 332.8 \times 10^3 (gR_{\text{earth}}^2) \quad \text{using Eq. (12.30)} \end{aligned}$$

Then

$$\begin{aligned} \theta_v &= \frac{t_{\text{probe}}}{r_v} \left[\frac{332.8 \times 10^3 (gR_{\text{earth}}^2)}{r_v} \right]^{1/2} \\ &= t_{\text{probe}} R_{\text{earth}} \frac{(332.8g \times 10^3)^{1/2}}{r_v^{3/2}} \end{aligned}$$

where

$$R_{\text{earth}} = 6370 \text{ km} = 6.37 \times 10^6 \text{ m}$$

and

$$r_v = 108 \times 10^6 \text{ km} = 108 \times 10^9 \text{ m}$$

Then

$$\begin{aligned} \theta_v &= (12.5841 \times 10^6 \text{ s})(6.37 \times 10^6 \text{ m}) \frac{(332.8 \times 10^3 \times 9.81 \text{ m/s}^2)^{1/2}}{(108 \times 10^9)^{3/2}} \\ &= 4.0809 \text{ rad} \\ &= 233.818^\circ \end{aligned}$$

Finally,

$$\phi = \theta_v - 180^\circ = 233.818^\circ - 180^\circ = 53.818^\circ$$

$$\phi = 53.8^\circ \blacktriangleleft$$

PROBLEM 12.111

Based on observations made during the 1996 sighting of comet Hyakutake, it was concluded that the trajectory of the comet is a highly elongated ellipse for which the eccentricity is approximately $\varepsilon = 0.999887$. Knowing that for the 1996 sighting, the minimum distance between the comet and the sun was $0.230R_E$, where R_E is the mean distance from the sun to the earth, determine the periodic time of the comet.

SOLUTION

For Earth's orbit about the sun,

$$v_0 = \sqrt{\frac{GM}{R_E}}, \quad \tau_0 = \frac{2\pi R_E}{v_0} = \frac{2\pi R_E^{3/2}}{\sqrt{GM}} \quad \text{or} \quad \sqrt{GM} = \frac{2\pi R_E^{3/2}}{\tau_0} \quad (1)$$

For the comet Hyakutake,

$$\frac{1}{r_0} = \frac{GM}{h^2} = (1 + \varepsilon), \quad \frac{1}{r_1} = \frac{GM}{h^2} (1 - \varepsilon), \quad r_1 = \frac{1 + \varepsilon}{1 - \varepsilon} r_0$$

$$a = \frac{1}{2}(r_0 + r_1) = \frac{r_0}{1 - \varepsilon}, \quad b = \sqrt{r_0 r_1} = \sqrt{\frac{1 + \varepsilon}{1 - \varepsilon}} r_0$$

$$h = \sqrt{GM r_0 (1 + \varepsilon)}$$

$$\tau = \frac{2\pi ab}{h} = \frac{2\pi r_0^2 (1 + \varepsilon)^{1/2}}{(1 - \varepsilon)^{3/2} \sqrt{GM r_0 (1 + \varepsilon)}}$$

$$= \frac{2\pi r_0^{3/2}}{\sqrt{GM (1 - \varepsilon)^{3/2}}} = \frac{2\pi r_0^{3/2} \tau_0}{2\pi R_E^3 (1 - \varepsilon)^{3/2}}$$

$$= \left(\frac{r_0}{R_E} \right)^{3/2} \frac{1}{(1 - \varepsilon)^{3/2}} \tau_0$$

$$= (0.230)^{3/2} \frac{1}{(1 - 0.999887)^{3/2}} \tau_0 = 91.8 \times 10^3 \tau_0$$

Since

$$\tau_0 = 1 \text{ yr}, \quad \tau = (91.8 \times 10^3)(1.000)$$

$$\tau = 91.8 \times 10^3 \text{ yr} \quad \blacktriangleleft$$

PROBLEM 12.112

Halley's comet travels in an elongated elliptic orbit for which the minimum distance from the sun is approximately $\frac{1}{2}r_E$, where $r_E = 150 \times 10^6$ km is the mean distance from the sun to the earth. Knowing that the periodic time of Halley's comet is about 76 years, determine the maximum distance from the sun reached by the comet.

SOLUTION

We apply Kepler's Third Law to the orbits and periodic times of earth and Halley's comet:

$$\left(\frac{\tau_H}{\tau_E}\right)^2 = \left(\frac{a_H}{a_E}\right)^3$$

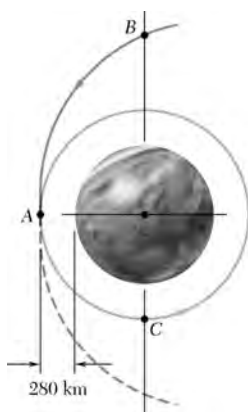
Thus

$$\begin{aligned}a_H &= a_E \left(\frac{\tau_H}{\tau_E}\right)^{2/3} \\&= \tau_E \left(\frac{76 \text{ years}}{1 \text{ year}}\right)^{2/3} \\&= 17.94\tau_E\end{aligned}$$

But

$$\begin{aligned}a_H &= \frac{1}{2}(\tau_{\min} + \tau_{\max}) \\17.94\tau_E &= \frac{1}{2}\left(\frac{1}{2}\tau_E + \tau_{\max}\right) \\\tau_{\max} &= 2(17.94\tau_E) - \frac{1}{2}\tau_E \\&= (35.88 - 0.5)\tau_E \\&= 35.38\tau_E\end{aligned}$$

$$\tau_{\max} = 150 \times 10^6 \text{ km} = 5.31 \times 10^9 \text{ km} \blacktriangleleft$$



PROBLEM 12.113

Determine the time needed for the space probe of Problem 12.99 to travel from B to C .

SOLUTION

From the solution to Problem 12.99, we have

$$(v_A)_{\text{par}} = 10,131.4 \text{ m/s}$$

and

$$(v_A)_{\text{circ}} = \frac{1}{\sqrt{2}}(v_A)_{\text{par}} = 7164.0 \text{ m/s}$$

Also,

$$r_A = (6052 + 280) \text{ km} = 6332 \text{ km}$$

For the parabolic trajectory BA , we have

$$\frac{1}{r} = \frac{GM_v}{h_{BA}^2}(1 + \varepsilon \cos \theta) \quad [\text{Eq. (12.39')}]$$

where $\varepsilon = 1$. Now

at A , $\theta = 0$:

$$\frac{1}{r_A} = \frac{GM_v}{h_{BA}^2}(1 + 1)$$

or

$$r_A = \frac{h_{BA}^2}{2GM_v}$$

at B , $\theta = -90^\circ$:

$$\frac{1}{r_B} = \frac{GM_v}{h_{BA}^2}(1 + 0)$$

or

$$r_B = \frac{h_{BA}^2}{GM_v}$$

$$r_B = 2r_A$$

PROBLEM 12.113 (Continued)

As the probe travels from B to A , the area swept out is the semiparabolic area defined by Vertex A and Point B . Thus,

$$(\text{Area swept out})_{BA} = A_{BA} = \frac{2}{3}(r_A)(r_B) = \frac{4}{3}r_A^2$$

Now $\frac{dA}{dt} = \frac{1}{2}h$

where $h = \text{constant}$

Then $A = \frac{1}{2}ht$ or $t_{BA} = \frac{2A_{BA}}{h_{BA}} \quad h_{BA} = r_A v_A$

$$= \frac{2 \times \frac{4}{3}r_A^2}{r_A v_A} = \frac{8}{3} \frac{r_A}{v_A}$$

$$= \frac{8 \, 6332 \times 10^3 \text{ m}}{3 \, 10,131.4 \text{ m/s}}$$

$$= 1666.63 \text{ s}$$

For the circular trajectory AC ,

$$t_{AC} = \frac{\frac{\pi}{2}r_A}{(v_A)_{\text{circ}}} = \frac{\pi \, 6332 \times 10^3 \text{ m}}{2 \, 7164.0 \text{ m/s}} = 1388.37 \text{ s}$$

Finally,

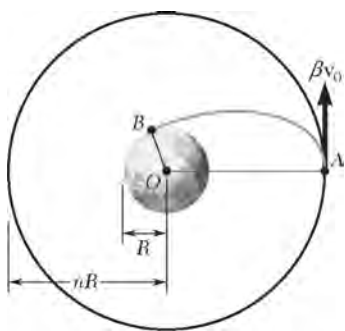
$$t_{BC} = t_{BA} + t_{AC}$$

$$= (1666.63 + 1388.37) \text{ s}$$

$$= 3055.0 \text{ s}$$

or

$$t_{BC} = 50 \text{ min } 55 \text{ s} \blacktriangleleft$$



PROBLEM 12.114

A space probe is describing a circular orbit of radius nR with a velocity v_0 about a planet of radius R and center O . As the probe passes through point A , its velocity is reduced from v_0 to βv_0 , where $\beta < 1$, to place the probe on a crash trajectory. Express in terms of n and β the angle AOB , where B denotes the point of impact of the probe on the planet.

SOLUTION

For the circular orbit,

$$r_0 = r_A = nR$$

$$v_0 = \sqrt{\frac{GM}{r_0}} = \sqrt{\frac{GM}{nR}}$$

The crash trajectory is elliptic.

$$v_A = \beta v_0 = \sqrt{\frac{\beta^2 GM}{nR}}$$

$$h = r_A v_A = nR v_A = \sqrt{\beta^2 n G M R}$$

$$\frac{GM}{h^2} = \frac{1}{\beta^2 n R}$$

$$\frac{1}{r} = \frac{GM}{h^2} (1 + \varepsilon \cos \theta) = \frac{1 + \varepsilon \cos \theta}{\beta^2 n R}$$

At Point A , $\theta = 180^\circ$

$$\frac{1}{r_A} = \frac{1}{nR} = \frac{1 - \varepsilon}{\beta^2 n R} \quad \text{or} \quad \beta^2 = 1 - \varepsilon \quad \text{or} \quad \varepsilon = 1 - \beta^2$$

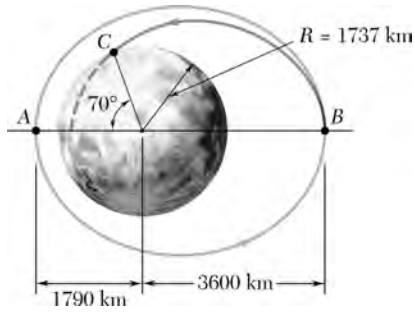
At impact Point B , $\theta = \pi - \phi$

$$\frac{1}{r_B} = \frac{1}{R}$$

$$\frac{1}{R} = \frac{1 + \varepsilon \cos(\pi - \phi)}{\beta^2 n R} = \frac{1 - \varepsilon \cos \phi}{\beta^2 n R}$$

$$\varepsilon \cos \phi = 1 - n\beta^2 \quad \text{or} \quad \cos \phi = \frac{1 - n\beta^2}{\varepsilon} = \frac{1 - n\beta^2}{1 - \beta^2}$$

$$\phi = \cos^{-1}[(1 - n\beta^2) / (1 - \beta^2)] \quad \blacktriangleleft$$



PROBLEM 12.115

Prior to the Apollo missions to the moon, several Lunar Orbiter spacecraft were used to photograph the lunar surface to obtain information regarding possible landing sites. At the conclusion of each mission, the trajectory of the spacecraft was adjusted so that the spacecraft would crash on the moon to further study the characteristics of the lunar surface. Shown is the elliptic orbit of Lunar Orbiter 2. Knowing that the mass of the moon is 0.01230 times the mass of the earth, determine the amount by which the speed of the orbiter should be reduced at Point B so that it impacts the lunar surface at Point C. (Hint: Point B is the apogee of the elliptic impact trajectory.)

SOLUTION

From the solution to Problem 12.102, we have for the elliptic orbit AB

$$\frac{1}{r_A} + \frac{1}{r_B} = \frac{2GM_{\text{moon}}}{h_{AB}^2}$$

where

$$h_{AB} = (h_B)_{AB} = r_B(v_B)_{AB}$$

and

$$\begin{aligned} GM_{\text{moon}} &= G(0.01230M_{\text{earth}}) \\ &= 0.01230gR_{\text{earth}}^2 \quad \text{using Eq. (12.30)} \end{aligned}$$

Then

$$\frac{1}{r_A} + \frac{1}{r_B} = \frac{2(0.01230gR_{\text{earth}}^2)}{[r_B(v_B)_{AB}]^2}$$

or

$$\begin{aligned} (v_B)_{AB} &= \frac{R_{\text{earth}}}{r_B} \left(\frac{0.0246g}{\frac{1}{r_A} + \frac{1}{r_B}} \right)^{1/2} \\ &= \frac{6.37 \times 10^6 \text{ m}}{3600 \times 10^3 \text{ m}} \left(\frac{0.0246 \times 9.81 \text{ m/s}^2}{\frac{1}{1790 \times 10^3 \text{ m}} + \frac{1}{3600 \times 10^3 \text{ m}}} \right)^{1/2} \\ &= 950.43 \text{ m/s} \end{aligned}$$

For the elliptic impact trajectory, we have

$$\frac{1}{r} = \frac{GM_{\text{moon}}}{h_{BC}^2} + C \cos \theta \quad [\text{Eq. (12.39)}]$$

where

$$h_{BC} = (h_B)_{BC} = r_B(v_B)_{BC}$$

PROBLEM 12.115 (Continued)

Noting that Point B is the apogee of this trajectory, we have

$$\text{at } B, \theta = 180^\circ: \quad \frac{1}{r_B} = \frac{GM_{\text{moon}}}{h_{BC}^2} - C$$

$$\text{or} \quad C = \frac{GM_{\text{moon}}}{h_{BC}^2} - \frac{1}{r_B}$$

$$\text{at } C, \theta = -70^\circ: \quad \frac{1}{R} = \frac{GM_{\text{moon}}}{h_{BC}^2} + C \cos(-70^\circ)$$

$$\text{or} \quad C = \frac{1}{\cos 70^\circ} \left(\frac{1}{R} - \frac{GM_{\text{moon}}}{h_{BC}^2} \right)$$

$$\text{Then} \quad \frac{GM_{\text{moon}}}{h_{BC}^2} - \frac{1}{r_B} = \frac{1}{\cos 70^\circ} \left(\frac{1}{R} - \frac{GM_{\text{moon}}}{h_{BC}^2} \right)$$

$$\text{or} \quad h_{BC}^2 = \frac{GM_{\text{moon}}(1 + \cos 70^\circ)}{\frac{1}{R} + \frac{\cos 70^\circ}{r_B}}$$

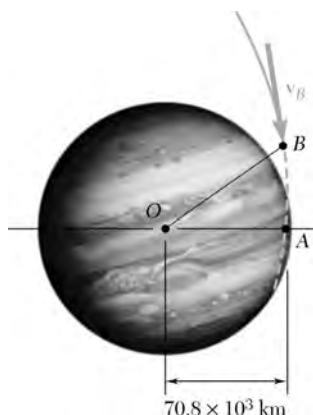
$$\text{or} \quad (v_B)_{BC} = \frac{R_{\text{earth}}}{r_B} \left[\frac{0.012309(1 + \cos 70^\circ)}{\frac{1}{R} + \frac{\cos 70^\circ}{r_B}} \right]^{1/2}$$

$$\begin{aligned} (v_B)_{BC} &= \frac{6.37 \times 10^6 \text{ m}}{3600 \times 10^3 \text{ m}} \left[\frac{0.01230(9.81 \text{ m/s}^2)(1 + \cos 70^\circ)}{\frac{1}{1737 \times 10^3 \text{ m}} + \frac{\cos 70^\circ}{3600 \times 10^3 \text{ m}}} \right]^{1/2} \\ &= 869.43 \text{ m/s} \end{aligned}$$

$$\text{Finally,} \quad (v_B)_{BC} = (v_B)_{AB} + \Delta v_B$$

$$\text{or} \quad \Delta v_B = (869.43 - 950.43) \text{ m/s}$$

$$\text{or} \quad |\Delta v_B| = 81.0 \text{ m/s} \quad \blacktriangleleft$$



PROBLEM 12.116

As a spacecraft approaches the planet Jupiter, it releases a probe which is to enter the planet's atmosphere at Point B at an altitude of 450 km above the surface of the planet. The trajectory of the probe is a hyperbola of eccentricity $\epsilon = 1.031$. Knowing that the radius and the mass of Jupiter are 71.492×10^3 km and 1.9×10^{27} kg, respectively, and that the velocity $\mathbf{v}_B$ of the probe at B forms an angle of 82.9° with the direction of OA , determine (a) the angle AOB , (b) the speed v_B of the probe at B .

SOLUTION

First we note

$$r_B = (71.492 \times 10^3 + 450) \text{ km} = 71.942 \times 10^3 \text{ km}$$

(a) We have

$$\frac{1}{r} = \frac{GM_j}{h^2} (1 + \epsilon \cos \theta) \quad [\text{Eq. (12.39')}]$$

At A , $\theta = 0$:

$$\frac{1}{r_A} = \frac{GM_j}{h^2} (1 + \epsilon)$$

or

$$\frac{h^2}{GM_j} = r_A (1 + \epsilon)$$

At B , $\theta = \theta_B = \angle AOB$:

$$\frac{1}{r_B} = \frac{GM_j}{h^2} (1 + \epsilon \cos \theta_B)$$

or

$$\frac{h^2}{GM_j} = r_B (1 + \epsilon \cos \theta_B)$$

Then

$$r_A (1 + \epsilon) = r_B (1 + \epsilon \cos \theta_B)$$

or

$$\begin{aligned} \cos \theta_B &= \frac{1}{\epsilon} \left[\frac{r_A}{r_B} (1 + \epsilon) - 1 \right] \\ &= \frac{1}{1.031} \left[\frac{70.8 \times 10^3 \text{ km}}{71.942 \times 10^3 \text{ km}} (1 + 1.031) - 1 \right] \\ &= 0.96873 \end{aligned}$$

or

$$\theta_B = 14.3661^\circ$$

$$\angle AOB = 14.37^\circ \blacktriangleleft$$

PROBLEM 12.116 (Continued)

(b) From above

$$h^2 = GM_j r_B (1 + \varepsilon \cos \theta_B)$$

where

$$h = \frac{1}{m} |\mathbf{r}_B \times m \mathbf{v}_B| = r_B v_B \sin \phi$$

$$\phi = (\theta_B + 82.9^\circ) = 97.2661^\circ$$

Then

$$(r_B v_B \sin \phi)^2 = GM_j r_B (1 + \varepsilon \cos \theta_B)$$

or

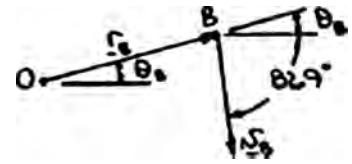
$$v_B = \frac{1}{\sin \phi} \left[\frac{GM_j}{r_B} (1 + \varepsilon \cos \theta_B) \right]^{1/2}$$

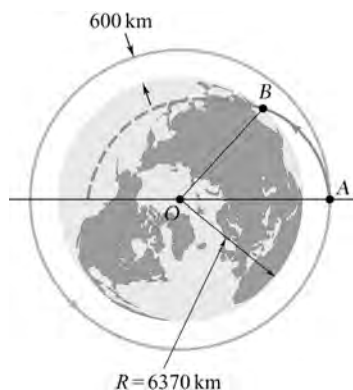
$$= \frac{1}{\sin 97.2661^\circ}$$

$$\left\{ \frac{66.73 \times 10^{-12} \text{ m}^2/\text{kg} \cdot \text{s}^2 \times 1.9 \times 10^{27} \text{ kg}}{71.942 \times 10^6 \text{ m}} \times [1 + (1.031)(0.96873)] \right\}^{1/2}$$

or

$$v_B = 59.8 \text{ km/s} \quad \blacktriangleleft$$





PROBLEM 12.117

A space shuttle is describing a circular orbit at an altitude of 600 km above the surface of the earth. As it passes through Point A, it fires its engine for a short interval of time to reduce its speed by 150 m/s and begin its descent toward the earth. Determine the angle AOB so that the altitude of the shuttle at Point B is 120 km. (*Hint: Point A is the apogee of the elliptic descent trajectory.*)

SOLUTION

First we note

$$\begin{aligned} R &= 6370 \text{ km} = 6.37 \times 10^6 \text{ m} \\ r_A &= (6370 + 600) \text{ km} = 6970 \text{ km} \\ &= 6.97 \times 10^6 \text{ m} \\ r_B &= (6370 + 125) \text{ km} = 6495 \text{ km} \end{aligned}$$

For the circular orbit, we have

$$\begin{aligned} v_{\text{circ}} &= \sqrt{\frac{gR^2}{r_A}} \quad [\text{Eq. (12.44)}] \\ &= 6.37 \times 10^6 \text{ m} \left(\frac{9.81 \text{ m/s}^2}{6.97 \times 10^6 \text{ m}} \right)^{1/2} \\ &= 7557.14 \text{ m/s} \end{aligned}$$

Now

$$\begin{aligned} (v_A)_{AB} &= v_{\text{circ}} + \Delta v_A = (7557.14 - 150) \text{ m/s} \\ &= 7407.14 \text{ m/s} \end{aligned}$$

For the elliptic descent trajectory, we have

$$\frac{1}{r} = \frac{GM}{h^2} + C \cos \theta \quad [\text{Eq. (12.39)}]$$

Noting that Point A is at the apogee of this trajectory, we have

$$\text{at } A, \theta = 180^\circ: \quad \frac{1}{r_A} = \frac{GM}{h^2} - C$$

or

$$C = \frac{GM}{h^2} - \frac{1}{r_A}$$

PROBLEM 12.117 (Continued)

at B, $\theta = \theta_B = 180^\circ - \angle AOB$:
$$\frac{1}{r_B} = \frac{GM}{h^2} + C \cos \theta_B$$

or
$$C = \frac{1}{\cos \theta_B} \left(\frac{1}{r_B} - \frac{GM}{h^2} \right)$$

Then
$$\frac{GM}{h^2} - \frac{1}{r_A} = \frac{1}{\cos \theta_B} \left(\frac{1}{r_B} - \frac{GM}{h^2} \right)$$

or
$$\cos \theta_B = \frac{\frac{1}{r_B} - \frac{GM}{h^2}}{\frac{GM}{h^2} - \frac{1}{r_A}}$$

Now
$$h = (h_A)_{AB} = r_A (v_A)_{AB}$$

and
$$GM = gR^2 \quad [\text{Eq. (12.30)}]$$

From above,
$$gR^2 = r_A (v_{\text{circ}})^2 \quad [\text{Eq. (12.44)}]$$

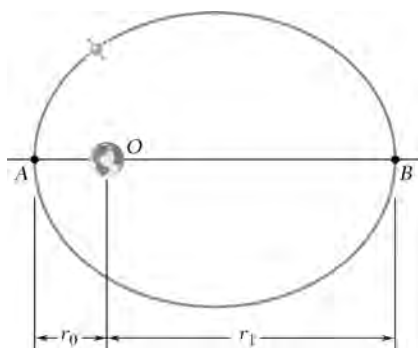
Then
$$\frac{GM}{h^2} = \frac{r_A (v_{\text{circ}})^2}{[r_A (v_A)_{AB}]^2} = \frac{1}{r_A} \left[\frac{v_{\text{circ}}}{(v_A)_{AB}} \right]^2$$

so that
$$\begin{aligned} \cos \theta_B &= \frac{\frac{1}{r_B} - \frac{1}{r_A} \left[\frac{v_{\text{circ}}}{(v_A)_{AB}} \right]^2}{\frac{1}{r_A} \left[\frac{v_{\text{circ}}}{(v_A)_{AB}} \right]^2 - \frac{1}{r_A}} = \frac{\frac{r_A}{r_B} - \left[\frac{v_{\text{circ}}}{(v_A)_{AB}} \right]^2}{\left[\frac{v_{\text{circ}}}{(v_A)_{AB}} \right]^2 - 1} \\ &= \frac{\frac{6970 \text{ km}}{6495 \text{ km}} - \left(\frac{7557.14}{7407.14} \right)^2}{\left(\frac{7557.14}{7407.14} \right)^2 - 1} \\ &= 0.78758 \end{aligned}$$

or
$$\theta_B = 38.04^\circ$$

Finally,
$$\angle AOB = 180^\circ - 38.04^\circ$$

or
$$\angle AOB = 142.0^\circ \blacktriangleleft$$



PROBLEM 12.118

A satellite describes an elliptic orbit about a planet. Denoting by r_0 and r_1 the distances corresponding, respectively, to the perigee and apogee of the orbit, show that the curvature of the orbit at each of these two points can be expressed as

$$\frac{1}{\rho} = \frac{1}{2} \left(\frac{1}{r_0} + \frac{1}{r_1} \right)$$

SOLUTION

Using Eq. (12.39),

$$\frac{1}{r_A} = \frac{GM}{h^2} + C \cos \theta_A$$

and

$$\frac{1}{r_B} = \frac{GM}{h^2} + C \cos \theta_B.$$

But

$$\theta_B = \theta_A + 180^\circ,$$

so that

$$\cos \theta_A = -\cos \theta_B$$

Adding,

$$\frac{1}{r_A} + \frac{1}{r_B} = \frac{2GM}{h^2}$$

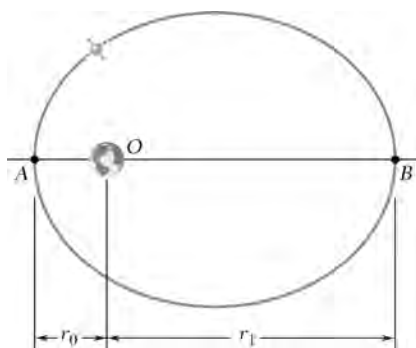
At Points A and B, the radial direction is normal to the path.

$$a_n = \frac{v^2}{\rho} = \frac{h^2}{r^2 \rho}$$

But

$$F_n = \frac{GMm}{r^2} = ma_n = \frac{mh^2}{r^2 \rho}$$

$$\frac{1}{\rho} = \frac{GM}{h^2} = \frac{1}{2} \left(\frac{1}{r_A} + \frac{1}{r_B} \right) \quad \blacktriangleleft$$



PROBLEM 12.119

- (a) Express the eccentricity ε of the elliptic orbit described by a satellite about a planet in terms of the distances r_0 and r_1 corresponding, respectively, to the perigee and apogee of the orbit.
- (b) Use the result obtained in Part a and the data given in Problem 12.111, where $R_E = 149.6 \times 10^6$ km, to determine the approximate maximum distance from the sun reached by comet Hyakutake.

SOLUTION

(a) We have
$$\frac{1}{r} = \frac{GM}{h^2} (1 + \varepsilon \cos \theta) \quad \text{Eq. (12.39')}$$

At A, $\theta = 0$:
$$\frac{1}{r_0} = \frac{GM}{h^2} (1 + \varepsilon)$$

or
$$\frac{h^2}{GM} = r_0 (1 + \varepsilon)$$

At B, $\theta = 180^\circ$:
$$\frac{1}{r_1} = \frac{GM}{h^2} (1 - \varepsilon)$$

or
$$\frac{h^2}{GM} = r_1 (1 - \varepsilon)$$

Then
$$r_0 (1 + \varepsilon) = r_1 (1 - \varepsilon)$$

or
$$\varepsilon = \frac{r_1 - r_0}{r_1 + r_0} \quad \blacktriangleleft$$

(b) From above,
$$r_1 = \frac{1 + \varepsilon}{1 - \varepsilon} r_0$$

where
$$r_0 = 0.230 R_E$$

Then
$$r_1 = \frac{1 + 0.999887}{1 - 0.999887} \times 0.230 (149.6 \times 10^9 \text{ m})$$

or
$$r_1 = 609 \times 10^{12} \text{ m} \quad \blacktriangleleft$$

Note: $r_1 = 4070 R_E$ or $r_1 = 0.064$ lightyears.

PROBLEM 12.120

Show that the angular momentum per unit mass h of a satellite describing an elliptic orbit of semimajor axis a and eccentricity ϵ about a planet of mass M can be expressed as

$$h = \sqrt{GMa(1 - \epsilon^2)}$$

SOLUTION

By Eq. (12.39'),

$$\frac{1}{r} = \frac{GM}{h^2}(1 + \epsilon \cos \theta)$$

At A, $\theta = 0^\circ$:

$$\frac{1}{r_A} = \frac{GM}{h^2} = (1 + \epsilon) \quad \text{or} \quad r_A = \frac{h^2}{GM(1 + \epsilon)}$$

At B, $\theta = 180^\circ$:

$$\frac{1}{r_B} = \frac{GM}{h^2} = (1 - \epsilon) \quad \text{or} \quad r_B = \frac{h^2}{GM(1 - \epsilon)}$$

Adding,

$$r_A + r_B = \frac{h^2}{GM} = \left(\frac{1}{1 + \epsilon} + \frac{1}{1 - \epsilon} \right) = \frac{2h^2}{GM(1 - \epsilon^2)}$$

But for an ellipse,

$$r_A + r_B = 2a$$

$$2a = \frac{2h^2}{GM(1 - \epsilon^2)}$$

$$h = \sqrt{GMa(1 - \epsilon^2)} \quad \blacktriangleleft$$

PROBLEM 12.121

Derive Kepler's third law of planetary motion from Eqs. (12.39) and (12.45).

SOLUTION

For an ellipse,

$$2a = r_A + r_B \quad \text{and} \quad b = \sqrt{r_A r_B}$$

Using Eq. (12.39),

$$\frac{1}{r_A} = \frac{GM}{h^2} + C \cos \theta_A$$

and

$$\frac{1}{r_B} = \frac{GM}{h^2} + C \cos \theta_B.$$

But

$$\theta_B = \theta_A + 180^\circ,$$

so that

$$\cos \theta_A = -\cos \theta_B.$$

Adding,

$$\frac{1}{r_A} + \frac{1}{r_B} = \frac{r_A + r_B}{r_A r_B} = \frac{2a}{b^2} = \frac{2GM}{h^2}$$

$$h = b \sqrt{\frac{GM}{a}}$$

By Eq. (12.45),

$$\tau = \frac{2\pi ab}{h} = \frac{2\pi ab \sqrt{a}}{b \sqrt{GM}} = \frac{2\pi a^{3/2}}{\sqrt{GM}}$$

$$\tau^2 = \frac{4\pi^2 a^3}{GM}$$

For Orbits 1 and 2 about the same large mass,

$$\tau_1^2 = \frac{4\pi^2 a_1^3}{GM}$$

and

$$\tau_2^2 = \frac{4\pi^2 a_2^3}{GM}$$

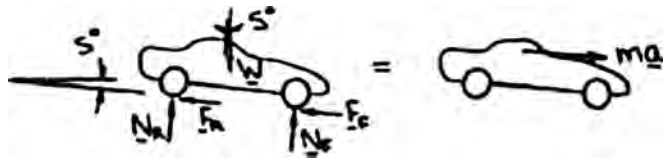
Forming the ratio,

$$\left(\frac{\tau_1}{\tau_2} \right)^2 = \left(\frac{a_1}{a_2} \right)^3 \quad \blacktriangleleft$$

PROBLEM 12.122

A 1000-kg automobile is being driven down a 5° incline at a speed of 80 km/h when the brakes are applied, causing a total braking force of 6000 N to be applied to the automobile. Determine the distance traveled by the automobile before it comes to a stop.

SOLUTION



We have
$$\rightarrow \Sigma F_x = ma: W \sin 5^\circ - (F_F + F_R) = \frac{W}{g} a$$

where
$$F_F + F_R = F_{\text{brake}}$$

Then
$$a = (9.81 \text{ m/s}^2) \left(\sin 5^\circ - \frac{6000 \text{ N}}{1000 \times 9.81} \right)$$

$$= -5.145 \text{ m/s}^2$$

For this uniformly decelerated motion, we have

$$v^2 = v_0^2 + 2a(x - 0)$$

where
$$v_0 = 80 \text{ km/h} = \frac{200}{9} \text{ m/s}$$

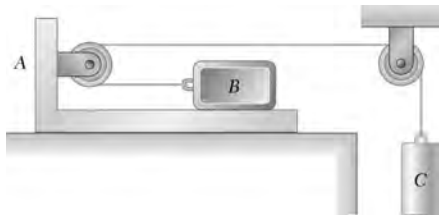
Then when $v = 0$,
$$0 = \left(\frac{200}{9} \right)^2 + 2(-5.145 \text{ m/s}^2)x$$

$$x = 47.991 \text{ m}$$

or

$$x = 48.0 \text{ m} \quad \blacktriangleleft$$

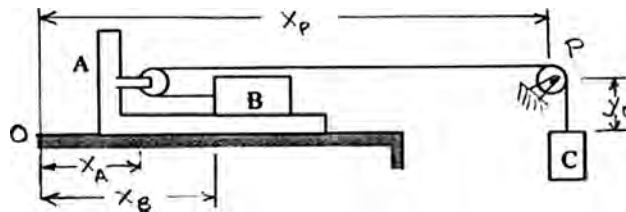
PROBLEM 12.123



A 6-kg block B rests as shown on a 10-kg bracket A . The coefficients of friction are $\mu_s = 0.30$ and $\mu_k = 0.25$ between block B and bracket A , and there is no friction in the pulley or between the bracket and the horizontal surface. (a) Determine the maximum mass of block C if block B is not to slide on bracket A . (b) If the mass of block C is 10% larger than the answer found in a, determine the accelerations of A , B and C .

SOLUTION

Kinematics. Let x_A and x_B be horizontal coordinates of A and B measured from a fixed vertical line to the left of A and B . Let y_C be the distance that block C is below the pulley. Note that y_C increases when C moves downward. See figure.



The cable length L is fixed.

$$L = (x_B - x_A) + (x_P - x_A) + y_C + \text{constant}$$

Differentiating and noting that $\dot{x}_P = 0$,

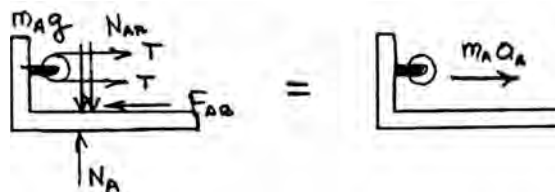
$$v_B - 2v_A + v_C = 0$$

$$-2a_A + a_B + a_C = 0 \quad (1)$$

Here, a_A and a_B are positive to the right, and a_C is positive downward.

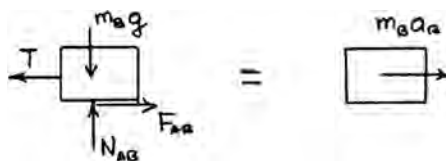
Kinetics. Let T be the tension in the cable and F_{AB} be the friction force between blocks A and B . The free body diagrams are:

Bracket A :



PROBLEM 12.123 (Continued)

Block B:



Block C:



Bracket A: $\rightarrow \Sigma F_x = ma_x: 2T - F_{AB} = m_A a_A$ (2)

Block B: $\rightarrow \Sigma F_x = ma_x: F_{AB} - T = m_B a_B$ (3)

$$+\uparrow \Sigma F_y = ma_y: N_{AB} - m_A g = 0$$

or

$$N_{AB} = m_A g$$

Block C: $+\downarrow \Sigma F_y = ma_y: m_C g - T = m_C a_C$ (4)

Adding Eqs. (2), (3), and (4), and transposing,

$$m_A a_A + m_B a_B + m_C a_C = m_C g$$
 (5)

Subtracting Eq. (4) from Eq. (3) and transposing,

$$m_B a_B - m_C a_C = F_{AB} - m_C g$$
 (6)

(a) No slip between A and B. $a_B = a_A$

From Eq. (1), $a_A = a_B = a_C = a$

From Eq. (5),
$$a = \frac{m_C g}{m_A + m_B + m_C}$$

For impending slip, $F_{AB} = \mu_s N_{AB} = \mu_s m_B g$

Substituting into Eq. (6),

$$\frac{(m_B - m_C)(m_C g)}{m_A + m_B + m_C} = \mu_s m_B g - m_C g$$

PROBLEM 12.123 (Continued)

Solving for m_C ,

$$\begin{aligned} m_C &= \frac{\mu_s m_B (m_A + m_B)}{m_A + 2m_B - \mu_s m_C} \\ &= \frac{(0.30)(6)(10 + 6)}{10 + (2)(6) - (0.30)(6)} \end{aligned}$$

$$m_C = 1.426 \text{ kg} \quad \blacktriangleleft$$

(b) m_C increased by 10%.

$$m_C = 1.568 \text{ kg}$$

Since slip is occurring,

$$F_{AB} = \mu_k N_{AB} = \mu_k m_B g$$

Eq. (6) becomes

$$m_B a_B - m_C a_C = (\mu_k m_B - m_C) g$$

or

$$6a_B - 1.568a_C = [(0.25)(6) - 1.568](9.81) \quad (7)$$

With numerical data, Eq. (5) becomes

$$10a_A + 6a_B + 1.568a_C = (1.568)(9.81) \quad (8)$$

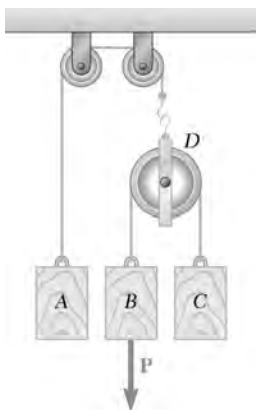
Solving Eqs. (1), (7), and (8) gives

$$a_A = 1.053 \text{ m/s}^2, \quad a_B = 0.348 \text{ m/s}^2, \quad a_C = 1.759 \text{ m/s}^2$$

$$\mathbf{a}_A = 1.503 \text{ m/s}^2 \rightarrow \blacktriangleleft$$

$$\mathbf{a}_B = 0.348 \text{ m/s}^2 \rightarrow \blacktriangleleft$$

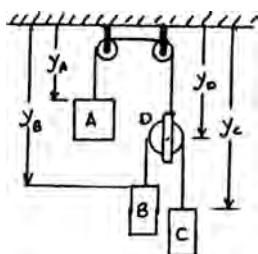
$$\mathbf{a}_C = 1.759 \text{ m/s}^2 \downarrow \blacktriangleleft$$



PROBLEM 12.124

Block A 10 kg, and blocks B and C weigh 5 kg each. Knowing that the blocks are initially at rest and that B moves 2.5 in 2 s, determine (a) the magnitude of the force **P**, (b) the tension in the cord AD. Neglect the masses of the pulleys and axle friction.

SOLUTION



Let the position coordinate y be positive downward.

Constraint of cord AD: $y_A + y_D = \text{constant}$

$$v_A + v_D = 0, \quad a_A + a_D = 0$$

Constraint of cord BC: $(y_B - y_D) + (y_C - y_D) = \text{constant}$

$$v_B + v_C - 2v_D = 0, \quad a_B + a_C - 2a_D = 0$$

Eliminate a_D . $2a_A + a_B + a_C = 0$ (1)

We have uniformly accelerated motion because all of the forces are constant.

$$y_B = (y_B)_0 + (v_B)_0 t + \frac{1}{2} a_B t^2, \quad (v_B)_0 = 0$$

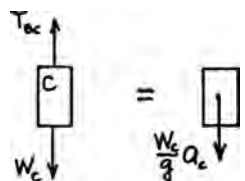
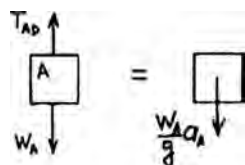
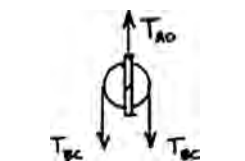
$$a_B = \frac{2[y_B - (y_B)_0]}{t^2} = \frac{2(2.5)}{(2)^2} = 1.25 \text{ m/s}^2$$

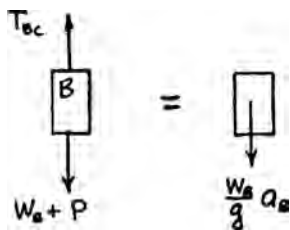
Pulley D: $+\downarrow \Sigma F_y = 0: 2T_{BC} - T_{AD} = 0$

$$T_{AD} = 2T_{BC}$$

Block A: $+\downarrow \Sigma F_y = ma_y: m_A g - T_{AD} = m_A a_A$

or
$$a_A = g - \frac{T_{AD}}{m_A} \quad (2)$$





PROBLEM 12.124 (Continued)

Block C: $+\downarrow \Sigma F_y = ma_y: m_c g - T_{BC} = m_c a_c$

or $a_c = g - \frac{T_{BC}}{m_c}$ (3)

Substituting the value for a_B and Eqs. (2) and (3) into Eq. (1), and solving for T_{BC} ,

$$2 \left(g - \frac{2T_{BC}}{m_A} \right) + a_B + \left(g - \frac{T_{BC}}{m_C} \right) = 0$$

$$2g - \frac{4T_{BC}}{10} + 1.25 + g - \frac{T_{BC}}{5} = 0$$

$$3g + 1.25 = \frac{3T_{BC}}{5} \quad T_{BC} = 51.133 \text{ N}$$

Block B: $+\downarrow \Sigma F_y = ma_y: P + m_B g - T_{BC} = m_B a_B$

(a) Magnitude of P .

$$P = T_{BC} - m_B g + m_B a_B$$

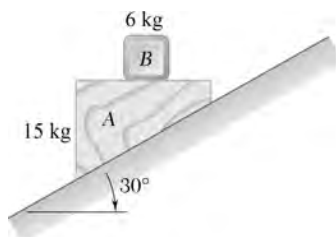
$$P = 51.133 - 5g + 5(1.25) \\ = 8.333 \text{ N}$$

$$P = 8.33 \text{ N} \blacktriangleleft$$

(b) Tension in cord AD.

$$T_{AD} = 2T_{BC} = (2)(51.133) \\ = 102.266 \text{ N}$$

$$T_{AD} = 102.3 \text{ N} \blacktriangleleft$$



PROBLEM 12.125

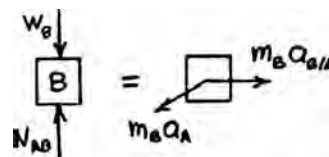
A 6-kg block B rests as shown on the upper surface of a 15 kg wedge A . Neglecting friction, determine immediately after the system is released from rest (a) the acceleration of A , (b) the acceleration of B relative to A .

SOLUTION

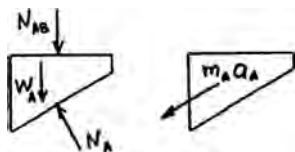
Acceleration vectors:

$$\mathbf{a}_A = a_A \nearrow 30^\circ, \quad \mathbf{a}_{B/A} = a_{B/A} \rightarrow$$

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$



Block B:



$$+\rightarrow \Sigma F_x = ma_x: \quad m_B a_{B/A} - m_B a_A \cos 30^\circ = 0$$

$$a_{B/A} = a_A \cos 30^\circ \quad (1)$$

$$+\downarrow \Sigma F_y = ma_y: \quad N_{AB} - W_B = -m_B a_A \sin 30^\circ$$

$$N_{AB} = W_B - (W_B \sin 30^\circ) \frac{a_A}{g} \quad (2)$$

Block A:

$$\nearrow \Sigma F = ma: \quad W_A \sin 30^\circ + N_{AB} \sin 30^\circ = W_A \frac{a_A}{g}$$

$$W_A \sin 30^\circ + W_B \sin 30^\circ - (W_B \sin^2 30^\circ) \frac{a_A}{g} = W_A \frac{a_A}{g}$$

$$a_A = \frac{(W_A + W_B) \sin 30^\circ}{W_A + W_B \sin^2 30^\circ} g = \frac{(m_A + m_B) \sin 30^\circ}{m_A + m_B \sin^2 30^\circ} g$$

$$= \frac{21 \sin 30^\circ}{15 + 6 \sin^2 30^\circ} \times (9.81)$$

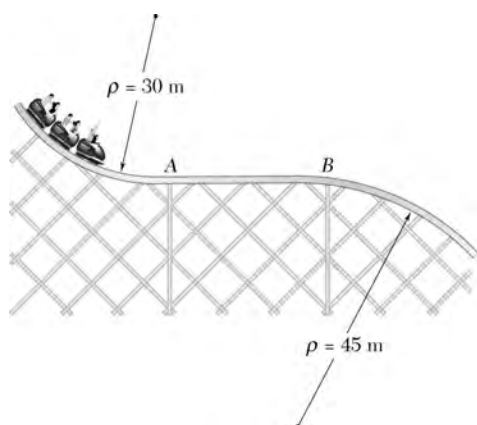
(a)

$$\mathbf{a}_A = 6.24 \text{ m/s}^2 \nearrow 30^\circ \blacktriangleleft$$

$$a_{B/A} = (6.24) \cos 30^\circ = 5.40 \text{ m/s}^2$$

(b)

$$\mathbf{a}_{B/A} = 5.40 \text{ m/s}^2 \rightarrow \blacktriangleleft$$



PROBLEM 12.126

The roller-coaster track shown is contained in a vertical plane. The portion of track between A and B is straight and horizontal, while the portions to the left of A and to the right of B have radii of curvature as indicated. A car is traveling at a speed of 72 km/h when the brakes are suddenly applied, causing the wheels of the car to slide on the track ($\mu_k = 0.25$). Determine the initial deceleration of the car if the brakes are applied as the car (a) has almost reached A, (b) is traveling between A and B, (c) has just passed B.

SOLUTION

$$v = 72 \text{ km/h} = 20 \text{ m/s}$$

(a) Almost reached Point A.

$$\rho = 30 \text{ m}$$

$$a_n = \frac{v^2}{\rho} = \frac{(20)^2}{30} = 13.333 \text{ m/s}^2 \uparrow$$

$$\Sigma F_y = ma_y: N_R + N_F - mg = ma_n$$

$$N_R + N_F = m(g + a_n)$$

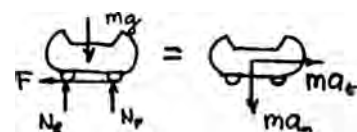
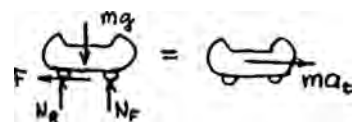
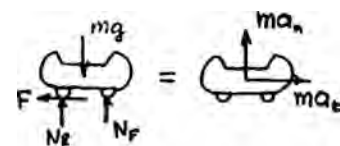
$$F = \mu_k (N_R + N_F) = \mu_k m(g + a_n)$$

$$\rightarrow \Sigma F_x = ma_x: -F = ma_t$$

$$a_t = -\frac{F}{m} = -\mu_k (g + a_n)$$

$$|a_t| = \mu_k (g + a_n) = 0.25(9.81 + 13.33)$$

$$|a_t| = 5.79 \text{ m/s}^2 \blacktriangleleft$$



(b) Between A and B.

$$\rho = \infty$$

$$a_n = 0$$

$$|a_t| = \mu_k g = (0.25)(9.81)$$

$$|a_t| = 2.45 \text{ m/s}^2 \blacktriangleleft$$

(c) Just passed Point B.

$$\rho = 40 \text{ m}$$

$$a_n = \frac{v^2}{\rho} = \frac{(20)^2}{45} = 8.8889 \text{ m/s}^2 \downarrow$$

$$\Sigma F_y = ma_y: N_R + N_F - mg = -ma_n$$

PROBLEM 12.126 (Continued)

or

$$N_R + N_F = m(g - a_n)$$

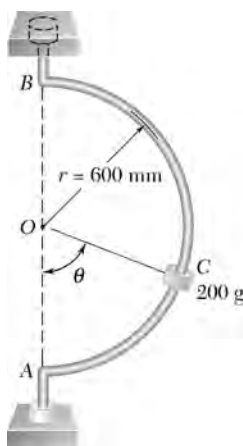
$$F = \mu_k(N_R + N_F) = \mu_k m(g - a_n)$$

$$\overset{+}{\rightarrow} \Sigma F_x = ma_x: \quad -F = ma_t$$

$$a_t = -\frac{F}{m} = -\mu_k(g - a_n)$$

$$|a_t| = \mu_k(g - a_n) = (0.25)(9.81 - 8.8889)$$

$$|a_t| = 0.230 \text{ m/s}^2 \quad \blacktriangleleft$$



PROBLEM 12.127

A small 200-g collar C can slide on a semicircular rod which is made to rotate about the vertical AB at the constant rate of 6 rad/s. Determine the minimum required value of the coefficient of static friction between the collar and the rod if the collar is not to slide when (a) $\theta = 90^\circ$, (b) $\theta = 75^\circ$, (c) $\theta = 45^\circ$. Indicate in each case the direction of the impending motion.

SOLUTION

First note

$$\begin{aligned} v_C &= (r \sin \theta) \dot{\phi}_{AB} \\ &= (0.6 \text{ m})(6 \text{ rad/s}) \sin \theta \\ &= (3.6 \text{ m/s}) \sin \theta \end{aligned}$$

(a) With $\theta = 90^\circ$,

$$v_C = 3.6 \text{ m/s}$$

$$+\uparrow \Sigma F_y = 0: F - W_C = 0$$

or

$$F = m_C g$$

Now

$$F = \mu_s N$$

or

$$N = \frac{1}{\mu_s} m_C g$$

$$\leftarrow + \Sigma F_n = m_C a_n: N = m_C \frac{v_C^2}{r}$$

or

$$\frac{1}{\mu_s} m_C g = m_C \frac{v_C^2}{r}$$

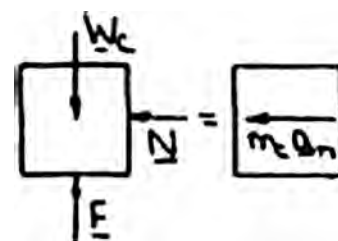
or

$$\mu_s = \frac{gr}{v_C^2} = \frac{(9.81 \text{ m/s}^2)(0.6 \text{ m})}{(3.6 \text{ m/s})^2}$$

or

$$(\mu_s)_{\min} = 0.454 \quad \blacktriangleleft$$

The direction of the impending motion is downward. $\blacktriangleleft$



PROBLEM 12.127 (Continued)

(b) and (c)

First observe that for an arbitrary value of θ , it is not known whether the impending motion will be upward or downward. To consider both possibilities for each value of θ , let F_{down} correspond to impending motion downward, F_{up} correspond to impending motion upward, then with the “top sign” corresponding to F_{down} , we have

$$+\uparrow \Sigma F_y = 0: \quad N \cos \theta \pm F \sin \theta - W_C = 0$$

Now

$$F = \mu_s N$$

Then

$$N \cos \theta \pm \mu_s N \sin \theta - m_C g = 0$$

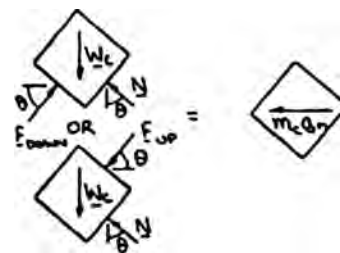
or

$$N = \frac{m_C g}{\cos \theta \pm \mu_s \sin \theta}$$

and

$$F = \frac{\mu_s m_C g}{\cos \theta \pm \mu_s \sin \theta}$$

$$\leftarrow + \Sigma F_n = m_C a_n: \quad N \sin \theta \mp F \cos \theta = m_C \frac{v_C^2}{\rho} \quad \rho = r \sin \theta$$



Substituting for N and F

$$\frac{m_C g}{\cos \theta \pm \mu_s \sin \theta} \sin \theta \mp \frac{\mu_s m_C g}{\cos \theta \pm \mu_s \sin \theta} \cos \theta = m_C \frac{v_C^2}{r \sin \theta}$$

or

$$\frac{\tan \theta}{1 \pm \mu_s \tan \theta} \mp \frac{\mu_s}{1 \pm \mu_s \tan \theta} = \frac{v_C^2}{gr \sin \theta}$$

or

$$\mu_s = \pm \frac{\tan \theta - \frac{v_C^2}{gr \sin \theta}}{1 + \frac{v_C^2}{gr \sin \theta} \tan \theta}$$

Now

$$\frac{v_C^2}{gr \sin \theta} = \frac{[(3.6 \text{ m/s}) \sin \theta]^2}{(9.81 \text{ m/s}^2)(0.6 \text{ m}) \sin \theta} = 2.2018 \sin \theta$$

Then

$$\mu_s = \pm \frac{\tan \theta - 2.2018 \sin \theta}{1 + 2.2018 \sin \theta \tan \theta}$$

(b) $\theta = 75^\circ$

$$\mu_s = \pm \frac{\tan 75^\circ - 2.2018 \sin 75^\circ}{1 + 2.2018 \sin 75^\circ \tan 75^\circ} = \pm 0.1796$$

PROBLEM 12.127 (Continued)

Then

downward: $\mu_s = +0.1796$

upward: $\mu_s < 0$ not possible

$$(\mu_s)_{\min} = 0.1796 \quad \blacktriangleleft$$

The direction of the impending motion is downward. $\blacktriangleleft$

(c) $\theta = 45^\circ$

$$\mu_s = \pm \frac{\tan 45^\circ - 2.2018 \sin 45^\circ}{1 + 2.2018 \sin 45^\circ \tan 45^\circ} = \pm(-0.218)$$

Then

downward: $\mu_s < 0$ not possible

upward: $\mu_s = 0.218$

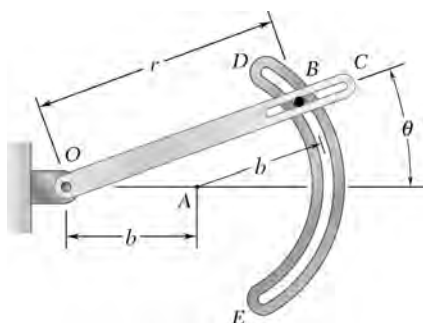
$$(\mu_s)_{\min} = 0.218 \quad \blacktriangleleft$$

The direction of the impending motion is upward. $\blacktriangleleft$

Note: When $\tan \theta - 2.2018 \sin \theta = 0$

or $\theta = 62.988^\circ$,

$\mu_s = 0$. Thus, for this value of θ , friction is not necessary to prevent the collar from sliding on the rod.



PROBLEM 12.128

Pin B weighs 100 g and is free to slide in a horizontal plane along the rotating arm OC and along the circular slot DE of radius $b = 500$ mm. Neglecting friction and assuming that $\dot{\theta} = 15$ rad/s and $\ddot{\theta} = 250$ rad/s² for the position $\theta = 20^\circ$, determine for that position (a) the radial and transverse components of the resultant force exerted on pin B , (b) the forces $\mathbf{P}$ and $\mathbf{Q}$ exerted on pin B , respectively, by rod OC and the wall of slot DE .

SOLUTION

Kinematics.

From the drawing of the system, we have

$$r = 2b \cos \theta$$

Then

$$\dot{r} = -(2b \sin \theta) \dot{\theta}$$

and

$$\ddot{r} = -2b(\ddot{\theta} \sin \theta + \dot{\theta}^2 \cos \theta)$$

Now

$$\begin{aligned} a_r &= \ddot{r} - r\dot{\theta}^2 = -2b(\ddot{\theta} \sin \theta + \dot{\theta}^2 \cos \theta) - (2b \cos \theta) \dot{\theta}^2 \\ &= -2b(\ddot{\theta} \sin \theta + 2\dot{\theta}^2 \cos \theta) \\ &= -2(0.5 \text{ m})[(250 \text{ rad/s}^2) \sin 20^\circ + 2(15 \text{ rad/s})^2 \cos 20^\circ] \\ &= -508.367 \text{ m/s}^2 \end{aligned}$$

and

$$\begin{aligned} a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} = (2b \cos \theta)\ddot{\theta} + 2(-2b\dot{\theta} \sin \theta)\dot{\theta} \\ &= 2b(\ddot{\theta} \cos \theta - 2\dot{\theta}^2 \sin \theta) \\ &= 2(0.5 \text{ m})[(250 \text{ rad/s}^2) \cos 20^\circ - 2(15 \text{ rad/s})^2 \sin 20^\circ] \\ &= 81.014 \text{ m/s}^2 \end{aligned}$$

Kinetics.

(a) We have

$$F_r = ma_r = 0.1 \text{ kg} \times (-508.367 \text{ m/s}^2) = -50.8367 \text{ N}$$

or

$$F_r = -50.8 \text{ N} \quad \blacktriangleleft$$

and

$$F_\theta = ma_\theta = 0.1 \text{ kg} \times (81.014 \text{ m/s}^2) = 8.1014 \text{ N}$$

or

$$F_\theta = 8.10 \text{ N} \quad \blacktriangleleft$$

PROBLEM 12.128 (Continued)

(b)

$$\rightarrow \Sigma F_r: -F_r = -Q \cos 20^\circ$$

or

$$Q = \frac{1}{\cos 20^\circ} (50.8367 \text{ N})$$

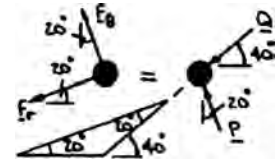
$$= 54.099 \text{ N}$$

$$\uparrow \Sigma F_\theta: F_\theta = P - Q \sin 20^\circ$$

or

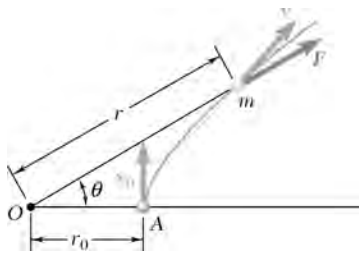
$$P = (8.1014 + 54.099 \sin 20^\circ) \text{ N}$$

$$= 26.604 \text{ N}$$



$$P = 26.6 \text{ N} \nearrow 70^\circ \blacktriangleleft$$

$$Q = 54.1 \text{ N} \nearrow 40^\circ \blacktriangleleft$$



PROBLEM 12.129

A particle of mass m is projected from Point A with an initial velocity $\mathbf{v}_0$ perpendicular to OA and moves under a central force $\mathbf{F}$ directed away from the center of force O . Knowing that the particle follows a path defined by the equation $r = r_0 / \cos 2\theta$, and using Eq. (12.27), express the radial and transverse components of the velocity $\mathbf{v}$ of the particle as functions of the angle θ .

SOLUTION

We have

$$r = \frac{r_0}{\cos 2\theta}$$

Then

$$\dot{r} = \frac{2r_0 \sin 2\theta}{\cos^2 2\theta} \dot{\theta}$$

Now

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta$$

so that at $t = 0$,

$$v_0 = r_0 \dot{\theta}_0$$

From Eq. (12.27):

$$r^2 \dot{\theta} = r_0^2 \dot{\theta}_0 = r_0 v_0$$

or

$$\dot{\theta} = \frac{r_0 v_0}{r^2} = r_0 v_0 \left(\frac{\cos 2\theta}{r_0} \right)^2 = \frac{v_0}{r_0} \cos^2 2\theta$$

Then

$$\dot{r} = \frac{2r_0 \sin 2\theta}{\cos^2 2\theta} \left(\frac{v_0}{r_0} \cos^2 2\theta \right) = 2v_0 \sin 2\theta$$

Now

$$v_r = \dot{r}$$

or

$$v_r = 2v_0 \sin 2\theta \quad \blacktriangleleft$$

and

$$v_\theta = r\dot{\theta} = \frac{r_0}{\cos 2\theta} = \frac{v_0}{r_0} \cos^2 2\theta$$

or

$$v_\theta = v_0 \cos 2\theta \quad \blacktriangleleft$$

PROBLEM 12.130

Show that the radius r of the moon's orbit can be determined from the radius R of the earth, the acceleration of gravity g at the surface of the earth, and the time τ required for the moon to complete one full revolution about the earth. Compute r knowing that $\tau = 27.3$ days.

SOLUTION

We have

$$F = G \frac{Mm}{r^2} \quad [\text{Eq. (12.28)}]$$

and

$$F = F_n = ma_n = m \frac{v^2}{r}$$

Then

$$G \frac{Mm}{r^2} = m \frac{v^2}{r}$$

or

$$v^2 = \frac{GM}{r}$$

Now

$$GM = gR^2 \quad [\text{Eq. (12.30)}]$$

so that

$$v^2 = \frac{gR^2}{r} \quad \text{or} \quad v = R \sqrt{\frac{g}{r}}$$

For one orbit,

$$\tau = \frac{2\pi r}{v} = \frac{2\pi r}{R \sqrt{\frac{g}{r}}}$$

or

$$r = \left(\frac{g\tau^2 R^2}{4\pi^2} \right)^{1/3} \quad \text{Q.E.D.} \quad \blacktriangleleft$$

Now

$$\tau = 27.3 \text{ days} = 2.35872 \times 10^6 \text{ s}$$

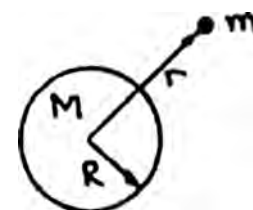
$$R = 3960 \text{ mi} = 20.9088 \times 10^6 \text{ ft}$$

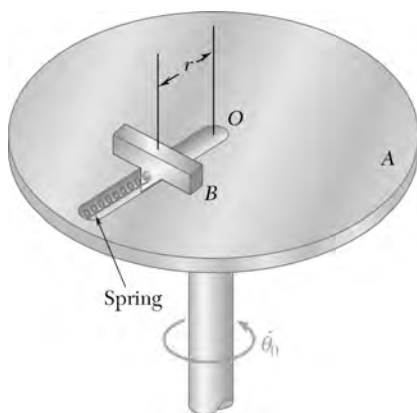
SI:

$$r = \left[\frac{9.81 \text{ m/s}^2 \times (2.35872 \times 10^6 \text{ s})^2 \times (6.37 \times 10^6 \text{ m})^2}{4\pi^2} \right]^{1/3} = 382.81 \times 10^6 \text{ m}$$

or

$$r = 383 \times 10^3 \text{ km} \quad \blacktriangleleft$$





PROBLEM 12.131*

Disk A rotates in a horizontal plane about a vertical axis at the constant rate $\dot{\theta}_0 = 10$ rad/s. Slider B weighs 240 g and moves in a frictionless slot cut in the disk. The slider is attached to a spring of constant k , which is undeformed when $r = 0$. Knowing that the slider is released with no radial velocity in the position $r = 400$ mm, determine the position of the slider and the horizontal force exerted on it by the disk at $t = 0.1$ s for (a) $k = 24$ N/m, (b) $k = 40$ N/m.

SOLUTION

First we note

$$\text{when } r = 0, \quad X_{sp} = 0 \Rightarrow F_{sp} = kr$$

$$r_0 = 400 \text{ mm} = 0.4 \text{ m}$$

and

$$\dot{\theta} = \dot{\theta}_0 = 10 \text{ rad/s}$$

then

$$\ddot{\theta} = 0$$

$$\leftarrow^+ \Sigma F_r = m_B a_r : -F_{sp} = m_B (\ddot{r} - r \dot{\theta}_0^2)$$

or

$$\ddot{r} + \left(\frac{k}{m_B} - \dot{\theta}_0^2 \right) r = 0 \quad (1)$$

$$\downarrow^+ \Sigma F_\theta = m_B a_\theta : F_A = m_B (0 + 2\dot{r}\dot{\theta}_0) \quad (2)$$

(a)

$$k = 24 \text{ N/m}$$

Substituting the given values into Eq. (1)

$$\ddot{r} + \left[\frac{24}{0.24} - (10)^2 \right] r = 0$$

or

$$\ddot{r} = 0$$

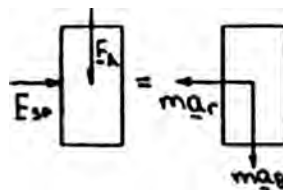
Then

$$\frac{d\dot{r}}{dt} = \ddot{r} = 0 \text{ and at } t = 0, \dot{r} = 0 :$$

$$\int_0^{\dot{r}} d\dot{r} = \int_0^{0.1} (0) dt$$

or

$$\dot{r} = 0$$



PROBLEM 12.131* (Continued)

and $\frac{dr}{dt} = \dot{r} = 0$ and at $t = 0$, $r_0 = 0.4$ m

$$\int_{r_0}^r dr = \int_0^{0.1} (0) dt$$

or $r = r_0$

$$r = 0.4 \text{ m} \quad \blacktriangleleft$$

Note: $\dot{r} = 0$ implies that the slider remains at its initial radial position.

With $\dot{r} = 0$, Eq. (2) implies

$$F_A = 0 \quad \blacktriangleleft$$

(b) $k = 40$ N/m

Substituting the given values into Eq. (1)

$$\ddot{r} + \left[\frac{40}{0.24} - 100 \right] r = 0$$

or $\ddot{r} + \frac{200}{3} r = 0$

Now $\ddot{r} = \frac{d}{dt}(\dot{r}) \quad \dot{r} = v_r \quad \frac{d}{dt} = \frac{dr}{dt} \frac{d}{dr} = v_r \frac{d}{dr}$

Then $\ddot{r} = v_r \frac{dv_r}{dr}$

so that $v_r \frac{dv_r}{dr} + \frac{200}{3} r = 0$

At $t = 0$, $v_r = 0$, $r = r_0$: $\int_0^{v_r} v_r dv_r = -\frac{200}{3} \int_{r_0}^r r dr$

or $v_r^2 = -\frac{200}{3} (r^2 - r_0^2)$

or $v_r = \frac{10\sqrt{6}}{3} \sqrt{r_0^2 - r^2}$

Now $v_r = \frac{dr}{dt} = \frac{10\sqrt{6}}{3} \sqrt{r_0^2 - r^2}$

At $t = 0$, $r = r_0$: $\int_{r_0}^r \frac{dr}{\sqrt{r_0^2 - r^2}} = \int_0^t \frac{10\sqrt{6}}{3} dt = \frac{10\sqrt{6}}{3} t$

PROBLEM 12.131* (Continued)

Let $r = r_0 \sin \phi, \quad dr = r_0 \cos \phi d\phi$

Then
$$\int_{\pi/2}^{\sin^{-1}(r/r_0)} \frac{r_0 \cos \phi d\phi}{\sqrt{r_0^2 - r_0^2 \sin^2 \phi}} = \frac{10\sqrt{6}}{3} t$$

or
$$\int_{\pi/2}^{\sin^{-1}(r/r_0)} d\phi = \frac{10\sqrt{6}}{3} t$$

or
$$\sin^{-1}\left(\frac{r}{r_0}\right) - \frac{\pi}{2} = \frac{10\sqrt{6}}{3} t$$

or
$$r = r_0 \sin\left(\frac{10\sqrt{6}}{3} t + \frac{\pi}{2}\right) = r_0 \cos \frac{10\sqrt{6}}{3} t = (0.4 \text{ m}) \cos \frac{10\sqrt{6}}{3} t$$

Then
$$\dot{r} = -\frac{4\sqrt{6}}{3} \sin \frac{10\sqrt{6}}{3} t$$

Finally, at $t = 0.1 \text{ s}$:

$$r = (0.4 \text{ m}) \cos\left(\frac{10\sqrt{6}}{3} \times 0.1\right)$$

or $r = 0.274 \text{ m} \quad \blacktriangleleft$

Eq. (2)
$$F_H = 0.24 \text{ kg} \times 2 \times \left[-\frac{4\sqrt{6}}{3} \sin\left(\frac{\sqrt{6}}{3}\right) \right] (10 \text{ rad/s})$$

or $F_H = -11.42 \text{ N} \quad \blacktriangleleft$

PROBLEM 12.132

It was observed that as the Voyager I spacecraft reached the point of its trajectory closest to the planet Saturn, it was at a distance of $185 \times 10^3 \text{ km}$ from the center of the planet and had a velocity of 21.0 km/s . Knowing that Tethys, one of Saturn's moons, describes a circular orbit of radius $295 \times 10^3 \text{ km}$ at a speed of 11.35 km/s , determine the eccentricity of the trajectory of Voyager I on its approach to Saturn.

SOLUTION

For a circular orbit,

Eq. (12.44)

$$v = \sqrt{\frac{GM}{r}}$$

For the orbit of Tethys,

$$GM = r_T v_T^2$$

For Voyager's trajectory, we have

$$\frac{1}{r} = \frac{GM}{h^2} (1 + \varepsilon \cos \theta)$$

where $h = r_0 v_0$

At O ,

$$r = r_0, \quad \theta = 0$$

Then

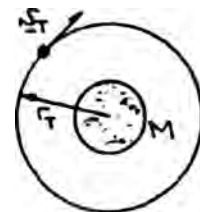
$$\frac{1}{r_0} = \frac{GM}{(r_0 v_0)^2} (1 + \varepsilon)$$

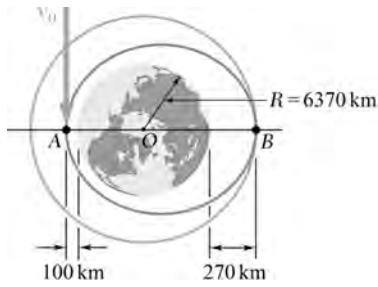
or

$$\begin{aligned} \varepsilon &= \frac{r_0 v_0^2}{GM} - 1 = \frac{r_0 v_0^2}{r_T v_T^2} - 1 \\ &= \frac{185 \times 10^3 \text{ km}}{295 \times 10^3 \text{ km}} \times \left(\frac{21.0 \text{ km/s}}{11.35 \text{ km/s}} \right)^2 - 1 \end{aligned}$$

or

$$\varepsilon = 1.147 \quad \blacktriangleleft$$





PROBLEM 12.133

At engine burnout on a mission, a shuttle had reached Point A at an altitude of 80 km above the surface of the earth and had a horizontal velocity v_0 . Knowing that its first orbit was elliptic and that the shuttle was transferred to a circular orbit as it passed through Point B at an altitude of 270 km, determine (a) the time needed for the shuttle to travel from A to B on its original elliptic orbit, (b) the periodic time of the shuttle on its final circular orbit.

SOLUTION

For Earth,

$$R = 6370 \text{ km} = 6.37 \times 10^6 \text{ m}, \quad g = 9.81 \text{ m/s}^2$$

$$GM = gR^2 = (9.81)(6.37 \times 10^6)^2 = 3.98059 \times 10^{14} \text{ m}^3/\text{s}^2$$

(a) For the elliptic orbit,

$$r_A = 6370 + 80 = 6450 \text{ km} = 6.45 \times 10^6 \text{ m}$$

$$r_B = 6370 + 270 = 6640 \text{ km} = 6.64 \times 10^6 \text{ m}$$

$$a = \frac{1}{2}(r_A + r_B) = 6.545 \times 10^6 \text{ m}$$

$$b = \sqrt{r_A r_B} = \sqrt{(6.45 \times 10^6)(6.545 \times 10^6)} = 6.49733 \times 10^6 \text{ m}$$

Using Eq. 12.39,
$$\frac{1}{r_A} = \frac{GM}{h^2} + C \cos \theta_A$$

and
$$\frac{1}{r_B} = \frac{GM}{h^2} + C \cos \theta_B$$

But $\theta_B = \theta_A + 180^\circ$, so that $\cos \theta_A = -\cos \theta_B$

Adding,
$$\frac{1}{r_A} + \frac{1}{r_B} = \frac{r_A + r_B}{r_A r_B} = \frac{2a}{b^2} = \frac{2GM}{h^2}$$

or
$$h = \sqrt{\frac{GMb^2}{a}}$$

Periodic time.
$$\tau = \frac{2\pi ab}{h} = \frac{2\pi ab\sqrt{a}}{\sqrt{GMb^2}} = \frac{2\pi a^{3/2}}{\sqrt{GM}}$$

$$\tau = \frac{2\pi(6.545 \times 10^6)^{3/2}}{\sqrt{3.98059 \times 10^{14}}} = 5273.156 \text{ s} = 1 \text{ hr } 27 \text{ min } 53 \text{ sec}$$

The time to travel from A to B is one half the periodic time

$$\tau_{AB} = 2636.58 \text{ s}$$

$$\tau_{AB} = 43 \text{ min } 56.6 \text{ sec} \quad \blacktriangleleft$$

PROBLEM 12.133 (Continued)

(b) For the circular orbit,

$$a = b = r_B = 6.64 \times 10^6 \text{ m}$$

$$\tau_{\text{circ}} = \frac{2\pi a^{3/2}}{\sqrt{GM}} = \frac{2\pi(6.64 \times 10^6)^{3/2}}{\sqrt{3.98059 \times 10^{14}}} = 5388 \text{ s}$$

$$\tau_{\text{circ}} = 1 \text{ h } 29 \text{ min } 48 \text{ sec}$$

$$\tau_{\text{circ}} = 1 \text{ h } 29 \text{ min } 48 \text{ sec} \blacktriangleleft$$